

SAINT MICHAEL COLLEGE OF CARAGA

INSTITUTIONAL SCHOLARSHIP MANAGEMENT SYSTEM FOR SAINT MICHAEL COLLEGE OF CARAGA

A Thesis Presented To
The Faculty of
College of Computing and Information Sciences

In Partial Requirements for the Degree
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY



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SAINT MICHAEL COLLEGE OF CARAGA

SAINT MICHAEL COLLEGE OF CARAGA
College Of Computing and Information Sciences

CERTIFICATE OF RESEARCH APPROVAL

This capstone entitled “**INSTITUTIONAL SCHOLARSHIP MANAGEMENT SYSTEM FOR SAINT MICHAEL COLLEGE OF CARAGA**” presented and submitted by **JHON IVAN C. TABANAO, MARK CHRISTIAN JOVEN P. BALATAYO, RONALD JHON N. REOJA, WARREN C. SALVA, and JOHN MARK G. SARABIA** in partial fulfillment of the requirements for the degree of **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY** is hereby accepted and recommended for Oral Examination.

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**ABSTRACT**

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I. OBJECTIVES

This study aims to design, develop, and implement a Web-Based Institutional Scholarship Management System for the Admission and Scholarship Office of Saint Michael College of Caraga. The system aims to make the scholarship process faster, easier, and more organized for both applicants and school personnel. Through this system, applicants can submit their applications and scholarship requirements online instead of going to the office in person, which helps reduce paper usage, delays, and lost documents. It also aims to automatically rank qualified applicants based on their test results, grades, and interview scores, making the selection process fairer and more accurate. In addition, the system provides an automatic review and reporting feature that allows heads of offices to submit reviews on time and helps the scholarship office generate reports more efficiently. Finally, the system will be evaluated using ISO 25010 standards to determine if it is functional, efficient, and easy to use. These goals are aligned with the study's goal of solving problems caused by manual scholarship processing, such as delays, errors, and difficulty tracking records.



II. METHODOLOGY

The Institutional Scholarship Management System is developed using the Software Development Life Cycle (SDLC) methodology with a three-tier architecture to ensure organized processing, secure access, and reliable system performance. System design used UML diagrams, including use case, activity, and sequence diagrams, along with an Entity Relationship Diagram (ERD) to define the structure of the database. It is built using PHP and MySQL, along with front-end technologies such as HTML, CSS, JavaScript, Tailwind, and jQuery. The system includes role-based access control (RBAC) for applicants, panelists, heads of offices, and administrators, and has been evaluated using ISO 25010 standards in terms of functionality, efficiency, and usability.

III. FINDINGS

The Institutional Scholarship Management System received high ratings in functional suitability, performance efficiency, and usability based on ISO 25010 standards. The system obtained a grand mean of 3.85, which is interpreted as Very Acceptable, showing that it is effective, reliable, efficient, and easy to use. These results indicate that the system can support the scholarship application, ranking, evaluation, and reporting processes of Saint Michael College of Caraga. However, further improvements are still recommended, such as improving notification and announcement features and adding support tools in the future to make the system more responsive, organized, and useful for its users.

IV. RECOMMENDATIONS

To improve the effectiveness and user experience of the Institutional Scholarship Management System, the following improvements are recommended: (1) add an announcement and notification feature to the homepage so that applicants, scholars, and administrators can easily receive updates about deadlines, interview schedules, requirements, and important



reminders; (2) integrate an AI-based scholar review feature that will help review scholar records, academic performance, renewal status, and other relevant data to support better decision-making; (3) add an AI chatbot that can answer common questions about scholarship requirements, application status, schedules, and qualifications; and (4) continually improve the features, security, and usability of the system based on user feedback to make it more reliable, organized, and useful for future use.

KEYWORDS: Web-Based System, Scholarship Management, Online Application, Automated Ranking, Evaluation And Reporting, Role-Based Access Control.



DEDICATION

This capstone research is sincerely dedicated to the people who provided their support, guidance, and encouragement throughout the completion of this study.

To the families of the researchers, they express their sincere gratitude for their love, patience, understanding, and continuous support. Their encouragement gave strength to the researchers to continue to work hard and overcome the challenges they faced in developing this study.

With their advisor, Ms. Rea Mie A. Omas-As, the researchers offer their deepest appreciation for her guidance, time, and valuable suggestions. His support and feedback helped improve the study and guided the researchers in completing their capstone project.

This study is also dedicated to the researchers themselves, whose cooperation, effort, patience, and commitment made this project possible. Each member's contribution helped turn the idea of an Institutional Scholarship Management System into a meaningful and functional system for Saint Michael College of Caraga.

Above all, the researchers dedicate this work to all who have been a part of their journey. This success would not have been possible without the people who supported and inspired them from start to finish.



ACKNOWLEDGEMENT

The researchers would like to begin by giving thanks to God for giving them strength, wisdom, patience, and guidance throughout the completion of this capstone project. Through His help, the researchers were able to continue despite the challenges they encountered in the study.

The researchers also express their deep gratitude to their advisor, Ms. Rea Mie A. Omas-As, for sharing her time, knowledge, and guidance. His comments and suggestions helped the researchers improve their work and complete the Institutional Scholarship Management System for Saint Michael College of Caraga.

The researchers also thank their instructors and panel members for their valuable feedback and recommendations. Their guidance helped the researchers to make the study clearer, more organized, and more meaningful.

To their families, the researchers offer sincere thanks for their love, patience, trust, and support. Their encouragement motivated the researchers to continue the effort until the study was completed.

The researchers are also grateful to their friends, classmates, and other people who provided help, advice, and encouragement in the development of this project. Their support made the journey easier and more memorable.

Finally, the researchers would like to thank everyone who has been a part of this capstone journey in any way. Every support, suggestion, and kind word helped the researchers to reach this success.



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CHAPTER 1 INTRODUCTION

1.1 Project Context

An institutional scholarship program plays an essential role in helping students continue their education by avoiding the financial pressure of tuition and school expenses. It is a hopeful and generous gift, especially for students who might not have much money, which can help them concentrate on their studies and learning in the future. However, despite its importance, many scholars are managed using an antiquated system that relies on a manual process. From applications to qualifications reviews and approvals, most methods are manual, leading to time delays, human error, and inefficiency for administrators and students alike. As a result, errors may occur during the ranking of the qualifiers, leading deserving students to lose their eligible places. In contrast, others who may not fully meet the standards are mistakenly placed first.

Global studies highlight the effectiveness of digitalization in the scholarship application process. According to Vaidya et al. [1], the traditional manual scholarship process remains ineffective, error-prone, and difficult to manage, especially as the number of applicants grows each academic year. The researchers developed a web-based scholarship management system to address the challenges of the traditional scholarship application process. However, gaps remain in extending these solutions to other specific contexts, including a lack of intelligent recommendation functionality and automated evaluation for scholars' retention and ranking, especially in interdepartmental evaluation.

In the local Context, Bagaporo and Espinosa [2] pointed out that paper-based systems are prone to typographical errors, insufficient information, or incorrect documents, which inadvertently disqualify eligible applicants. They also consume a lot of resources and time. For Camarines Sur Polytechnic Colleges, researchers have developed an online scholarship management system to overcome the



challenges posed by the paper-based process. As in the global context, these solutions should be enhanced for other private institutions with specific needs, such as Saint Michael College of Caraga, which provides many institutional scholarship grants in addition to monitoring the semester for awards.

Over the years, the institutional scholarship department has dealt with traditional paper-based applications, especially with the Saint Michael College of Caraga Admission and Scholarship Office. It is in this context that Saint Michael College of Caraga's scholarship program became one of the institution's essential tools for helping poor but deserving students gain access to quality education. However, its management process greatly relies on manual operations. Lack of performance-based, prospective maintenance automation often leads to lag, inconsistencies, and opacity in decision-making. As the number of scholars increases, administrators are facing greater difficulty in monitoring compliance and accurately and precisely evaluating academic performance. Therefore, a reliable digital solution is essential to ensure that the institution can promote an efficient and equitable scholarship program.

To solve this problem, the researchers propose developing a web-based management system using PHP, JavaScript, and MySQL. This system will create a faster and more organized process for handling scholarship applications through a digital platform. It also aims to improve the accuracy and security of confidential data, reducing errors and risks with manual handling [2]. In addition, in the study of a web-based scholarship information system by Prijayanti, Artha, and Arumi, the researchers created an online system to replace the manual and paper-based process used in universities for scholarship applications. Before this system, students had to submit printed forms, grades, and approval papers, which were stored in folders. This process often resulted in lost documents, delays, and problems in evaluating scholars' performance. The new web-based system makes it possible for students to submit



their requirements online, and for staff to review, verify, and approve applications using a central database [3].

This study is important to many stakeholders. For scholars, it promotes a fair, merit-based evaluation process that motivates them to maintain high academic performance and comply with institutional requirements. For administrators and evaluators, the system provides an easy-to-use tool for monitoring, reporting, and evaluating. This helps reduce their workload and reduces the chances of bias or error. For the institution, the system strengthens the scholarship program by ensuring that deserving students meet the criteria and receive the resources they need. Finally, for future researchers and other educational institutions, this project can serve as a model or guide for creating similar systems that improve scholarly management, promote transparency, and support performance-based maintenance process



1.2 Review of Related Literature

This section presents a collection of related literature that helps researchers understand and address the problems identified in the study. It examines the concept of digitalization and its effects on processes across different systems, especially in scholarly management. The discussion is organized around three main themes: efficiency, transparency, and fairness, which are essential to any digital transformation effort. These themes explain how technology can ease administrative work, promote accountability, and ensure that decisions are made objectively. Besides highlighting the positive effects of digitalization, this section also includes studies that present critical perspectives. Some researchers warn that overreliance on digital platforms may create new challenges, including system errors, privacy risks, and unequal access among users. By considering both supportive and critical perspectives, this review provides a balanced discussion that situates the current study within a broader academic context. This allows researchers to build on existing knowledge, identify gaps in previous studies, and justify the need for a more reliable, performance-based scholarly management system.

Digital Process

Bagaporo and Espinosa [1] developed an online scholarship management system for Camarines Sur Polytechnic Colleges that simplified the application, verification, and tracking of scholarship records. Their study highlighted how digitization improved transparency, reduced errors, and increased administrative efficiency. Similarly, Al-Ayyubi and Maulana [2] developed a web-based information system that consolidated records and reduced data redundancy. Both studies align with the theme of leveraging technology to optimize access and efficiency, but they also note gaps such as limited integration and scalability in existing systems. Synthesizing these studies, both works relate to the



theme that seizure technology improves access, accuracy, and efficiency in scholarly management, providing a strong case for digitization as a means of modernizing administrative processes.

However, not all studies support full digitization without proper preparation. Gkrimpizi et al. [4] reviewed 44 research studies from an initial 768 articles and identified 20 common barriers to digital innovation in higher education. The most frequent problems are resistance to change, lack of digital skills, poor IT infrastructure, and limited budgets. The study also showed that internet access is uneven, with 81% access in developed countries, but only 45% in developing countries, which affects how digital systems perform. Similarly, Robertsons and Lapina [5] warned that digital systems are not a one-size-fits-all solution and can increase inefficiencies and inequities when implemented without proper planning. Collectively, these studies show that while digitization can be helpful, it also carries risks if institutions are not ready. Therefore, digitization must be carefully planned, well supported, and user-friendly to ensure that its benefits are realized and not undermined by poor implementation.

Email Notification

Using email notifications in scholarship systems helps make communication between students and the office faster and easier. In the study of a comprehensive scholarship and management system (SCHOGMS) by RSI International [6], the system includes an email feature that automatically sends messages to applicants about their application status, approval, or renewal. The study found that this feature reduced staff workload and helped applicants make updates quickly and correctly. It also made the process more transparent as students could easily track what was happening with their scholarship. Another study, the paperless scholarship disbursement system of IJirset [7], showed the same results. The system used both email and SMS notifications to tell applicants when their application was received, approved, or released. The researchers explained that this saved time for both students and



staff and made communication smoother. The scholars became more satisfied as they did not have to visit the office just to ask for updates. In summary, studies show that automated email notifications make scholarly communication faster, clearer, and more reliable. They also reduce errors and help build trust between students and school offices because updates are sent immediately.

In contrast, some scholarship programs still use manual communication methods instead of email notifications. In a review of TESDA scholarship programs targeting the poor [8], it was found that many local offices still use phone calls, printed notices, or face-to-face announcements to inform applicants about their scholarship status. The traditional way of updating students often leads to delays and confusion, especially for those who live far from the office. The study also noted that without email notifications, it becomes more difficult for students to stay informed about deadlines and results.

In another report by the Japan International Cooperation Agency (JICA) titled review of evaluation methods for scholarship programs [9], the same issue is observed. Many schools and agencies still send messages manually using personal or office emails instead of automated systems. Although it keeps communication more personal, it also takes longer and can cause mistakes like sending messages to the wrong person or forgetting to attach documents. Jica explained that the lack of a strong internet connection, limited technology, and privacy policies are some of the reasons why full automation is still not used in many institutions. In summary, these studies show that manual or traditional methods of sending notifications are still common. While they preserve personal communication, it is slower and less efficient, proving that automatic email notifications are better for faster and more organized communication.



1.3 Project Objectives

The main objective of this study is to design, develop, and implement a Web-Based Institutional Scholarship Management System for Saint Michael College of Caraga's Admission and Scholarship Office.

Specifically, it aims to:

1. Allow applicants to submit scholarship applications and requirements online, eliminating the need for walk-in submissions.
2. Automate the ranking of qualifiers based on the exam result, grades, and interview scores.
3. Provide an automated evaluation and reporting module that will facilitate the timely submission of evaluations from heads of offices and generate accurate reports efficiently.
4. Evaluate the system using ISO 25010 Standards to ensure its functionality, efficiency, and usability.

1.4 Scope and Limitations

This study focuses on the creation of the Institutional Scholarship Management System For Saint Michael College of Caraga. The system aims to make the scholarship process easier and faster by allowing students to submit applications online and by automatically ranking qualified applicants based on their test results, grades, and interview scores. It also includes a module where department heads can evaluate the performance of scholars and submit their assessments online. To keep data safe, all user passwords are encrypted before being saved, and the Role-Based Access Control (RBAC) feature ensures that only authorized users administrators, reviewers, and scholars can access and perform tasks according to their assigned roles.



The system will only be used for SMCC's Institutional Scholarship Program. It will not include scholarships funded by the government or private organizations. The system will also not have features for handling financial transactions, mobile app access, or other functions outside of scholarship review and maintenance. The pilot test will be conducted only among selected administrators and scholars from SMCC. Because of this limited testing, results may vary if the system is used in larger or different programs.

1.5. Definition of Terms

Automation - This means using technology or a computer system to do tasks automatically instead of doing them manually. In this system, the use of automation is to help speed up scholarship processes such as application submission, review, and report generation. This reduces human errors and saves time for both students and staff.

Encryption - This is the process of converting data into a secret or unreadable form to keep it safe from unauthorized access. In this study, encryption is used to protect important information such as student records, login details, and scholarly data from being viewed or stolen by others.

Hash Password – This is a security process that changes or "scrambles" plain-text passwords into a special code before saving them in the database. Once a password is hashed, it cannot be changed back to its original form, making it a one-way and irreversible process. This means that even if someone gains access to the database, they cannot easily read or steal the actual passwords. In the scholarship system, password hashing helps keep the accounts of scholars, administrators, and authorized personnel secure.

ISO 25010 - This is a quality standard used to measure how efficient and effective a system is. It lists important features such as usability, functionality, and efficiency. In this study, ISO 25010 is used to



evaluate the Institutional Scholarship Management System of Saint Michael College of Caraga to evaluate how easy it is to use, how well it works, and how quickly it performs.

JavaScript – This is a programming language used to make the system more interactive and dynamic. It allows web pages to quickly respond to user actions without reloading the entire page. In the scholarly system, JavaScript helps create a smoother and more engaging user experience by handling real-time form validation, interactive dashboards, and easy navigation.

MySQL – This is an open-source database management system used to store, organize, and manage all information in the scholarly system. It serves as the main storage area for securely maintaining records, including applicant details, academic performance, examination results, and retention decisions. MySQL enables the system to handle large amounts of data efficiently while keeping everything safe and well-organized. In the scholarship system, MySQL enables users to easily save, update, and retrieve data as needed.

PHP (Hypertext Preprocessor) – It is a server-side scripting language used to develop and manage the back-end functions of the system. It works behind the scenes to ensure that all processes run smoothly. PHP acts as the main foundation or "backbone" of the system's logic, handling important tasks such as processing form submissions, checking user inputs, and communicating with the database to save, update, or retrieve data. The use of PHP in the system is to connect the user interface to the database, allowing scholars, administrators, and authorized personnel to perform their tasks smoothly.

RBAC (Role-Based Access Control) – is a security model that grants users access to specific features or information based on their assigned roles. Instead of allowing everyone to use all parts of the system, RBAC limits access based on each user's role. This makes the system more efficient and secure. In the scholar system, RBAC helps to manage the actions of different users. Students can simply submit their scholarship applications and check the status of their submissions.



SMCC (Saint Michael College of Caraga) – Saint Michael College of Caraga, or SMCC, is a private Catholic educational institution located in Nasipit, Agusan del Norte. It is known for its commitment to academic excellence, values development, and community service. In this study, SMCC serves as the primary responder and primary setting where the proposed scholarship system is developed and implemented.

Web-Based System – This refers to an application that can be accessed through a web browser using the Internet, without having to install any software on a computer. It allows users to connect to the system anytime and anywhere, as long as they have an internet connection. A web-based system is convenient and efficient because it can be easily updated, ensures data consistency, and provides secure access for all users.

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CHAPTER 2

METHODOLOGY

2.1 Research Design

This study will employ a Design and Development Research (DDR) approach, which emphasizes both the systematic creation and evaluation of innovative solutions. Specifically, the DDR approach will guide the development of the Institutional Scholarship Management System for Saint Michael College of Caraga. Through this method, the research aims not only to design and build the system but also to assess its effectiveness in enhancing the operation of the SMCC Admission and Scholarship Office, particularly in monitoring scholars and supporting performance-based retention decisions.

Software Development Life Cycle

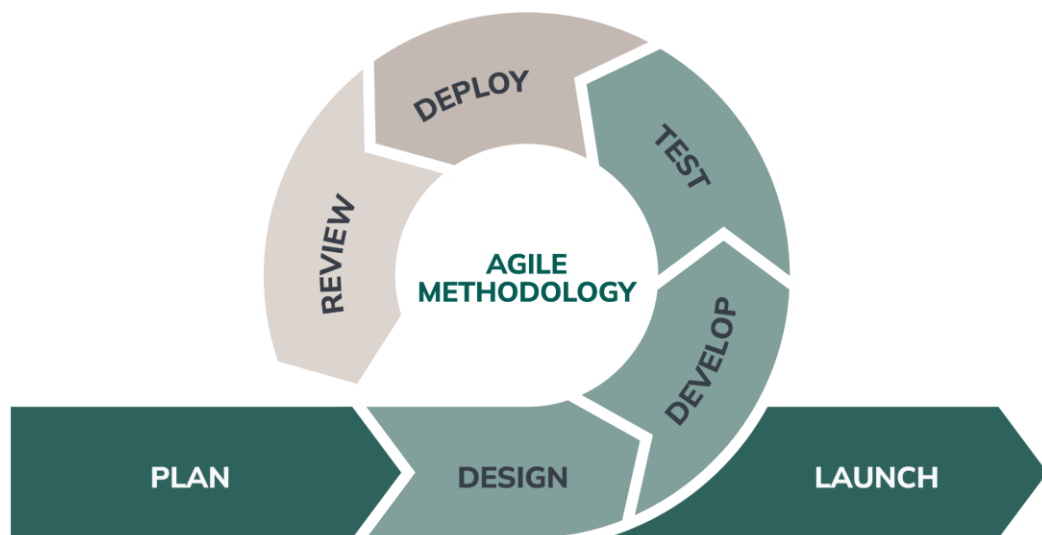


Figure 1. Agile Methodology

Figure 1 illustrates the development of the Institutional Scholarship Management System for Saint Michael College of Caraga that follows the Agile Software Development Life Cycle (SDLC). Agile focuses on flexibility, teamwork, and continuous improvement, making it ideal for systems that must adapt to user needs. The process begins with planning, where the team gathers system requirements



from administrators and scholars. It then moves into design and development, where system features are built step by step. After development, the system is tested to make sure it works properly. It is then deployed, and user feedback is collected for improvements. Finally, the system is launched and constantly updated to better serve its users. Through this cycle, the system improves over time and becomes more responsive to the needs of its users.

2.2 System Architecture

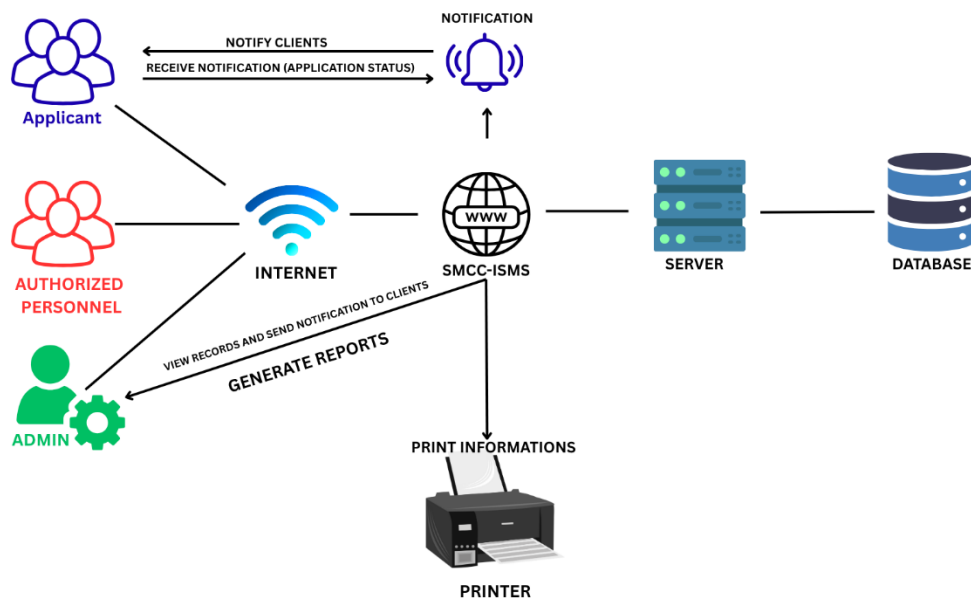


Figure 2. System Architecture

Figure 2 shows the system architecture of the Institutional Scholarship Management System For Saint Michael College of Caraga. It relates to three leading user roles: clients, authorized personnel, and admin. All users interact with the system through the internet, where their requests are handled by the application server and stored in the database server. The system sends notifications to keep users updated, while admin and authorized personnel can manage, verify, and track records. It also



supports printing reports or documents through a connected printer and displaying information on a screen, ensuring smooth data flow and coordination among all users.

2.3 Conceptual Diagram

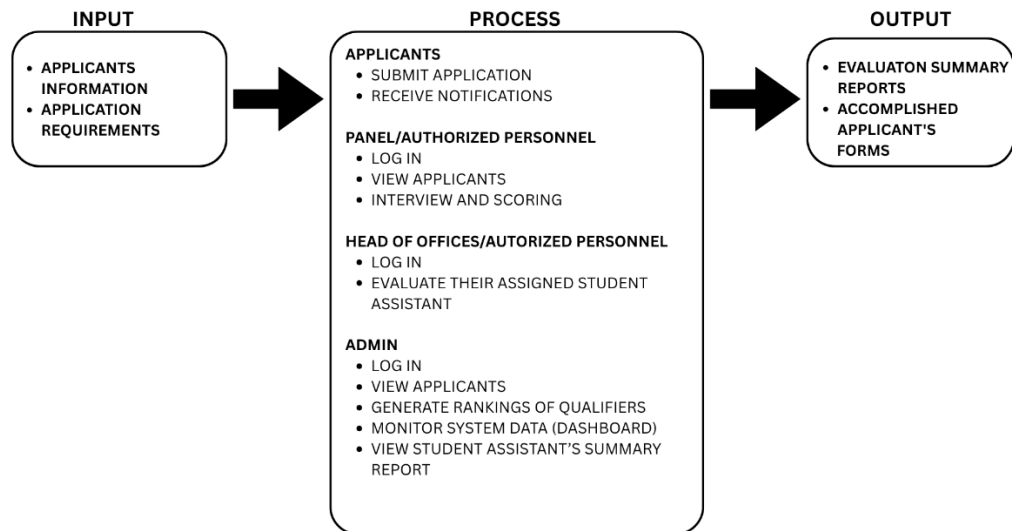


Figure 3. Conceptual Diagram

Figure 3 shows the Institutional Scholarship Management System For Saint Michael College of Caraga, where each user has specific roles and tasks. Applicants fill out application forms, upload required documents, and receive notifications about their status. Panel members or authorized personnel log into the system, review applicants, and conduct scoring interviews. Heads of offices review their assigned student assistants, while admins have full access to manage the system. Admins can log in, view applicants, generate rankings of qualified scholars, print information, create reports, and track data through the dashboard. All these actions work together to produce the system's main results - evaluation and maintenance decisions.



2.4 Use Case Diagram



Figure 4. Use Case Diagram

Figure 4 shows how the four main users of the system—applications, panelists, heads of offices, and admins—interact with it. Applicants can fill out application forms, upload required documents, and receive notifications. Panelists log in, view applicant information, and assign interview scores. Heads of offices log in to review their designated applicants. Admins have the most access; they can log in, view, and review applicants and qualifying rankings, generate reports, and send notifications. Overall, the diagram shows how all users work together to manage applications, reviews, and decisions effectively.



2.5 Activity Diagram

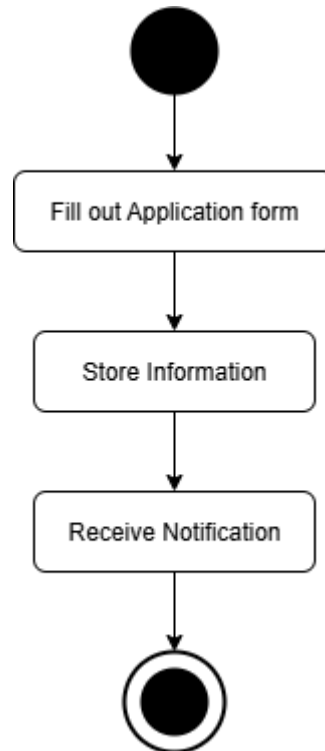


Figure 5. Activity Diagram - Applicant

Figure 5 illustrates the simple and user-friendly process an applicant follows when applying through the system. It begins with the applicant filling out the application form, where they provide their personal and required details. Once completed, the system stores all submitted information securely in the database to ensure proper recordkeeping and easy access for review. After the information is successfully stored, the system automatically processes and updates the applicant's status. Finally, the applicant receives a notification confirming that their application has been received and is under evaluation. This straightforward flow ensures efficiency, reduces manual errors, and keeps the applicant informed every step of the way.

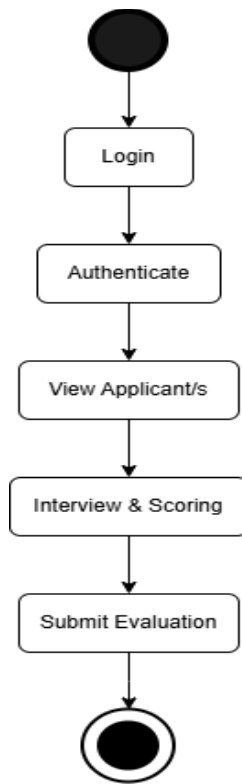


Figure 6 Activity Diagram - Panel

Figure 6 shows the steps a panel member follows when using the system. The process starts with the panel logging in and being verified through authentication. After successfully logging in, the panel can view the list of applicants to be interviewed. The panel then conducts the interview and gives scores based on the applicant's performance. Once the evaluation is complete, the panel submits the scores and feedback through the system. This organized process helps make the evaluation faster, more accurate, and ensures that all applicants data is properly recorded.

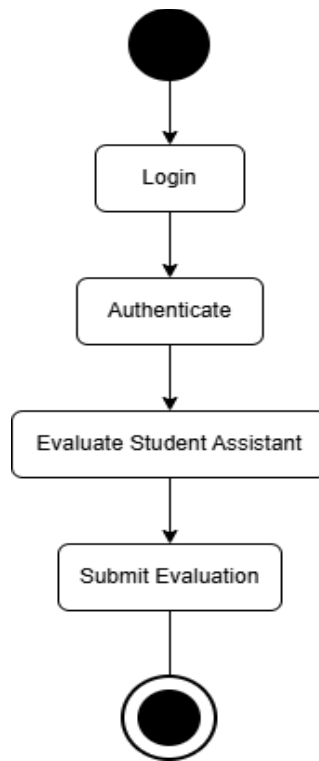


Figure 7. Activity Diagram - Head of Office

Figure 7 shows the simple process that a head of office follows when using the system. It starts when the head of office logs into their account and the system checks their credentials for authentication. Once verified, the head of office can evaluate the performance of a student assistant based on their assigned tasks or duties. After completing the evaluation, the head of office submits the results through the system. This flow ensures that evaluations are done efficiently, securely, and in a consistent way that helps support fair assessment of student assistants.

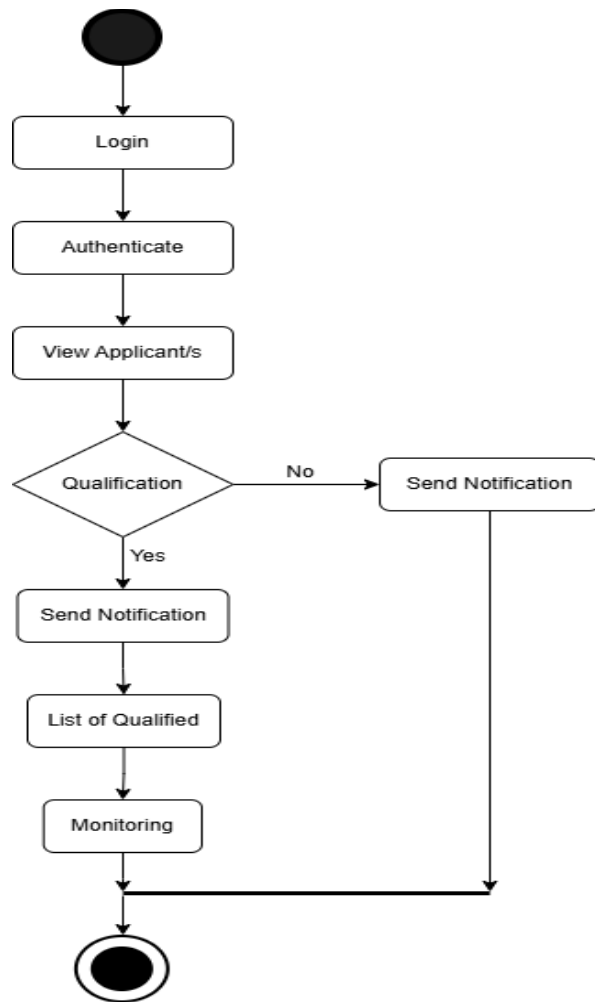


Figure 8. Activity Diagram - Admin

Figure 8 shows how the admin manages applicants in the system. It starts with the admin logging in and being authenticated to ensure secure access. Once logged in, the admin can view the list of applicants and check their qualifications. If an applicant meets the requirements, the admin sends a notification confirming their qualification and adds them to the list of qualified applicants. If not, the admin also sends notification informing them of the result. Afterward, the admin continues monitoring the list and updates as needed. This process helps the admin maintain fairness, transparency, and efficiency in handling application results.



2.6 Sequence Diagram

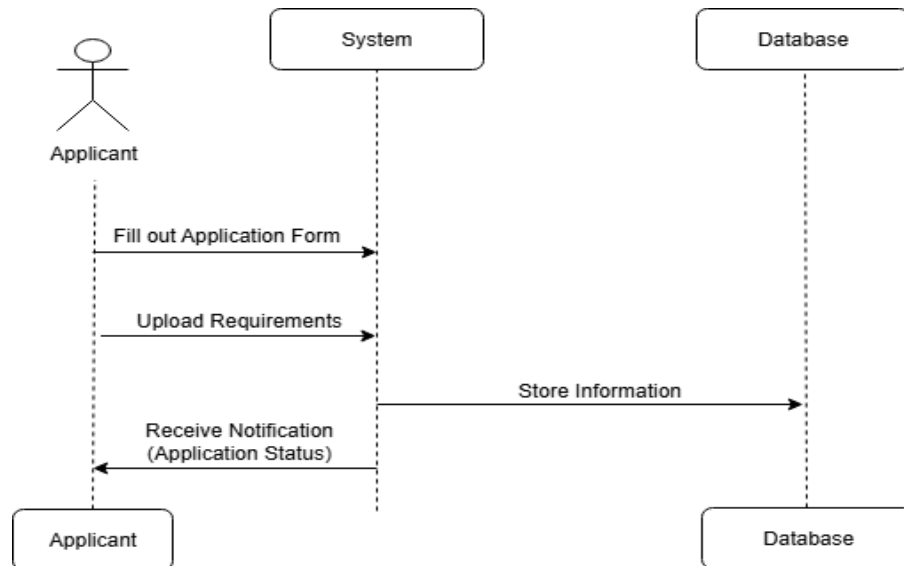


Figure 9. Sequence Diagram - Applicants

Figure 9. shows how an applicant uses the system through a web page and gets updates through a notification or text message. The process begins when the applicant fills the online form and uploads the required documents. After sending the application, the system checks and processes the submission. There are two possible outcomes. If the application is accepted, the system sends a message to tell the applicant about the approval. If it is rejected, a message is also sent to inform the applicant. The applicant can log into the system to see the status of the application at any time. This process ensures that communication is clear, prompt, and fair for each applicant.

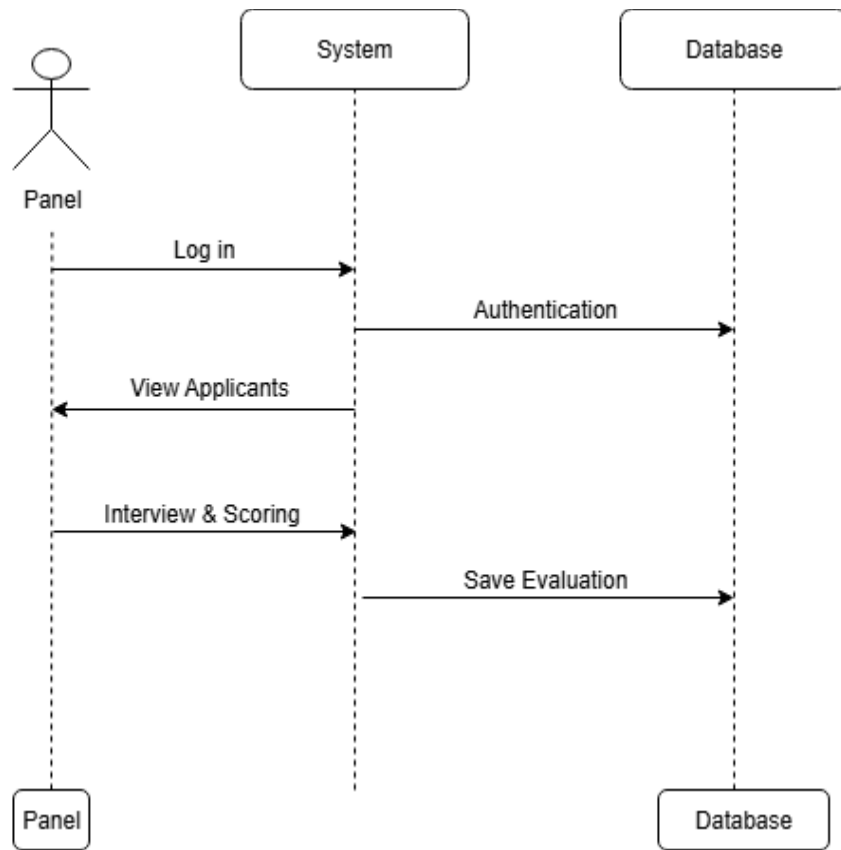


Figure 10. Sequence Diagram - Panel

Figure 10 illustrates how the panel members interact with the system and database during the applicant review process. The process begins when a panel member logs into the system, and their credentials are verified for security. After successfully logging in, the panel member can view applicant information, conduct interviews, and record their evaluation scores. All recorded assessment data is automatically stored in the database to ensure that each interview result is properly saved and accessed when needed. This process helps maintain accuracy, consistency, and transparency in the evaluation of applicants. It also supports data-driven decision making by allowing administrators to review reliable and organized interview records during the selection process.

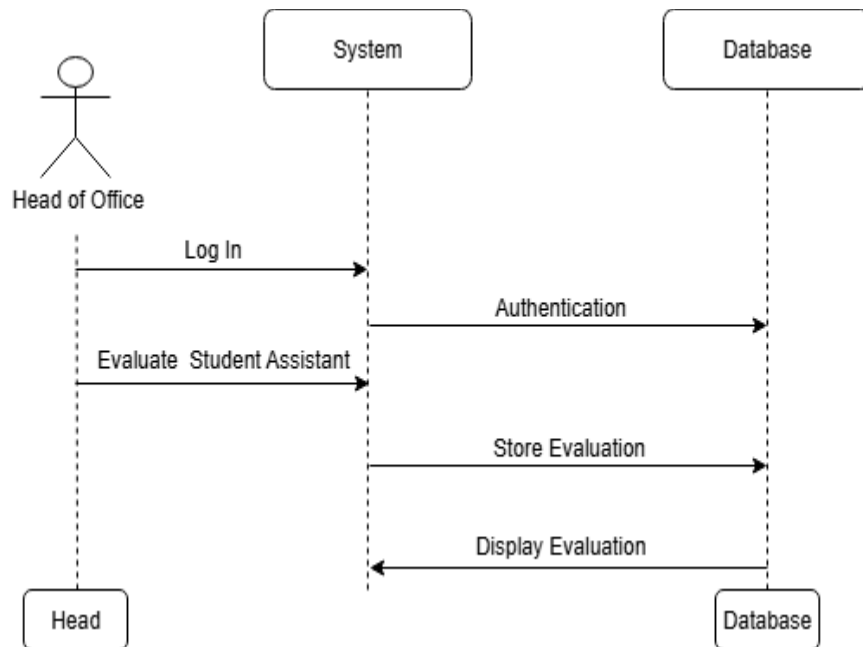


Figure 11. Sequence Diagram - Head Of Office

Figure 11 shows how the head of office or authorized personnel use the system to evaluate their assigned student assistants. The process begins when the head logs are in the system and the analysis begins. After completing the analysis, the system automatically saves the results to the database to keep the information safe and organized. Later, the system retrieves and displays the saved data for the head to check and confirm the results. This process helps ensure that all evaluations are recorded correctly and supports accurate tracking and fair decision making.

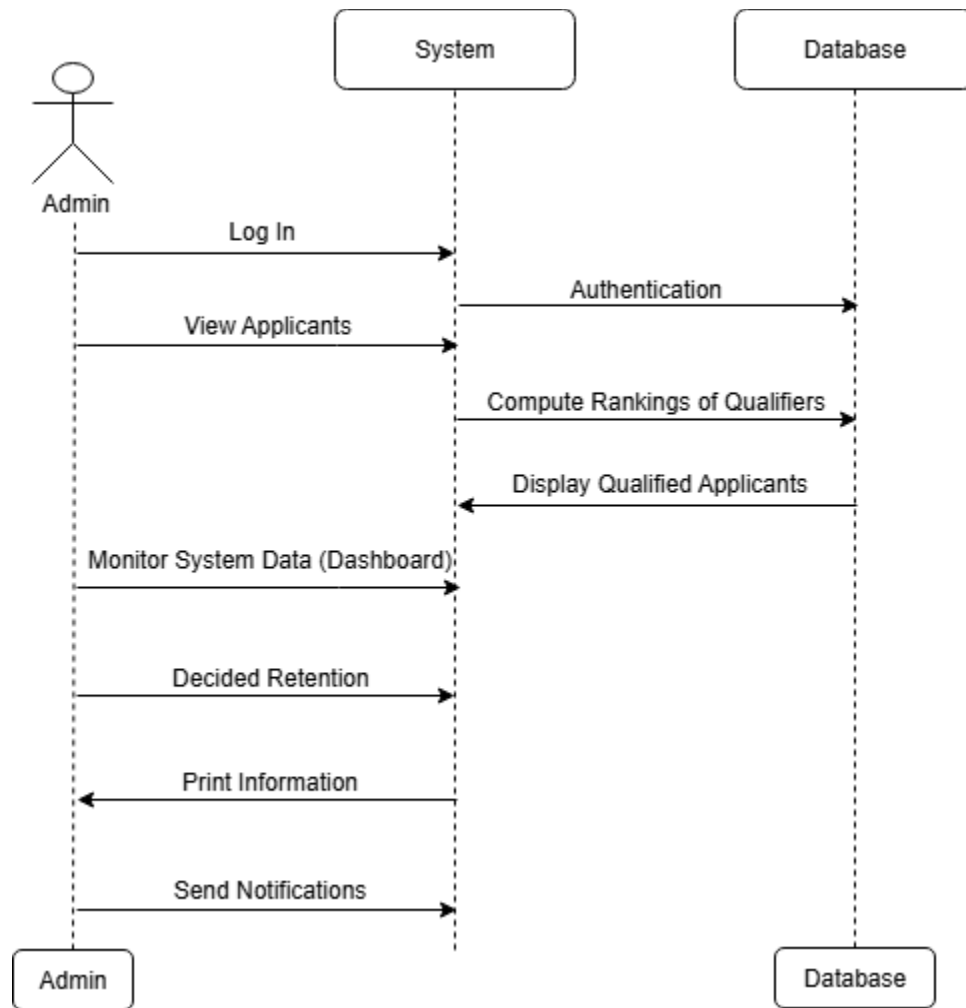


Figure 12. Sequence Diagram - Admin

Figure 12 illustrates the main tasks that an admin can perform in the system. The process starts when the admin logs into their account. After logging in, the admin can view applicant profiles, rank qualified scholars, print applicant information, check system data on the dashboard, and send notifications to users. All these actions follow the maintenance decision process and are saved in the database. This ensures that every review and final decision is properly recorded and can be reviewed as needed. The process helps keep the system organized, secure, and manageable for administrators.



2.7 ERD (Database Design)

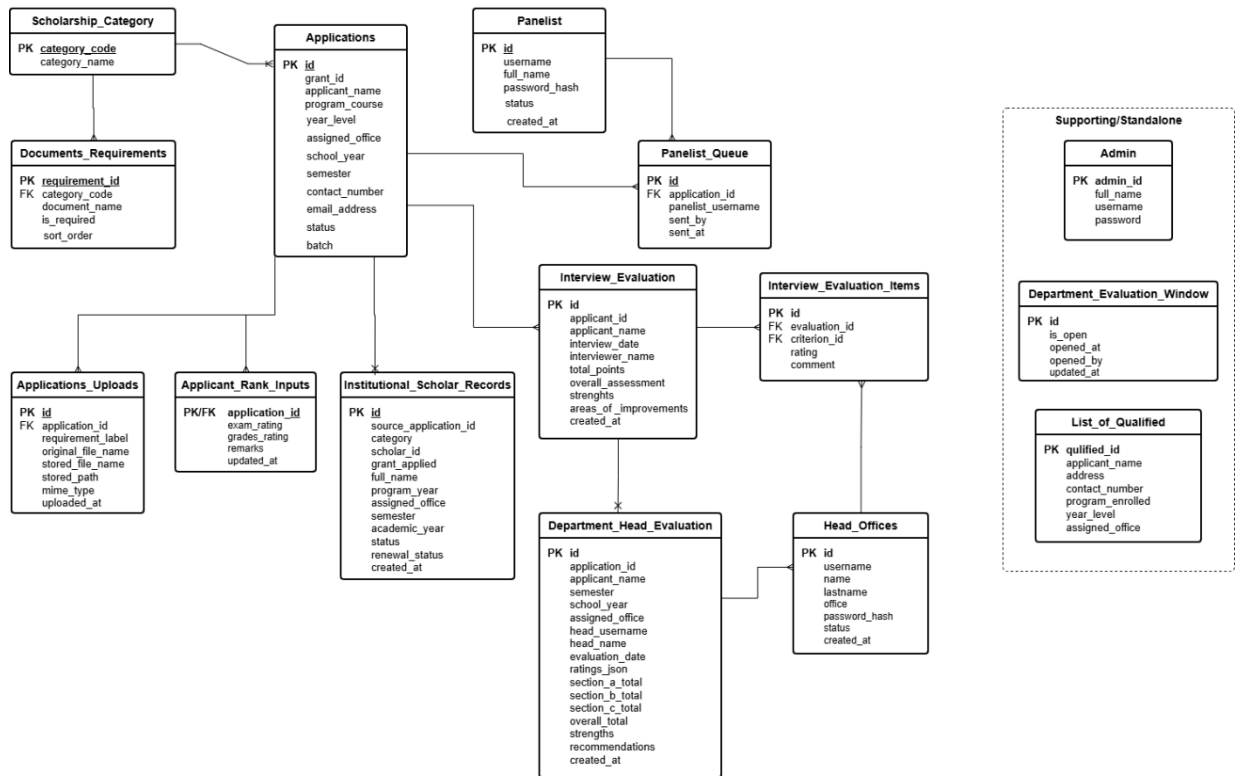


Figure 13. Entity Relationship Diagram

Figure 13 shows the design of the SMCC Institutional Scholarship Management System database, including a performance-based maintenance decision mechanism. It includes several connected entities that manage information about applicants, evaluations, and retention results. The main components of the system are applicant information, applicant ranking, interview evaluation, interview results, documents, and a list of qualifiers, which are linked by primary and foreign keys. Other entities, such as admin, panelists, and heads of offices, support the process by managing evaluations and results. Additional evaluation tables, such as head evaluation, functional skills, criterion, and H-evaluation summary report record, detailed scores, performance data, and comments. The relationships among these entities ensure that all applicant information—academic



performance, interview results, and evaluation decisions—is organized and connected. This structure provides a strong and transparent foundation for scholar selection, ranking, and retention decisions.

2.8 User Interface Design (Prototype)

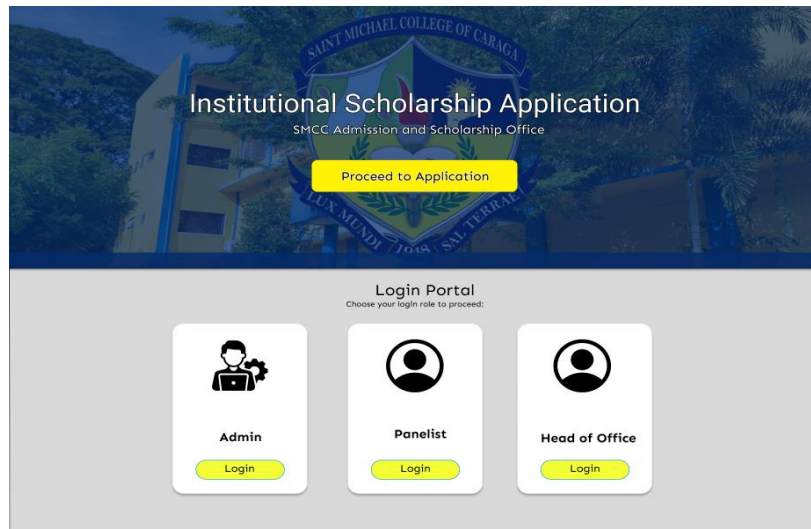


Figure 14. Landing Page

Figure 14 illustrates the official entry point for all users of the platform. It was designed to be simple, intuitive, and accessible to ensure that both administrators and scholarship participants can easily navigate the system. A prominent “Proceed to application” button immediately directs aspiring scholars to the online application form. Below the banner, the login portal provides secure access for different user roles and has a dedicated login button that leads to their respective dashboards, allowing users to perform functions related to their roles and responsibilities.



ISG REQUIREMENTS

Note: Please read and submit the requirements of the Institutional Scholarship Grants that you wanted to apply. Non-compliance with any of the requirements stated below or any provisions of the scholarship contract will constitute a ground for the forfeiture of the scholarship program.

1. STUDENT ASSISTANT

- Accomplished Application Form (to be secured from the Office of Admission & Scholarship)
- 1 copy of 2x2 ID picture
- Application Letter
- Resume
- Photocopy of Form 138 or Report of Grades
- Certificate of Indigency
- Certificate of Good Moral Character

Programs need to enroll

Bachelor Programs

- Bachelor of Arts in English Language (AB English)
- Bachelor of Technical Vocational Teacher Education (BTVTED)
- Bachelor of Secondary Education
 - major in Mathematics
 - major in Social Studies
- Bachelor of Science in Computer Science (BSCS)
- Bachelor of Library and Information Science (BLIS)
- Bachelor of Science in Information Systems (BSIS)
- Bachelor of Public Administration (BPA)
- Bachelor of Science in Entrepreneurship (BSE)
- Bachelor of Science in Accounting Information System (BSAIS)

New Program Offerings

- Bachelor of Science in Business Administration (BSBA)
 - major in Operations Management
 - major in Business Economics
- Bachelor in Human Services (BHumserv)

UPLOAD DOCUMENTS

Figure 15. Applicant Side (Viewing and Uploading Requirements)

Figure 15 shows the first step for applicants who wish to apply for the Institutional Scholarship grant at Saint Michael College of Caraga. Upon accessing the page, applicants will be guided through the complete list of requirements they must prepare before submitting their Scholarship application. The interface is designed to be clear, friendly, and easy to follow, helping applicants to understand exactly what documents they need to submit. This section allows the applicants to view what kind of Institutional Scholarship grant they want to apply. Once all the required files are ready, applicants can conveniently upload their documents directly through the upload documents button provided on the page. This upload feature aims to make the process paperless, faster, and removing the need for physical submission and reducing the risk of misplaced or incomplete application.



SMCC Admission and Scholarship Office

← Back

Upload Your Documents Here

Drag and Drop Files Or Browse

File Name: Choose File No file chosen Delete

File Name: Choose File No file chosen Delete

File Name: Choose File No file chosen Delete

Proceed Next Step

Figure 16. Applicant Side (Upload Requirements)

Figure 16 illustrate applicants to upload their scholarship requirements online. Instead of submitting paperwork in person, students can easily upload their digital files, such as grade reports, letters of recommendation, and other required documents. The upload area accepts via drag-and-drop or by clicking the "Choose File" button. Each uploaded file can be reviewed or rejected if the applicant wishes to change it. Once all the documents are uploaded, the applicant can click "Proceed to Next Step" to proceed to the next part of the scholarship application process. This feature helps the SMCC Admission and Scholarship Office manage and organize requirements more efficiently, reduce paperwork, and make the application process faster and more convenient for both students and staff.

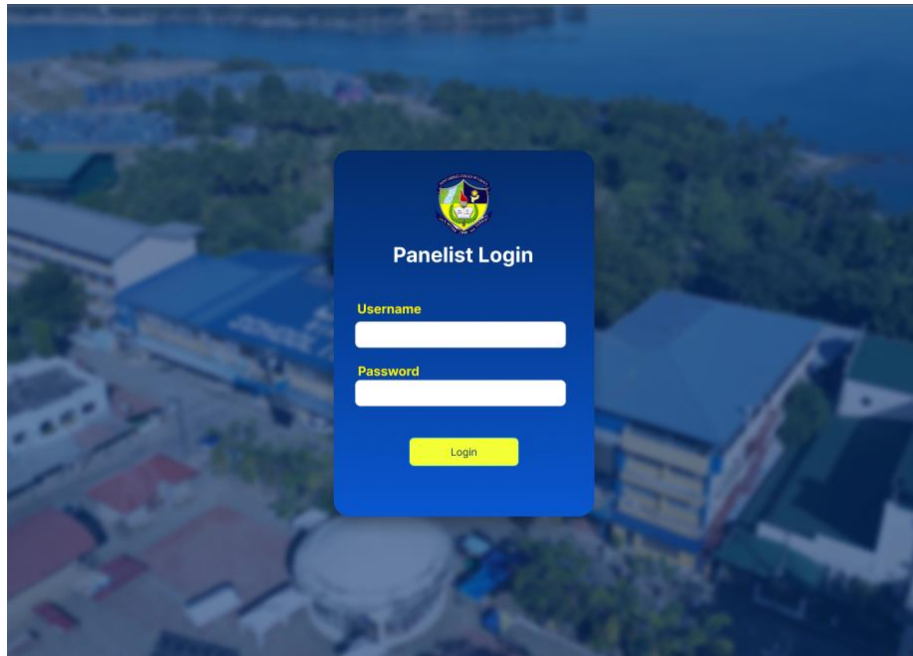


Figure 17. Login Page - Panelist

Figure 17 illustrates the secure entry point for authorized evaluators of the scholarship program. Each panelist will be provided with a unique username and password to ensure that only verified users can access applicant data. Upon entering their credentials, panelists will be authenticated through the system before gaining access to their dashboard. This login process will protect the integrity of applicant information and prevent unauthorized access, aligning with the institution's data privacy and confidentiality standards. The page will be designed with a clean and professional interface, reflecting the identity of Saint Michael College of Caraga while ensuring that users can log in easily and securely.



The screenshot shows the 'Interview Evaluation Sheet' within the 'Panelist Dashboard' for the 'STUDENT ASSISTANCE SCHOLARSHIP PROGRAM'. The interface includes a sidebar with navigation links: Home, Applicants (3), and Evaluated SA. The main content area has a header with the program name and a sub-header 'Interview Evaluation Sheet'. Below this, there are input fields for 'Applicant's Name', 'Date of Interview' (with a date picker), and 'Interviewer's Name'. A 'DIRECTIONS' section explains the rating scale. The evaluation table has columns for criteria and ratings from 4 to 1.

CRITERIA	4	3	2	1
Academic performance	☐	☐	☐	☐
Related work experience	☐	☐	☐	☐

Figure 18. Panel Side (Interview Evaluation Sheet)

Figure 18 shows the Interview Evaluation Sheet, they will be able to record their observations and rate each applicant based on specific performance criteria. The evaluation form will include rating scales from 1(Need Improvement) to 4(Excellent), covering factors such as academic performance, communication skills, related work experience, and overall suitability for the scholarship. This digital form will ensure consistency and fairness in scoring, as all panelists will follow the same structured evaluation process. Once the assessment is completed, the data will be securely stored in the database and automatically linked to the applicant's record. This feature will not only make evaluations faster and more organized but will also provide a reliable foundation for generating reports and making informed retention or qualification decisions.

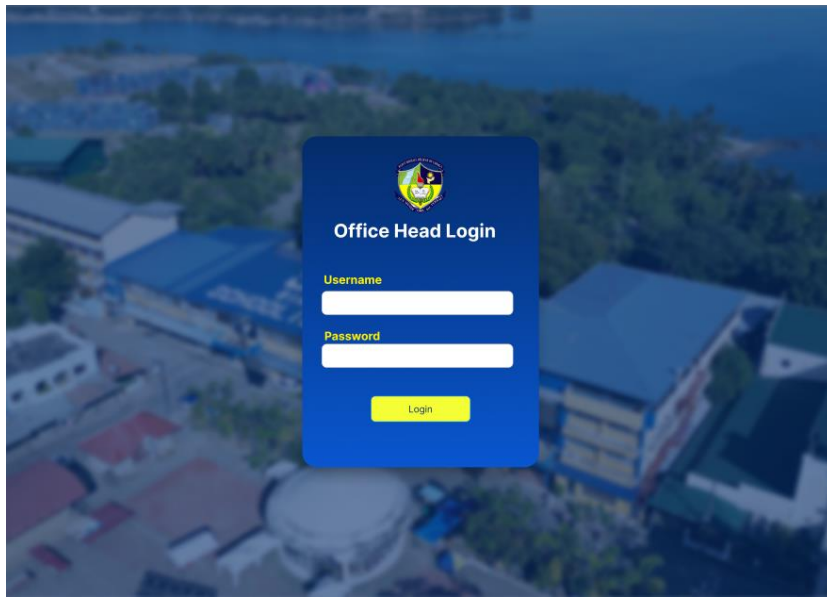


Figure 19. Head of Offices Side (Login Page)

Figure 19 illustrates a secure and user-friendly entry point for department heads or supervisors who are responsible for evaluating the performance of student assistants under their office. Each office head will be provided with a unique username and password to ensure that only authorized users can access the system, protecting sensitive data and maintaining confidentiality in line with institutional policies. The page will feature a simple and professional interface that reflects the identity of Saint Michael College of Caraga, allowing office heads to log in easily and proceed directly to their personalized dashboard. Through this access point, they will later be able to view assigned student assistants, conduct evaluations, and submit performance reports efficiently. Overall, this login page will ensure both security and accessibility, enabling office heads to perform their supervisory tasks smoothly and confidently within the digital platform.

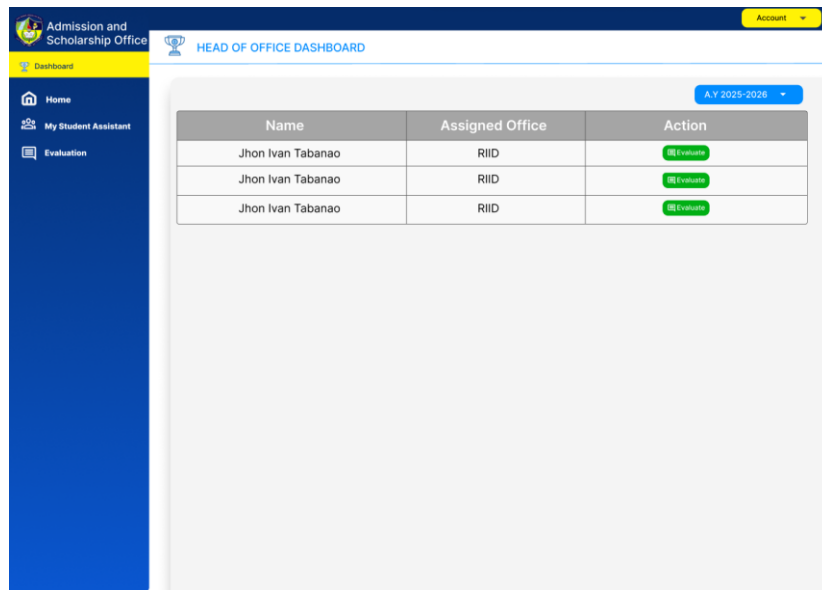


Figure 20. Head of Offices Side (Dashboard)

Figure 20 shows the main interface for department heads or supervisors once they log into the system. Through this dashboard, office heads will be able to view a complete list of their assigned student assistants, including the students' names and designated offices. The interface will provide a clear and organized table format that makes monitoring easier and more efficient. Beside each record is an "Evaluate" button, allowing office heads to proceed directly to the evaluation form for each student assistant. This setup will simplify the process of tracking and assessing performance, ensuring that evaluations are conducted promptly and accurately. The design of the dashboard will focus on clarity, usability, and professionalism, helping office heads carry out their supervisory tasks without confusion or delay. Overall, this page will provide a convenient and secure workspace where evaluations can be managed digitally, promoting fairness, accountability, and consistency in the scholarship supervision process.



Scale	Verbal Description	Verbal Interpretation
4	Excellent	This rating is given to student assistants who consistently exceed expectations and demonstrate outstanding performance in their assigned tasks.
3	Good	This rating is given to student assistants who consistently meet expectations and perform their assigned tasks satisfactorily.
2	Fair	This rating is given to student assistants who occasionally meet expectations but may have some areas for improvement.
1	Poor	This rating is given to student assistants who consistently fail to meet expectations and demonstrate unsatisfactory performance in their assigned tasks.

Criteria	Excellent (4)	Good (3)	Fair (2)	Poor (1)
A. Quality and Quantity of Work				
A.1 Accurate at work assigned	☐	☐	☐	☐
A.2 Always completes tasks	☐	☐	☐	☐
A.3 Works in a timely manner	☐	☐	☐	☐
A.4 Asks for more work when assigned tasks are done	☐	☐	☐	☐
A.5 Easily accepts new responsibilities	☐	☐	☐	☐
Total				

Figure 21. Head of Offices Side (Student Assistants' Evaluation Form)

Figure 21 illustrates the digital workspace for department heads or supervisors once they click the “Evaluate” button from their dashboard. This form will allow the Head of Office to assess each assigned student assistant based on their actual work performance, attendance, and overall behavior in the workplace. It will display the student’s basic information such as name, area of assignment, semester, and evaluation date for proper documentation. The form will use a standardized rating scale ranging from 4(Excellent) to 1(Poor), ensuring that all student assistants are evaluated fairly and consistently. Each criterion such as accuracy of work, timeliness, task completion, and acceptance of responsibilities will reflect the student’s work ethics and productivity. By using this system, office heads will be able to provide clear, objective, and data-driven evaluations without relying on paper forms. The automated process will promote transparency, efficiency, and accountability in performance assessment, allowing the institution to maintain high standards in its student assistantship program while recognizing students who demonstrate dedication and excellence in their assigned duties.

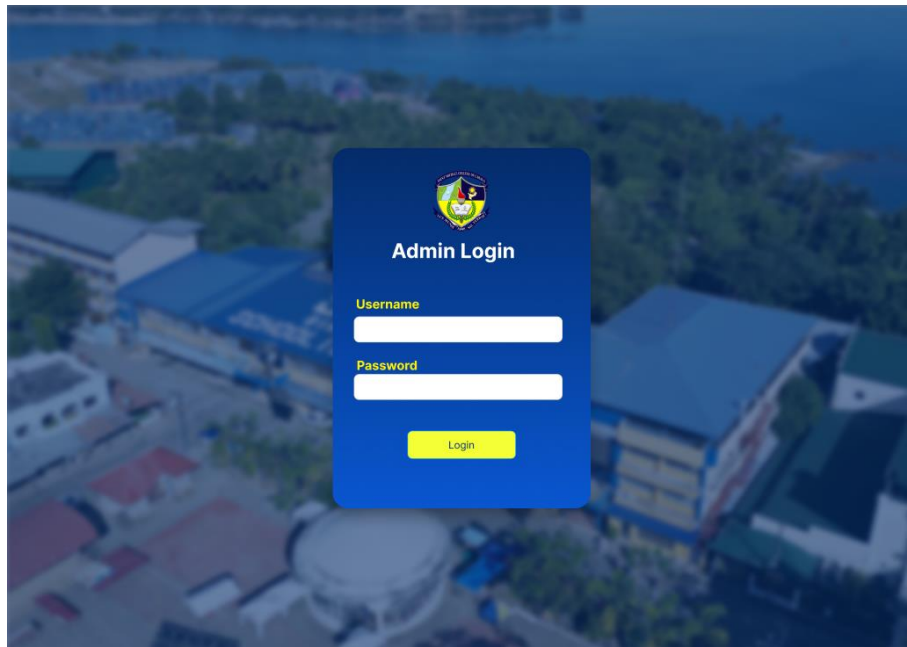


Figure 22. Admin Side (Login Page)

Figure 22 shows the secure entry point for the system administrator, who will be responsible for managing and overseeing the entire scholarship supervisory system. Each admin will have a unique username and password to ensure that only authorized personnel can access and modify sensitive data within the platform. This page will be designed with a clean and professional interface that reflects the institutional identity of Saint Michael College Caraga, providing a sense of trust and formality. Through this login page, administrators will be able to safely access the system's back-end functions, such as managing user accounts, viewing applicant records, monitoring evaluations, and generating reports. The security of the login process will be prioritized to protect confidential scholarship data and ensure that all administrative activities remain well-documented and traceable. Overall, this page will provide a simple yet vital gateway for administrators to perform their duties efficiently while upholding the principles of data privacy, security, and transparency within the scholarship management process.

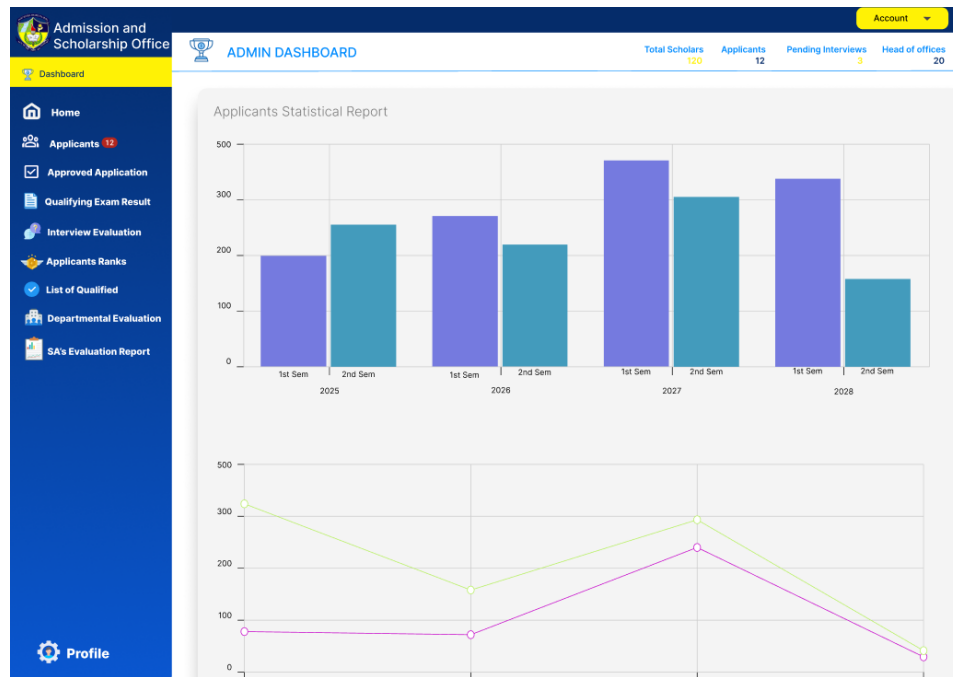


Figure 23. Admin Side (Applicants Statistical Report)

Figure 23 illustrates the central control panel for the scholarship administrator, providing a clear and visual overview of all scholarship-related data and activities. One of its key features will be the Applicant Statistical Reports, which will display graphical summaries of the number of applicants, qualified scholars, and evaluation trends across different semesters and school years. Through bar and line charts, the administrators will be able to easily track fluctuations in scholarship applications and performance over time, helping identify patterns and make data-driven decisions. This visual report will enhance monitoring efficiency by presenting complex information in a simple and understandable format. It will also allow administrators to evaluate the effectiveness of the scholarship program and determine areas that may require improvement or policy adjustment. Overall, the Admin Dashboard will function as an intelligent monitoring hub helping the scholarship office maintain transparency, accuracy, and efficiency in managing institutional scholarship operations.



Admission and Scholarship Office **ADMIN DASHBOARD** Account

Total Scholars 120 Applicants 12 Pending Interviews 3 Head of offices 20

Dashboard

Home **Applicants 12** **Approved Application** **Qualifying Exam Result** **Interview Evaluation** **Applicants Ranks** **List of Qualified** **Departmental Evaluation** **SA's Evaluation Report** **Profile**

Student Assistants 3 **Kabayani Scholarship 3** **Academic Scholarship 3** **Others 3**

APPLICANTS Search Here

Application Id	Name	Course	ISO Grant	Status	Action
0001	Jhon Ivan Tabanao	Bachelor of Science in Information Technology	Student Assistant	Pending	View Accept Reject
0002	Mark Christian Joven Balatayo	Bachelor of Science in Information Technology	Kabayani	Pending	View Accept Reject
0003	John Doe	Bachelor of Science in Criminology	Others	Pending	View Accept Reject
0004	Jhon Ivan Tabanao	Bachelor of Science in Information Technology	Academic Scholar	Pending	View Accept Reject
0005	Mark Christian Joven Balatayo	Bachelor of Science in Information Technology	Kabayani	Pending	View Accept Reject
0006	John Doe	Bachelor of Science in Criminology	kabayani	Pending	View Accept Reject
0007	Jhon Ivan Tabanao	Bachelor of Science in Information Technology	Others	Pending	View Accept Reject
0008	Mark Christian Joven Balatayo	Bachelor of Science in Information Technology	Others	Pending	View Accept Reject
0009	John Doe	Bachelor of Science in Criminology	Student Assistant	Pending	View Accept Reject
0010	Jhon Ivan Tabanao	Bachelor of Science in Information Technology	Academic	Pending	View Accept Reject
0011	Mark Christian Joven Balatayo	Bachelor of Science in Information Technology	Academic	Pending	View Accept Reject
0012	John Doe	Bachelor of Science in Criminology	Student Assistant	Pending	View Accept Reject

Figure 24. Admin Side (Applicant Management Page)

Figure 24 shows the core features of the system, designed to help the administrator efficiently monitor, verify, and manage all scholarship applications. This page will display a complete list of applicants with important details such as their application ID, full name, course, type of scholarship grant, and current application status. The administrator will also be provided with quick action buttons View, Accept, and Decline to review an applicant's profile, approve qualified applicants, or reject incomplete or ineligible submissions. A built-in search bar category tabs, such as Student Assistant, Kabayanihan Scholarship, Academic Scholarship, Others, will make navigation easier, allowing the admin to filter and locate applications quickly. The interface will be designed to be intuitive and organized, reducing administrative workload and minimizing the chance of oversight or error. Through this feature, the scholarship office will be able to maintain a fair, transparent and systematic approach to screening applicants, ensuring that every decision made is properly documented and based on accurate and verified information.



Admission and Scholarship Office

ADMIN DASHBOARD

Total Scholars: 120 Applicants: 12 Pending Interviews: 3 Head of Offices: 20

Print

Saint Michael College of Caraga
 Brgy. 4, Nasipit, Agusan del Norte, Philippines
 District 8, Brgy. Triangulo, Nasipit, Agusan del Norte, Philippines
 Tel. Nos. +63 085 343-3251 / +63 085 283-3113
www.smcnasipit.edu.ph

APPLICATION FORM
 For Institutional Scholars/Grantees

Type of Scholarship/Grant:

☒ Academic ☐ Student Assistance
☐ Kabayani ☐ Others

Please specify: _____

Scholar/Grantee's Profile:

Name of Applicant : Juan Dela Cruz
 Program/Course Enrolled : Bachelor of Science in Information Technology
 Year Level : 3rd Year
 School Year : 2024-2025
 Permanent Address : P-1 Brgy. 4, Nasipit, Agusan del Norte Gender : Male
 Date of Birth : January 15, 2004 Age : 20
 Contact Number : 0917-120-4567
 Estimated Gross Income of the Family/Month : PHP 25,000

Mother's Name : Maria Santos Dela Cruz Age: 45
 Contact Number : 0918-234-5678 Occupation: Elementary School Teacher
 Company's Name : Nasipit Central Elementary School
 Company's Address : Roxas St., Nasipit, Agusan del Norte

Figure 25. Admin Side (Applicant Profile and Application Form Page)

Figure 25 shows the applicant's chosen type of scholarship grant, whether Academic, Kabayanihan, Student Assistant, or Others, along with their personal and family background, educational program, and financial details. By having access to this digital form, administrators will be able to easily verify the accuracy of the applicant's submitted data and supporting documents before making approval decisions. This page will also include a Print button, enabling the admin to generate a hard copy of the application form when needed for documentation or filling purposes. Designed to reflect the official format of Saint Michael College of Caraga, this page will ensure that all records are consistent, complete, and securely stored within the system. Overall, this feature will simplify the verification process and promote efficiency, transparency, and data integrity in managing applications.



Admission and Scholarship Office

Dashboard

Home

Applicants 12

Approved Application

Qualifying Exam Result

Interview Evaluation

Applicants Ranks

List of Qualified

Departmental Evaluation

SA's Evaluation Report

Profile

ADMIN DASHBOARD

Total Scholars 120 Applicants 12 Pending Interviews 7 Head of Offices 20

Print

Saint Michael College of Caraga
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www.smcnasipit.edu.ph

STUDENT ASSISTANTS' EVALUATION FORM

Name of Student Assistant: John Michael Santos

Semester & School Year: 2nd Semester, S.Y. 2024-2025

Area of Assignment: Learning Resource Center

Head of Office: Mary Ann L. Rosales

Date of Evaluation: March 15, 2025

Directions: Please rate each item below to determine the performance of the assigned student assistant of your respective office/department. Put a check (✓) to rate their performance.

SCALE	VERBAL DESCRIPTION	VERBAL INTERPRETATION
4	Excellent	This rating is given to student assistants who consistently exceed expectations and demonstrate outstanding performance in their assigned tasks.
3	Good	This rating is given to student assistants who consistently meet expectations and perform their assigned tasks satisfactorily.
2	Fair	This rating is given to student assistants who occasionally meet expectations but may have areas for improvement.
1	Poor	This rating is given to student assistants who consistently fail to meet expectations and demonstrate unsatisfactory performance in their assigned tasks.

PERFORMANCE INDICATORS	RATING			
	EXCELLENT (4)	GOOD (3)	FAIR (2)	POOR (1)
A. QUALITY AND QUANTITY OF WORK				
A.1 Accurate at work assigned		✓		
A.2 Always completes tasks		✓		
A.3 Works in a timely manner	✓			
A.4 Asks for more work when assigned tasks are done		✓		
A.5 Easily accepts new responsibilities		✓		

Figure 26. Admin Side (Student Assistant's Evaluation Form Page)

Figure 26 illustrate the key feature within the Admin Dashboard, allowing administrators to view and monitor the performance evaluations submitted by the Heads of Offices. This page will display the official evaluation form of Saint Michael College of Caraga, containing the student assistant's details such as name, semester, area of assignment, head of office, and date of evaluation. It will also present the performance indicators and rating scale ranging from 4(Excellent) to 1(Poor), along with the verbal interpretation of each score. Through this feature, administrators will be able to review the accuracy and completeness of the evaluation, ensuring that assessments are properly aligned with institutional standards. The Print option will enable the admin to generate hard copies for documentation and reporting purposes when needed. Designed with simplicity and clarity, this page will promote transparency and accountability in evaluating student assistants, ensuring that performance records are systematically stored and easily accessible for retention decisions and future reference.

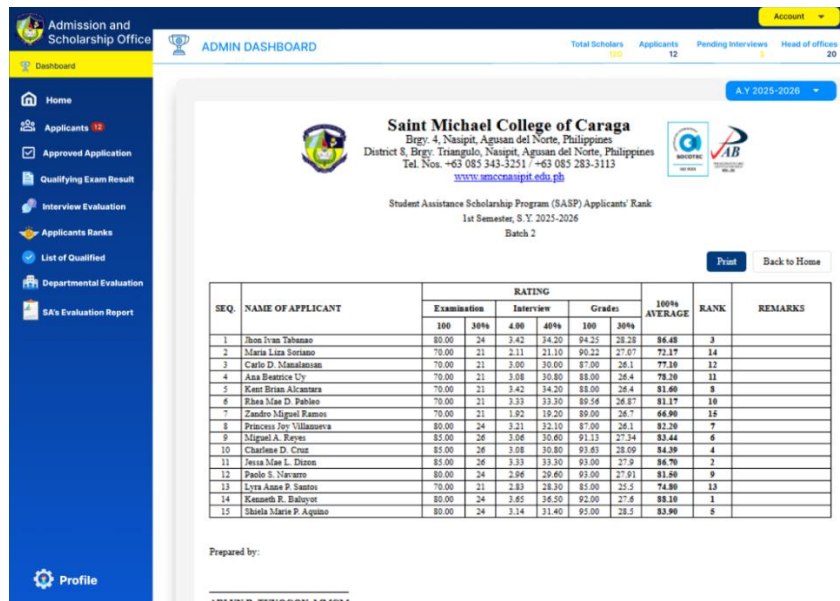


Figure 27. Admin Side (Student Assistants rankings)

Figure 27 shows the generated list of scholarship applicants ranked based on their total evaluation scores in the institutional scholarship management system of Saint Michael College of Caraga. The table shows each applicant's test result, interview score, and academic scores, automatically calculated by the system according to their assigned percentage weight. This feature helps administrators easily identify top performing applicants who qualify for the scholarship program. It also allows for transparent and consistent evaluation, as all scores and feedback are recorded and displayed digitally within the admin dashboard.





Admission and Scholarship Office

Dashboard

Home

Applicants 12

Approved Application

Qualifying Exam Result

Interview Evaluation

Applicants Ranks

List of Qualified

Departmental Evaluation

SA's Evaluation Report

Profile



ADMIN DASHBOARD

Total Scholars120

Applicants12

Pending Interviews3

Head of offices20

A.Y 2025-2026

1st Semester



Saint Michael College of Caraga

Brgy. 4, Nasipit, Agusan del Norte, Philippines

Tel. Nos. +63 085 343-3251 / +63 085 283-3113

www.smcadmission.edu.ph





OFFICE OF THE ADMISSION & SCHOLARSHIP

List of Qualified Applicants for Student Assistance Scholarship Program

1st Semester, S.Y. 2025-2026

Batch 1

#	NAME	ADDRESS	CONTACT NUMBER	PROGRAM ENROLLED	YEAR LEVEL	ASSIGNED OFFICE
1	Maria L. Santos	Brgy. 1, Nasipit	0917 201 1123	BSIT	2	ICT Office
2	John P. Dela Cruz	Brgy. 7, Nasipit	0908 334 5566	BSBA	3	Registrar
3	Angela R. Gomez	Brgy. 4, Butuan City	0920 889 7733	BSHM	1	Hospitality Lab
4	Kevin J. Ramos	Brgy. 3, Nasipit	0945 123 0044	BSED English	2	Library
5	Ella Mae P. Rivera	Magallanes, Agusan del Norte	0910 555 8877	BEED	4	SAS Office
6	Mark Adrian T. Uy	Brgy. 5, Nasipit	0936 778 9922	BSCRIM	3	Security
7	Rose Ann G. Casing	Carmen, Agusan del Norte	0918 667 4411	BSN	1	Clinic
8	Jasper K. Lim	Brgy. Triangulo, Nasipit	0921 903 2205	BSIT	3	ICT Office
9	Patricia D. Villanueva	Bumarista, Agusan del Norte	0956 441 0033	BSAcc	2	Finance
10	Vincent R. Alonzo	Brgy. 2, Nasipit	0995 812 6677	BSCE	2	Engineering
11	Hazel Joy B. Ramos	Las Nieves, Agusan del Norte	0917 330 9911	BSHM	4	Guidance
12	Ryan G. Mondejar	Butuan City	0906 421 5570	BSBA	1	HRMDO
13	Shaira M. Quizon	Brgy. 8, Nasipit	0927 665 3101	BSAIS	2	Accounting
14	Louise C. Bayla	Jabonga, Agusan del Norte	0915 220 1188	BSIT	4	Research
15	Kathleen S. Domingo	Kitcharao, Agusan del Norte	0991 702 4410	BSED Math	3	Basic Ed
16	Alfredo T. Maslangit	RTR, Agusan del Norte	0938 770 5521	BSCS	1	NSTP

Figure 28. Admin Side (List of Qualified)

Figure 28 presents the automatically generated list of qualified applicants for the Saint Michael College of Caraga Student Scholarship Program. This list is displayed through the admin dashboard of the Institutional Scholarship Management System, which shows the important details of the applicant such as name, address, contact number, course enrolled, year level, and their designated office for extension. This feature helps the admissions and scholarship office manage eligible scholars more efficiently. Instead of preparing manual lists, the system organizes and presents data immediately after analysis.



2.9 Software Platforms, Development Environment and Tools

Components	Specification	Usage
Internet Browser	Any	A web browser enables users to easily access and display content from the internet. It is responsible for loading the SMCC Institutional Scholars Supervisory System interface allowing users access to the system's features after a successful login.
Front-end	HTML CSS JavaScript Tailwind	HTML defines the structural layout of the SMCC Institutional Scholars Supervisory System's interface, arranging elements like forms, texts, and other important elements. CSS is used for improving the aesthetics and designs of the HTML structure providing a more engaging and responsive website. JavaScript has the ability to make the website interactive and provide more functionality for the users, JavaScript is used to handle all client-side functionalities. Tailwind is a framework that allows developers to easily design responsive and mobile-



	JQuery	first web pages by providing a collection of pre-designed components, grids, and styles to provide a more responsive website regardless of the screen size and device. jQuery is a quick and efficient JavaScript library that simplifies HTML document manipulation, event handling, animations, and Ajax interactions. It enables developers to write less code for common web tasks and ensures cross-browser compatibility.
Back-end	XXAMP	XXAMP is preferred in back-end development that bundles Apache and PHP, allowing the developers to easily set up a local server environment without complicated installation or configuration. It provides a user-friendly interface and



	PHP	ready-to-use tools that make testing, debugging, and database management more efficient. On the other hand, PHP is a widely used server-side scripting language that is valued for its simplicity, flexibility, and compatibility with various databases and web servers. Its large community support, extensive libraries, and ability to create dynamic, data-driven web applications make it a reliable choice for building the logic and functionalities of a system's back-end.
Server	Apache	Utilizing the XAMPP control panel, developers can manage the behavioral state of its components ensuring smooth and easy access to



		local host servers like Apache and Database.
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2.10 Hardware Requirements

Components	Specification	Usage
Device	RAM (8 GB) Processor (Intel Core i5) SSD (128 GB) Flash Storage (8 GB)	RAM facilitates fast and temporary data storage. It is used to enhance the speed and performance of devices in accessing information. The processor or the CPU, acts as the central component of the computer, handling the execution of instructions sent from the website. It also facilitates the processing of data. Devices employ SSD for storing large amounts of audit files, system software, and reports, while flash storage is commonly used in mobile devices for lightweight data storage. These storage solutions ensure quick retrieval and secure long-term data handling.



Printer	Any Ink Jet Printer Units	Inkjet printers are utilized for printing audit reports, data summaries, and compliance documents, producing clear and high-quality output. These printers are versatile and affordable, making them suitable for regular printing tasks related to audit management, such as generating official documents for internal use or distribution to stakeholders.
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System Evaluation

This section describes the different evaluation procedures done to the system to assess its functional suitability, performance efficiency, and usability. The survey instrument is adapted from ISO 25010 Software Product Quality Standards; the test scale gives four options to the respondents and will rate the system based on the question with 4 as the highest and 1 as the lowest.

Data were analyzed and interpreted based on the following parameters:



Table 3
Functional Suitability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Functional
3	2.50 – 3.49	Functional
2	1.50 – 2.49	Moderately Functional
1	1.0 – 1.49	Poor Functional

Table 3 presents the criteria used to evaluate a product or system's functional suitability according to the ISO survey.

Table 4
Performance Efficiency

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Efficient
3	2.50 – 3.49	Efficient
2	1.50 – 2.49	Moderately Efficient
1	1.0 – 1.49	Poor Efficient

Table 4 outlines the scoring parameters used to assess the system's overall performance efficiency in executing its functions and tasks.



Table 5

Usability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Usable
3	2.50 – 3.49	Usable
2	1.50 – 2.49	Moderately Usable
1	1.0 – 1.49	Poor Usable

Table 5 presents the scoring parameters used to the usability of the system.

Table 6

Respondent's Distribution

INSTITUTIONAL SCHOLARSHIP MANAGEMENT SYSTEM FOR SAINT MICHAEL COLLEGE OF CARAGA		
Position	N (Population)	Percentage (%)
Admission Staff	1	12.5%
Panel	1	12.5%
Personnel	2	25%
Student	4	50%
Total	8	100

Table 6 presents the distribution of respondents who participated in the system review within the SMCC Admission and Scholarship Office. Respondents were grouped according to their positions,



with 12.5% admissions staff, 12.5% panel members, 25% staff, and 50% students. The results show that most of the feedback comes from students, who serve as the main users of the system. Their insights are particularly important because they interact directly with the platform, providing practical insights into functionality, efficiency, and user-friendliness in real-world scenarios.

2.11 Research Ethical Standards

The researchers affirm their full commitment to upholding the ethical standards of Saint Michael College of Caraga throughout the conduct of this study entitled “Institutional Scholarship Management System for Saint Michael College of Caraga”. This study will be carried out with deep respect for human dignity, data privacy, and intellectual honesty. The principles of integrity, transparency, beneficence, justice, and respect for persons will guide all stages of the research process.

A. Protection of Intellectual Property Rights (IPR)

The system that will be developed is an original work of researchers. All codes, designs, documents, and outputs are recognized as the intellectual property of Saint Michael College of Caraga, in accordance with the terms of RA 8293. We also ensure that no copyrighted or protected material is used without proper permission. The project follows the ISO/IEC 25010 software quality standard to guide the development of the system, helping to ensure that it is functional, efficient, and easy to use. Since this system is made for internal use only and does not create any new logos or brands, no trademark or patent registration is required.

B. Informed Consent

In accordance with the ethical standards of Saint Michael College of Caraga, all participants will first receive a clear explanation about the purpose, method, and any risks of the study, and will be asked to participate. A consent form will be provided either in printed or digital format, using plain English or Filipino to ensure understanding. The form will clearly state that participation is voluntary,



that anyone can withdraw at any time, and that there are no negative consequences for choosing not to continue. All signed consent forms will be kept secure and confidential, stored either in a secure digital folder or in a locked cabinet, and accessible only to the research team.

C. Data Privacy and Confidentiality

This study complies with the Data Privacy Act of 2012 (RA 10173) and the Data Privacy Policy of Saint Michael College of Caraga. All information collected will be kept confidential and anonymous by removing names and any details that could identify participants. Data will be securely stored in an encrypted, password-protected digital folder and in a locked cabinet for printed files, with access limited to the research team only. No anonymized or identifiable data will be shared with third parties. All collected data will be kept for one year after completion of the study and then permanently deleted or destroyed to ensure that no personal information is recovered.

D. Voluntary Participation and Freedom to Withdraw

The voluntary nature of the study was clearly explained during participant orientation and written on the consent form, ensuring that everyone understood that participation was their personal choice. All invited participants are free to join or not, and they can withdraw at any time without penalty, pressure, or loss of privileges. No form of coercion or influence will be used, and each decision to participate is based solely on the person's own convenience and free will.

E. Minimization of Harm and Risk Management

This study involves only light activities such as surveys, interviews, and system testing, and participants will be told in advance that they can stop at any time if they feel uncomfortable. If a participant experiences any distress, the research team will immediately pause the activity, provide reassurance, and refer the person to the SMCC Guidance Office or the designated support contact for appropriate assistance. A simple risk management procedure will also take place, including clear steps on how to report concerns and how researchers will respond quickly to protect participant well-being.



F. Beneficence and Contribution to Knowledge

This study aims to benefit the Michaelinian community by improving fairness, transparency, and efficiency in scholarly management through a more organized and reliable digital system. When the study is completed, a summary of the results and the final output of the system will be shared with the institution and available to participants who wish to view the findings. The project also hopes to add new knowledge in the field of educational management and digital management, supporting the goals and values of Saint Michael College of Caraga.

G. Justice and Fair Participant Selection

Participants will be selected fairly and only from SMCC because the study is focused on improving the institution's scholarship processes, making administrators, department heads, evaluators, and student scholars the most directly involved and affected groups. No animals will be used in this study, and no one will be treated differently based on gender, age, religion, socioeconomic background, or academic status. The selection process follows SMCC's commitment to fairness, equality, and respect for each individual.

H. Data Integrity and Accuracy

The study will use standardized and consistent methods during data collection and analysis to ensure that all information gathered is accurate, reliable, and aligned with the ethical standards of Saint Michael College of Caraga. Any possible errors, biases, or limitations that may arise during the research are honestly documented and clearly explained in the final report. All findings will be recorded truthfully and cross-checked to maintain data integrity and ensure that the study reflects true and reliable results.

I. Transparency and Honesty in Reporting

The study will present all results truthfully, without changing, hiding, or distorting any information, in full alignment with the ethical standards of Saint Michael College of Caraga. All



external resources, software modules, and third-party materials, including open-source tools under licenses such as MIT, Apache 2.0, or GPL, are properly acknowledged and cited to give credit to their original creators. The final manuscript will reflect the true outcomes of the research, ensuring transparency, accountability, and honesty for the institution and future researchers. Additionally, it will be declared that there is no conflict of interest in conducting this study, guaranteeing that the research is free from any biases or external influence.

J. Use of Patented and Copyrighted Materials

The study will follow on RA 8293, the Intellectual Property Code of the Philippines, which covers the proper use of copyrighted works, patented technology, and all other protected materials. Any software, code libraries, or tools used in the system, including those under open source licenses such as MIT, Apache 2.0, or GPL, will be used according to their terms, with clear acknowledgment of their creators. All academic frameworks, documents, or digital resources borrowed or adapted will be properly cited to ensure full respect for the intellectual property rights of others and full compliance with SMCC's ethical standards.

K. Ethical Considerations for Animals and Human Trials

This study involved only human participants; No animal testing is conducted. All participants are clearly informed of any possible risks or benefits before taking part, and their rights and welfare are prioritized throughout the process. Ethical clearance from the Saint Michael College of Caraga Ethics Review Committee will be secured to ensure that all procedures meet institutional standards for the safe and responsible conduct of human research.

L. Responsible Use of AI and Other Related Technologies

AI tools were used only to help improve the clarity and organization of the manuscript—Chatgpt refined the structure, scispace assisted in summarizing cited references, and grammar corrected language and readability without changing the content. No AI tool was used to create the



primary data, results, or analysis, and all interpretations and decisions were made by human judgment to maintain accuracy and academic integrity. These tools served only as support, and the research process remained fully aligned with the ethical standards of Saint Michael College of Caraga.

M. Ethical Clearance and Institutional Approval

Formal ethical clearance will be obtained from the Saint Michael College of Caraga Ethics Review committee before any data collection or system testing begins, and no research activity will proceed without official approval. Once approved, the section will be updated to state that the study has obtained ethical clearance along with the date of approval. This process ensures that all research activities adhere to SMCC ethical standards for studies involving human participants.



CHAPTER III

RESULT AND DISCUSSION

3.1 Requirement Analysis Result

The requirement analysis is based on the project objectives of the Institutional Scholarship Management System project for Saint Michael College of Caraga. The analysis helped the researchers identify the main needs of the Admission and Scholarship Office, applicants, panelists, heads of offices, and administrators. The results showed that the system should support online application submission, automatic ranking, automatic analysis and reporting, and system quality evaluation using ISO 25010 standards.

3.1.1 Online Submission of Scholarship Applications and Requirements

Based on the first objective, the system needs to allow applicants to submit their scholarship applications and requirements online. This need was identified because the manual process required applicants to visit the office in person and submit printed documents. This process can cause delays, lost paperwork, incomplete submissions, and additional workload for the Admissions and Scholarship Office.

To meet this need, the system provides an online application form where applicants can enter their personal information, family background, education details, and selected scholarship grant. It also includes an upload feature where applicants can submit the required documents digitally. With this feature, applicants no longer need to rely solely on walk-in submission. The system helps make the application process faster, more organized, and easier for both applicants and office staff.

This requirement also supports better record keeping as submitted applications and uploaded files are stored in the system. As a result, the administrator can easily view, review, approve, or reject applications based on the submitted information and requirements.



3.1.2 Automated Ranking of Qualified Applicants

Based on the second objective, the system needs to automate the ranking of qualified applicants using test results, grades, and interview scores. This need was identified because manual ranking can take more time and can lead to calculation errors. It can also cause confusion if scores are not properly organized and calculated.

To meet this need, the system includes a ranking module that collects and processes the applicant's test score, academic score, and interview score. The system automatically calculates the total result and ranks the applicants based on their final marks. This helps the administrator identify qualified applicants more easily and fairly.

The automatic ranking feature also helps promote transparency and consistency in the selection process. Since the ranking is based on recorded scores, the scholarship office can reduce manual errors and make better decisions with clear and organized data.

3.1.3 Automated Evaluation and Reporting Modules

Based on the third objective, the system needs to provide an automatic analysis and reporting module for heads of offices and administrators. This need was identified because the manual evaluation process can be slow, difficult to monitor, and difficult to adjust, especially when several student assistants need to be evaluated.

To meet this need, the system provides a digital evaluation form for heads of offices. Through this feature, the office head can evaluate the assigned student assistant based on performance, work habits, task completion, and other required criteria. Once the review is submitted, the system stores the result and makes it available for administrator review.

The system also includes reporting features that allow the administrator to view test results, ranking reports, qualified applicant lists, and scholar records. These reports help the Admission and



Scholarship Office prepare accurate information faster and with less manual effort. This requirement helps improve tracking, reporting, and decision-making in the scholarship process.

3.1.4 Evaluation of The System Using ISO 25010 Standards

Based on the fourth objective, the system needs to be evaluated using ISO 25010 standards in terms of functionality, efficiency, and usability. This requirement is included to determine if the developed system works well, performs well, and is easy for users to understand and operate.

To meet this need, the researchers used an evaluation instrument based on the ISO 25010 Software Product Quality Standards. The system was evaluated by selected respondents according to functional suitability, performance efficiency, and usability. Functional suitability is used to check if the system provides the required features and correct results. Performance efficiency is used to evaluate whether the system responds quickly and performs tasks well. Usability was used to evaluate whether the system is easy to learn, easy to use, and suitable for users.

This requirement helped the researchers measure the quality and acceptability of the system. The result of the analysis served as the basis for determining whether the Institutional Scholarship Management System meets its intended purpose and supports the needs of its users.

3.2 System Development Output

The output of the system development showed that the proposed scholarship system has successfully become a functional web-based platform with complete pages and features for its users. It includes interfaces for applicants, panel members, heads of offices, and administrators, allowing them to perform their tasks more easily and in a more organized manner. The developed system provides pages for login, application submission, requirement upload, interview review, performance assessment, applicant management, ranking, report, and qualified scholarship lists. This shows that the system has been able to transform the manual process into a more efficient, transparent, and easy-to-use digital system.

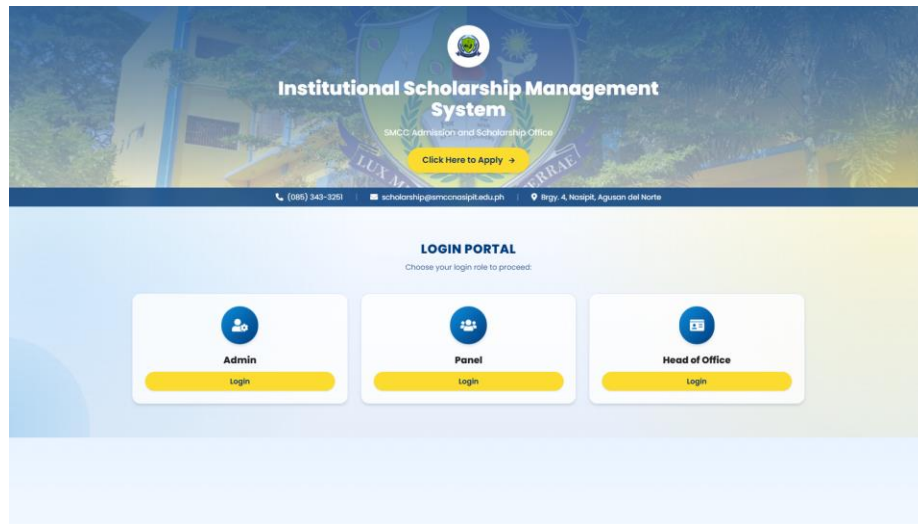


Figure 30. Landing Page

Figure 30 shows the landing page of the Institutional Scholarship Management System. This page is the main entrance to the system where users can quickly find the system title, school contact details, and the scholarship application button. It also provides a login portal for Admin, Panel, and Head of Office, so that each user can access the system based on their role. The page is designed to be simple, clear, and easy to use for all users.



```

<!-- Background image (subtle) -->
<div class="absolute inset-0 -z-20 opacity-30">
  
</div>

<!-- Semi-transparent overlay to blend image -->
<div class="absolute inset-0 -z-10 bg-gradient-to-b from-[#003b7d]/70 via-[#003b7d]/40 to-transparent"></div>

<!-- Content -->
<div class="relative z-10 max-w-3xl mx-auto animate-fade-in-up">
  

  <h1 class="text-white font-extrabold text-3xl md:text-4xl leading-tight mt-5 drop-shadow-sm">
    Institutional Scholarship Management System
  </h1>

  <p class="text-white mt-3 text-sm md:text-base max-w-md mx-auto">
    SMCC Admission and Scholarship Office
  </p>

  <button
    id="applyBtn"
    class="mt-3 md:mt-4 bg-[#fcdc2f] text-[#052c6a] font-semibold rounded-full px-6 py-2.5 md:px-7 md:py-3 text"
    type="button"
  >
    Click Here to Apply
    <i class="fas fa-arrow-right ml-2 text-xs"></i>
  </button>
</div>

<!-- Login Portal Title -->
<section class="text-center max-w-xl mx-auto mb-10 animate-fade-in-up delay-1">
  <h2 class="text-[#003b7d] font-extrabold text-2xl mb-2 uppercase tracking-wide">
    Login Portal
  </h2>
  <p class="text-sm text-[#052c6a]/80">
    Choose your login role to proceed:
  </p>
</section>

<!-- Login Cards -->
<section
  class="grid grid-cols-1 sm:grid-cols-3 gap-6 animate-fade-in-up delay-2"
>

```

Figure 31. Code Snippet for Landing Page

Figure 31 shows the HTML code snippet for the landing page and login portal of the Institutional Scholarship Management System. It creates application button, login cards for Admin, Panel, and Head of Office, with each button linked to its assigned login page. This code helps make the page simple, clear, and easy to use.



SAINT MICHAEL COLLEGE OF CARAGA
OFFICE OF ADMISSION AND SCHOLARSHIP SERVICES
Institutional Scholarship Grants (Step 3 of 3)
Application Form for Institutional Scholarships / Grants

APPLICATION FORM
For Institutional Scholars / Grantees of Saint Michael College of Caraga

Step 1: List & Requirements Step 2: Application Step 3: Application Step 4: Upload Requirements

SELECTED SCHOLARSHIP / GRANT
Student Assistant (Grant #1)
(This is based on what you chose in Step 1) < Back to Step 1

Type of Scholarship / Grant
Please select the type of scholarship or grant you are applying for. This will help the office categorize your application.

☐ Academic ☒ Student Assistant
☐ Kabayani ☐ Others / Discounts

Please specify (if Kabayani):
e.g. Division Scholar, School President's Scholar, etc.

Please specify (if Others / Discounts):
Specify other grant/discount (e.g. PWT, Alumni, DSDM-FM)

Scholar / Grantee's Profile

Name of Applicant *
Full name

Program / Course Studied *
Select program / course

Year Level *
Select year level

Program / Course Studied *
Select program / course

Year Level *
Select year level

School Year *
Select school year

Semester *
Select semester

Permanent Address *
Complete house address

Gender *
Select gender

Age *
e.g. 20

Date of Birth *
mm/dd/yyyy

Contact Number *
09000000000

Email Address *
name@gmail.com

Estimated Gross Income of the Family per Month *
e.g. 25000

Mother's Name *
Full name

Contact Number *
09000000000

Occupation *
Occupation

Company's Name *
Company name

Company's Address *
Company address

Age *
e.g. 45

Father's Information

Father's Name *
Full name

Contact Number *
09000000000

Occupation *
Occupation

Company's Name *
Company name

Company's Address *
Company address

Age *
e.g. 45

After submitting this form, you will proceed to **Step 4: Upload Requirements** for the grant you selected.

Submit & Go to Step 4

Figure 32. Applicant's Application Form

Figure 32 shows the application form page of the Institutional Scholarship Management System. It contains the key fields required for the scholarship application, including the selected grant, applicant details, and parent information. It also guides the user to the next step of the application process.



```

if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $grantId = $currentGrantId;

    if ($grantId <= 0) {
        $errors[] = "Missing grant selection.";
    }

    $requiredFields = [
        "scholarshipType" => "Scholarship type is required.",
        "applicantName" => "Applicant name is required.",
        "programCourse" => "Program/course is required.",
        "yearLevel" => "Year level is required.",
        "schoolYear" => "School year is required.",
        "semester" => "Semester is required.",
        "permanentAddress" => "Permanent address is required.",
        "gender" => "Gender is required.",
        "age" => "Age is required.",
        "dateOfBirth" => "Date of birth is required.",
        "contactNumber" => "Contact number is required.",
        "emailAddress" => "Email address is required.",
        "estimatedIncome" => "Estimated income is required.",
        "motherName" => "Mother's name is required.",
        "fatherName" => "Father's name is required."
    ];

    foreach ($requiredFields as $field => $message) {
        if (read_post_field($field) === "") {
            $errors[] = $message;
        }
    }

    if (
        uses_student_assistant_program_options($grantId, read_post_field("scholarshipType")) &&
        $programCourseValue !== "" &&
        !in_array($programCourseValue, $studentAssistantPrograms, true)
    ) {
        $errors[] = "Please select a valid Student Assistant program/course.";
    }

    $emailAddress = read_post_field("emailAddress");
    if ($emailAddress !== "") {
        $isValidEmail = filter_var($emailAddress, FILTER_VALIDATE_EMAIL);
        $isGmail = (bool)preg_match('/@gmail\.com$/i', $emailAddress);
        if (!$isValidEmail || !$isGmail) {
            $errors[] = "Email address must be a valid @gmail.com address.";
        }
    }
}

```

Figure 33. Code Snippet of Application Form

Figure 33 shows the code snippet for the Institutional Scholarship Management System application form. It is used to check the information entered by the applicant, ensure that the required



fields are complete, save the submitted details, and guide the user to the next step of the application process.

Upload Requirements

Please upload all the required documents for the selected scholarship grant. Make sure files are clear and readable.

Student Assistant
STEP 3 - UPLOAD REQUIREMENTS

Reminder: Accepted file types: PDF, JPG, JPEG, PNG. Upload one file per requirement.

Requirements to upload:

- 2x2 ID Picture
- Application Letter
- Resume
- Form 135 / Report of Grades
- Certificate of Indigency
- Certificate of Good Moral Character

2x2 ID Picture
Choose File No file chosen Clear File

Application Letter
Choose File No file chosen Clear File

Resume
Choose File No file chosen Clear File

Form 135 / Report of Grades
Choose File No file chosen Clear File

Certificate of Indigency
Choose File No file chosen Clear File

Certificate of Good Moral Character
Choose File No file chosen Clear File

Submit Requirements

Figure 34. Applicant's Upload Requirements

Figure 34 shows the Institutional Scholarship Management System upload requirements page. This allows the applicant to submit the necessary documents for the selected scholarship or grant. The page lists the required files and provides upload fields where the applicant can attach each document before submitting the requirements.



```
$grants = [  
  1 => [  
    "title" => "Student Assistant",  
    "requirements" => [  
      "2x2 ID Picture",  
      "Application Letter",  
      "Resume",  
      "Form 138 / Report of Grades",  
      "Certificate of Indigency",  
      "Certificate of Good Moral Character"  
    ]  
  ],  
  
  2 => [  
    "title" => "Academic Scholarship Program",  
    "requirements" => [  
      "Certification (Top 1 or Top 2)",  
      "2x2 ID Picture",  
      "Form 138 / Grades",  
      "Certificate of Indigency",  
      "Certificate of Good Moral Character"  
    ]  
  ],  
  
  3 => [  
    "title" => "ESG President Scholarship",  
    "requirements" => [  
      "2x2 ID Picture",  
      "Endorsement Letter from OSAS"  
    ]  
  ],  
  
  4 => [  
    "title" => "Kabayani Scholarship Program",  
    "requirements" => [  
      "2x2 ID Picture",  
      "Endorsement Letter"  
    ]  
  ],  
  
  5 => [  
    "title" => "Kabayani Loyalty Grant",  
    "requirements" => [  
      "2x2 ID Picture",  
      "Certification from External Linkages Coordinator",  
      "Proof of Relationship",  
      "Endorsement Letter from Retiree"  
    ]  
  ]  
]
```

Figure 35. Code Snippet of Upload Requirements

Figure 35 shows the code snippet used for the Institutional Scholarship Management System upload requirements page. It identifies the scholarship or grant options and lists the required documents for each. This helps the system to display the correct files that the applicant needs to upload.

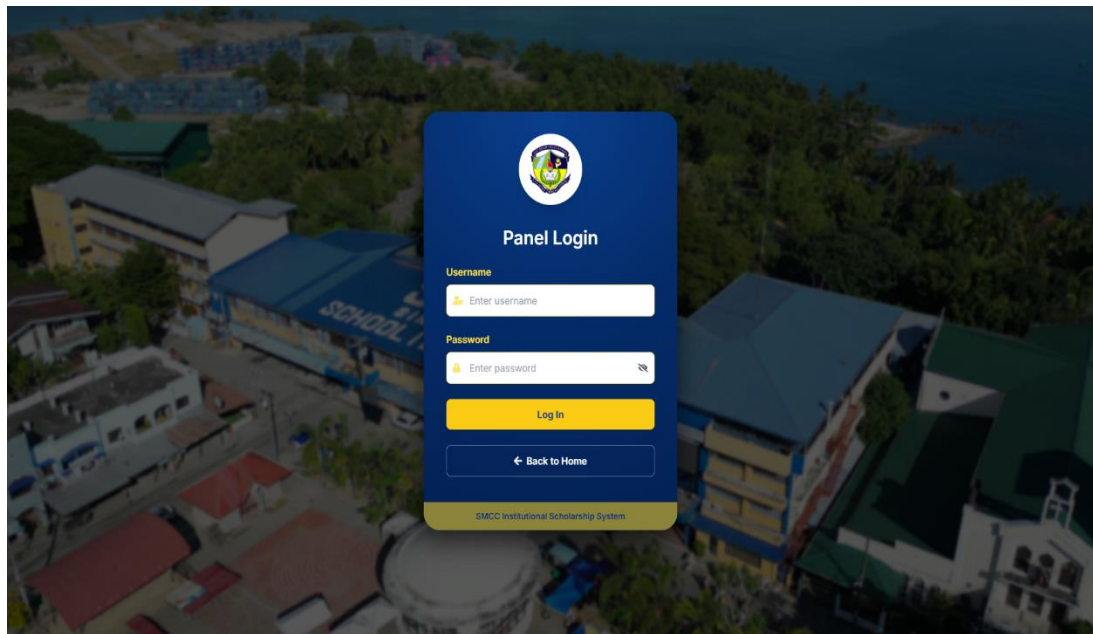


Figure 36. Panel Login Page

Figure 36 shows the panel login page of the Institutional Scholarship Management System. It allows the panel member to enter a username and password to access the system. It also includes a button to log in and an option to return to the home page.



```

if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $panelistName = trim((string)($_POST["panelist_name"] ?? ""));
    $panelistPassword = trim((string)($_POST["panelist_password"] ?? ""));

    if ($panelistName === "" || $panelistPassword === "") {
        $loginError = "Please enter your username and password.";
    } else {
        $stmt = $conn->prepare("SELECT full_name, password_hash, status FROM panelists WHERE username = ? LIMIT 1");
        if ($stmt) {
            $stmt->bind_param("s", $panelistName);
            $stmt->execute();
            $result = $stmt->get_result();
            $row = $result ? $result->fetch_assoc() : null;
            $stmt->close();

            if (!$row || (string)$row["status"] ?? "" !== "active") {
                $loginError = "Invalid username or password.";
            } else {
                $storedHash = (string)$row["password_hash"] ?? "";
                $verified = false;
                if ($storedHash !== "") {
                    if (strpos($storedHash, "$2y$") === 0 || strpos($storedHash, "$argon2") === 0) {
                        $verified = password_verify($panelistPassword, $storedHash);
                    } else {
                        $verified = hash("sha256", $panelistPassword) === $storedHash;
                    }
                }

                if ($verified) {
                    $displayName = trim((string)$row["full_name"] ?? "");
                    if ($displayName === "") {
                        $displayName = $panelistName;
                    }
                    $_SESSION["panelist_username"] = $panelistName;
                    $_SESSION["panelist_name"] = $displayName;
                    $_SESSION["panelist_login_success"] = true;
                    header("Location: " . panelistConsumeRedirectTarget("panelistDashboard.php"));
                    exit();
                }
                $loginError = "Invalid username or password.";
            }
        }
    }
}

```

Figure 37. Code Snippet of Panel Login

Figure 37 shows the code snippet used for the panel login page of the Institutional Scholarship Management System. It checks the username and password entered by the panel member, ensures that the account is valid and active, and then allows access to the panel dashboard if the login details are correct.

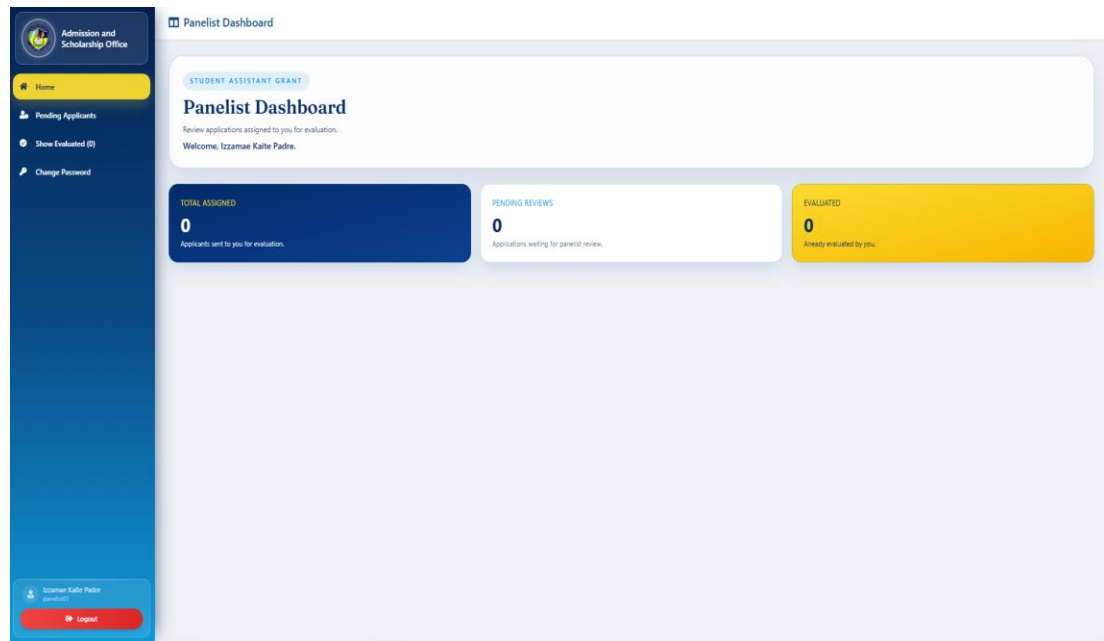


Figure 38. Panelist Dashboard

Figure 38 shows the panelist dashboard of the Institutional Scholarship Management System. This serves as the main page for panel members after logging in. The dashboard displays the total assigned applicants, pending reviews, and reviewed applications, and it also provides menu options for viewing applicants, checking reviewed records, and changing passwords.

Figure 39. Evaluation Form - Panel



Figure 39 shows the interview evaluation sheet of the Institutional Scholarship Management System. It is used by the panel member to evaluate the applicant based on the given criteria and rating scale. It also shows the applicant's name, interview date, interviewer's name, and space for review details and comments.

```
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $applicant_name = $prefillApplicantName;
    $interview_date = date('Y-m-d');
    $interviewer_name = $panelistName;
    $overall_assessment = $_POST['overall_assessment'] ?? null;
    $strengths = trim($_POST['strengths']) ?? '';
    $areas_for_improvement = trim($_POST['areas_for_improvement']) ?? '';
    $signature_data = $_POST['signature_data'] ?? null;

    $ratings = $_POST['rating'] ?? [];
    $comments = $_POST['comment'] ?? [];
    $computed_total = 0;
    foreach (range(1, 12) as $criterion_id) {
        if (!isset($ratings[$criterion_id])) {
            continue;
        }
        $rating_value = (int) $ratings[$criterion_id];
        if ($rating_value < 1 || $rating_value > 5) {
            continue;
        }
        $computed_total += $rating_value;
    }
    $total_points = (string) $computed_total;
    $overall_assessment = ($overall_assessment === '' || $overall_assessment === null) ? null : $overall_assessment;
    $strengths = $strengths === '' ? null : $strengths;
    $areas_for_improvement = $areas_for_improvement === '' ? null : $areas_for_improvement;
    $signature_data = $signature_data === '' ? null : $signature_data;

    if ($loadError !== '' || $applicant_name === '' || $interviewer_name === '') {
        $save_error = $loadError !== '' ? $loadError : 'Applicant or panelist details are missing.';
    } else {
        try {
            $conn->begin_transaction();

            $stmt = $conn->prepare(
                'INSERT INTO interview_evaluations (applicant_id, applicant_name, interview_date, interviewer_name, total_points, overall_assessment, strengths, areas_for_improvement, signature_data) VALUES (?, ?, ?, ?, ?, ?, ?, ?)'
            );
            $stmt->bind_param(
                'isssssss',
                $applicant_id,
                $applicant_name,
                $interview_date,
                $interviewer_name,
                $total_points,
                $overall_assessment,
                $strengths,
                $areas_for_improvement,
                $signature_data
            );
        }
    }
}
```

Figure 40. Code Snippet for Evaluation Sheet

Figure 40 shows the code snippet used for the interview evaluation sheet of the Institutional Scholarship Management System. It is used to capture the ratings and comments entered by the panel member, calculate the total score, and save the applicant's evaluation details in the system.

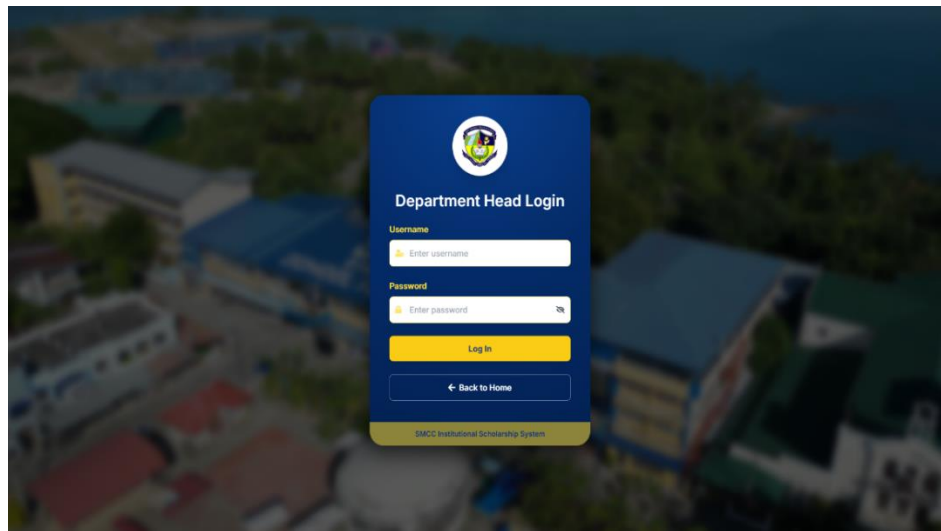


Figure 41. login page -Department head

Figure 41 shows the Institutional Scholarship Management System Department Head login page. It allows the Head of Office to enter a username and password to access the system. It also provides buttons for logging in and returning to the home page.

```
if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $username = trim((string)($_POST["head_username"] ?? ""));
    $password = trim((string)($_POST["head_password"] ?? ""));

    if ($username === "" || $password === "") {
        $loginError = "Please enter your username and password.";
    } else {
        $stmt = $conn->prepare("SELECT name, lastname, office, password_hash, status FROM head_offices WHERE username = ? LIMIT 1");
        if ($stmt) {
            $stmt->bind_param("s", $username);
            $stmt->execute();
            $result = $stmt->get_result();
            $row = $result ? $result->fetch_assoc() : null;
            $stmt->close();

            if (!$row || (string)$row["status"] ?? "" !== "active") {
                $loginError = "Invalid username or password.";
            } else {
                $storedHash = (string)$row["password_hash"] ?? "";
                $verified = false;
                if ($storedHash !== "") {
                    if (strpos($storedHash, "$2y$") === 0 || strpos($storedHash, "$argon2") === 0) {
                        $verified = password_verify($password, $storedHash);
                    } else {
                        $verified = hash("sha256", $password) === $storedHash;
                    }
                }
            }
        }
    }
}
```

Figure 42. Code Snippet for Login Page – Department Head

Figure 42 shows the code snippet used for the Department Head login page of the Institutional Scholarship Management System. It checks the username and password entered by the Head of



Office, ensures that the account is valid and active, and allows access to the system when the login details are correct.

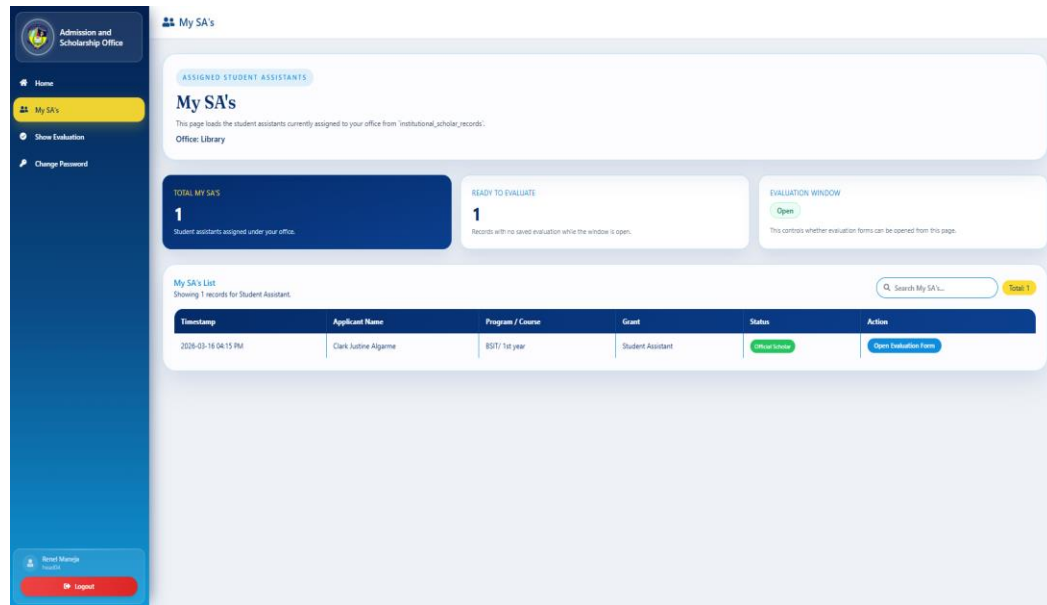


Figure 43. Department Head Dashboard

Figure 43 shows the Department Dashboard page of the Institutional Scholarship Management System. It shows the list of student assistants assigned to the Head of Office, along with their basic details and status. It also provides an option to open the evaluation form for each assigned student assistant.



```

if ($headOffice === "") {
    $loadError = "No office is assigned to this head account.";
    $evaluationLoadError = $loadError;
} else {
    $headOfficeKey = strtolower(trim($headOffice));
    $scholarTableResult = $conn->query("SHOW TABLES LIKE 'institutional_scholar_records'");
    $hasScholarTable = $scholarTableResult instanceof mysqli_result && $scholarTableResult->num_rows > 0;
    if ($scholarTableResult instanceof mysqli_result) {
        $scholarTableResult->free();
    }

    if (!$hasScholarTable) {
        $loadError = "Scholar records table is not available.";
        $evaluationLoadError = $loadError;
    } else {
        $approvedApplicantMap = [];
        $studentAssistantQuery = "SELECT
            id,
            scholar_id,
            full_name,
            program_year,
            semester,
            academic_year,
            assigned_office,
            category,
            grant_applied,
            status,
            created_at
        FROM institutional_scholar_records
        WHERE (
            LOWER(TRIM(COALESCE(category, ''))) = 'student_assistant'
            OR (
                LOWER(TRIM(COALESCE(category, ''))) = 'official'
                AND LOWER(TRIM(COALESCE(grant_applied, ''))) LIKE '%assistant%'
            )
        )
        AND LOWER(TRIM(COALESCE(assigned_office, ''))) = ?
        AND COALESCE(contract_ended, 0) = 0
        ORDER BY
        CASE
            WHEN LOWER(TRIM(COALESCE(category, ''))) = 'official'
            AND LOWER(TRIM(COALESCE(grant_applied, ''))) LIKE '%assistant%'
            THEN 0
            ELSE 1
        END,
        created_at DESC,
        id DESC";

```

Figure 44. Code Snippet for Department Head Dashboard

Figure 44 shows the code snippet used for the My SA page of the Institutional Scholarship Management System. It is used to retrieve the list of student assistants assigned to the Head of Office and display their records in the system.

Scale	Verbal Description	Verbal Interpretation
4	Excellent	This rating is given to student assistants who consistently exceed expectations and demonstrate outstanding performance in their assigned tasks.
3	Good	This rating is given to student assistants who consistently meet expectations and perform their assigned tasks satisfactorily.
2	Fair	This rating is given to student assistants who occasionally meet expectations but may have some areas for improvement.
1	Poor	This rating is given to student assistants who consistently fail to meet expectations and demonstrate unsatisfactory performance in their assigned tasks.

Criteria	Excellent (4)	Good (3)	Fair (2)	Poor (1)
A. Quality and Quantity of Work				
A.1 Accurate at work assigned	☐	☐	☐	☐
A.2 Always completes tasks	☐	☐	☐	☐
A.3 Works in a timely manner	☐	☐	☐	☐
A.4 Asks for more work when assigned tasks are done	☐	☐	☐	☐
A.5 Easily accepts new responsibilities	☐	☐	☐	☐
Total				0

Figure 45. Evaluation Form – Department Head



Figure 45 shows the student assistant evaluation form of the Institutional Scholarship Management System. It is used by the Head of Office to rate the performance of the assigned student assistant based on the given criteria. The form also shows the student assistant's name, assignment area, semester and year of study, date of evaluation, and the rating scale used in the evaluation.

```

if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $strengthsInput = trim((string)($_POST["strengths"] ?? ""));
    $recommendationsInput = trim((string)($_POST["recommendations"] ?? ""));
    $signatureDataInput = trim((string)($_POST["signature_data"] ?? ""));
    $evaluationDateValue = date("Y-m-d");

    $invalidRatings = [];
    foreach ($formRatings as $fieldName => $unusedValue) {
        $ratingValue = isset($_POST[$fieldName]) ? (int)$_POST[$fieldName] : 0;
        if ($ratingValue < 1 || $ratingValue > 4) {
            $formRatings[$fieldName] = null;
            $invalidRatings[] = $fieldName;
        } else {
            $formRatings[$fieldName] = $ratingValue;
        }
    }

    if ($loadError !== "") {
        $saveError = $loadError;
    } elseif (!$hasEvaluationTable) {
        $saveError = "Evaluation storage is unavailable.";
    } elseif (!$isEvaluationWindowOpen) {
        $saveError = "Evaluation window is currently closed.";
    } elseif (!empty($invalidRatings)) {
        $saveError = "Please complete all ratings before submitting.";
    } elseif ($signatureDataInput === "" || strpos($signatureDataInput, "data:image/") !== 0) {
        $saveError = "Signature is required before saving.";
    } else {
        $sectionATotal = 0;
        foreach ($ratingGroups["a"] as $fieldName) {
            $sectionATotal += (int)($formRatings[$fieldName] ?? 0);
        }
        $sectionBTotat = 0;
        foreach ($ratingGroups["b"] as $fieldName) {
            $sectionBTotat += (int)($formRatings[$fieldName] ?? 0);
        }
        $sectionCTotat = 0;
        foreach ($ratingGroups["c"] as $fieldName) {
            $sectionCTotat += (int)($formRatings[$fieldName] ?? 0);
        }
        $overallTotal = $sectionATotal + $sectionBTotat + $sectionCTotat;

        $ratingsJson = json_encode($formRatings, JSON_UNESCAPED_SLASHES);
        $strengthsValue = $strengthsInput !== "" ? $strengthsInput : null;
        $recommendationsValue = $recommendationsInput !== "" ? $recommendationsInput : null;
        $assignedOfficeValue = $applicationProfile["assigned_office"] !== ""
            ? $applicationProfile["assigned_office"]
            : $headOffice;
    }
}

```

Figure 46. Code Snippet for Evaluation Form

Figure 46 shows the code snippet used for the student assistant evaluation form of the Institutional Scholarship Management System. It checks the ratings entered by the Head of Office,



ensures that the required information is complete, calculates the total score, and saves the evaluation details in the system.

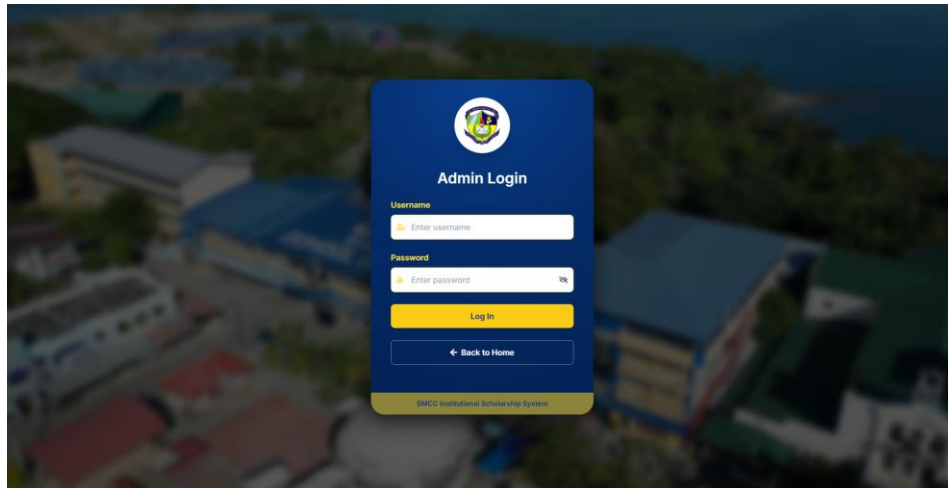


Figure 47. Login Page - Admin

Figure 47 shows the Admin login page of the Institutional Scholarship Management System. It allows the administrator to enter a username and password to access the system. It also provides buttons for logging in and returning to the home page.



```
$loginError = "";
if (adminIsAuthenticated()) {
    header("Location: " . adminPath("adminDashboard.php"));
    exit();
}

if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $username = trim((string)$_POST["admin_username"] ?? "");
    $password = trim((string)$_POST["admin_password"] ?? "");

    if ($username === "" || $password === "") {
        $loginError = "Please enter your username and password.";
    } else {
        // For demonstration, we will just check against hardcoded credentials.
        // In a real application, you would query the database here.
        if ($username === "admin" && $password === "admin123") {
            $_SESSION["admin_username"] = $username;
            $_SESSION["admin_name"] = $username;
            header("Location: " . adminConsumeRedirectTarget("adminDashboard.php"));
            exit();
        } else {
            $loginError = "Invalid username or password.";
        }
    }
}
?>
```

Figure 48. Code Snippet for Admin Login

Figure 48 shows the code snippet used for the Admin login page of the Institutional Scholarship Management System. It checks the username and password entered by the administrator and allows access to the system when the login details are correct.

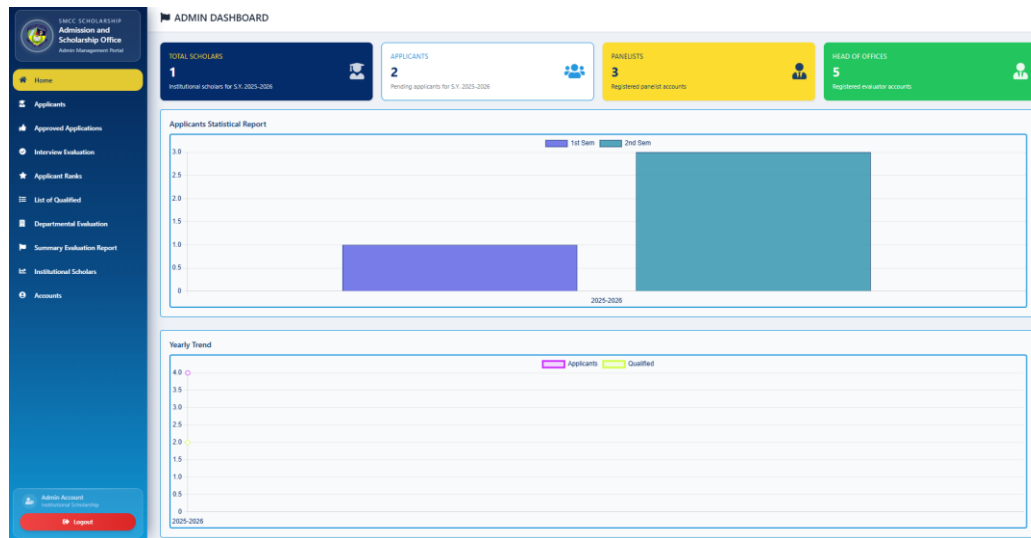


Figure 49. Admin Dashboard

Figure 49 shows the admin dashboard of the Institutional Scholarship Management System. It serves as the main page for the administrator after logging in. The dashboard displays important summary information such as total scholars, applicants, panelists, and Heads of Offices, along with reports and charts that help track scholarship records.



```
function adminDashboardScholarCountKey(array $row): string
{
    $scholarTableResult = $conn->query("SHOW TABLES LIKE 'institutional_scholar_records'");
    if ($scholarTableResult instanceof mysqli_result) {
        $hasScholarTable = $scholarTableResult->num_rows > 0;
        $scholarTableResult->free();
        if ($hasScholarTable) {
            $hasContractEndedColumn = false;
            $contractEndedColumnResult = $conn->query("SHOW COLUMNS FROM institutional_scholar_records LIKE 'contract_ended'");
            if ($contractEndedColumnResult instanceof mysqli_result) {
                $hasContractEndedColumn = $contractEndedColumnResult->num_rows > 0;
                $contractEndedColumnResult->free();
            }

            $scholarWhereClauses = ["TRIM(COALESCE(academic_year, '')) = ?"];
            if ($hasContractEndedColumn) {
                $scholarWhereClauses[] = "COALESCE(contract_ended, 0) = 0";
            }

            $scholarCountSql = "
            SELECT
                source_application_id,
                scholar_id,
                full_name,
                program_year,
                academic_year
            FROM institutional_scholar_records
            WHERE " . implode(" AND ", $scholarWhereClauses) . "
            ORDER BY id DESC
            ";
            $scholarCountStmt = $conn->prepare($scholarCountSql);
            if ($scholarCountStmt) {
                $scholarCountStmt->bind_param("s", $dashboardSchoolYear);
                if ($scholarCountStmt->execute()) {
                    $scholarCountResult = $scholarCountStmt->get_result();
                    $uniqueScholarKeys = [];
                    while ($scholarRow = $scholarCountResult->fetch_assoc()) {
                        $rowKey = adminDashboardScholarCountKey($scholarRow);
                        if ($rowKey != "") {
                            $uniqueScholarKeys[$rowKey] = true;
                        }
                    }
                    $totalScholarsCount = count($uniqueScholarKeys);
                    if ($scholarCountResult instanceof mysqli_result) {
                        $scholarCountResult->free();
                    }
                }
                $scholarCountStmt->close();
            }
        }
    }
}
```

Figure 50. Code Snippet for Admin Dashboard

Figure 50 shows the code snippet used for the admin dashboard of the Institutional Scholarship Management System. It is used to retrieve the scholarship records for the selected academic year and count the total number of scholarships to be displayed on the dashboard.

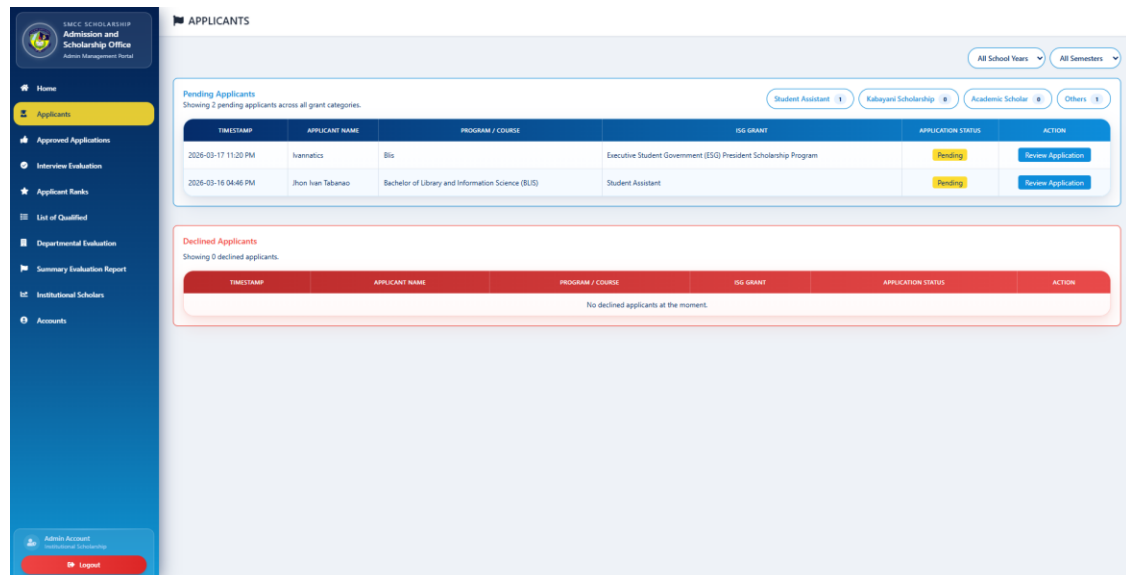




Figure 51. Pending Applicants

Figure 51 shows the Institutional Scholarship Management System applicants page. It shows the list of scholarship applicants and their basic details, such as name, program or course, grant applied for, and application status. It also allows the administrator to review applications and use school year and semester filters to sort records.

```
$pendingApplicants = [];
$pendingQuery = "SELECT id, created_at, applicant_name, program_course, grant_id, status
FROM applications
WHERE (status IS NULL
OR TRIM(status) = ''
OR LOWER(TRIM(status)) = 'pending')";
if (!empty($filterClauses)) {
    $pendingQuery .= " AND " . implode(" AND ", $filterClauses);
}
$pendingQuery .= " ORDER BY created_at DESC";
if ($stmt = $conn->prepare($pendingQuery)) {
    if (!empty($filterParams)) {
        $stmt->bind_param($filterTypes, ...$filterParams);
    }
    $stmt->execute();
    $result = $stmt->get_result();
    if ($result) {
        while ($row = $result->fetch_assoc()) {
            $applicationId = (int)$row["id"] ?? 0;
            $grantId = (int)$row["grant_id"] ?? 0;
            $grantLabel = $grantLabels[$grantId] ?? "Others";
            $submittedAtRaw = $row["created_at"] ?? "";
            $submittedAt = $submittedAtRaw ? date("Y-m-d h:i A", strtotime($submittedAtRaw)) : "";
            $status = isset($row["status"]) ? trim((string)$row["status"]) : "";
            if ($status === "") {
                $status = "Pending";
            }
        }

        $pendingApplicants[] = [
            "id" => $applicationId,
            "submitted_at" => $submittedAt,
            "name" => $row["applicant_name"] ?? "",
            "program_course" => $row["program_course"] ?? "",
            "grant" => $grantLabel,
            "status" => $status,
        ];
    }
    $result->free();
}
$stmt->close();
}
```

Figure 52. Code Snippet for Pending Applicants

Figure 52 shows the code snippet used for the Institutional Scholarship Management System applicants page. It retrieves the list of pending applicants from the system and displays their important details, such as name, program or course, grant applied for, date of submission, and application status.



Figure 53. Application Form

Figure 53 shows the application form view of the Institutional Scholarship Management System. It presents complete details of the applicant's application in a paper-style format for review. This page helps the administrator to review the submitted information clearly and provides an option to print the form.

```


Scholar/Grantee's Profile:



Name of Applicant



<?= htmlspecialchars(app_value($application, "applicant_name")) ?>



Program/Course Enrolled



<?= htmlspecialchars(app_value($application, "program_course")) ?>



Year Level



<?= htmlspecialchars(app_value($application, "year_level")) ?>



School Year



<?= htmlspecialchars(app_value($application, "school_year")) ?>



Semester



<?= htmlspecialchars(app_value($application, "semester")) ?>



Permanent Address



<?= htmlspecialchars(app_value($application, "permanent_address")) ?>


```



Figure 54. Code Snippet for Application Form

Figure 54 shows the code snippet used for the application form view of the Institutional Scholarship Management System. It displays applicant profile details in a paper-style application form.

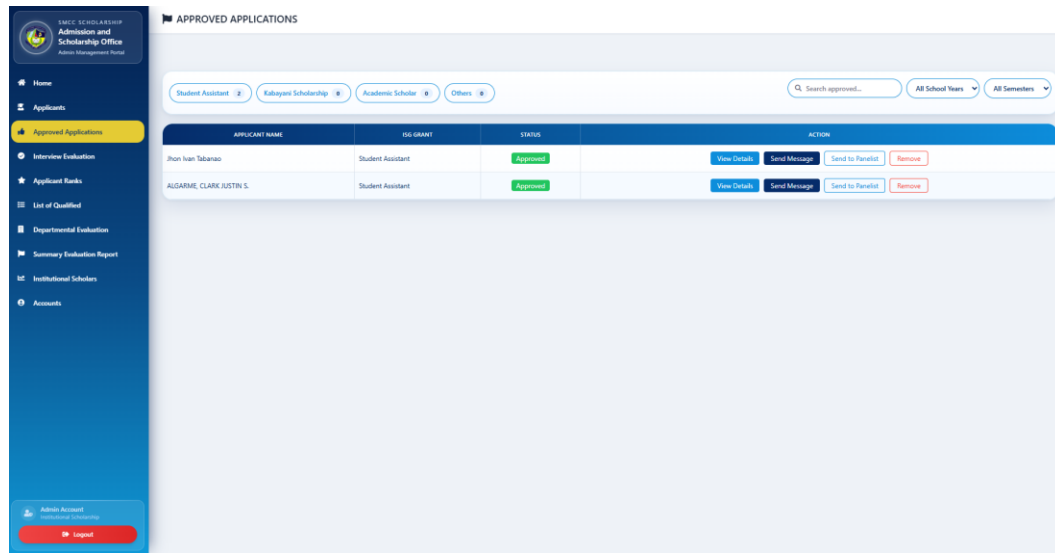


Figure 55. Approved Applicants

Figure 55 shows the approved application page of the Institutional Scholarship Management System. It shows the list of applicants whose scholarship applications have been approved. It also provides action buttons for viewing details, sending a message, assigning the applicant to a panelist, and removing a note.

```
$approvedApplicants = [];
$approvedQuery = "SELECT id, applicant_name, email_address, grant_id, status FROM applications WHERE status = 'Approved'";
if (empty($filterClauses)) {
    $approvedQuery .= " AND " . implode(" AND ", $filterClauses);
}
$approvedQuery .= " ORDER BY created_at DESC";
if ($stmt = $conn->prepare($approvedQuery)) {
    if (empty($filterParams)) {
        $stmt->bind_param($filterTypes, ...$filterParams);
    }
    $stmt->execute();
    $result = $stmt->get_result();
    if ($result) {
        while ($row = $result->fetch_assoc()) {
            $grantId = (int)$row["grant_id"] ?? 0;
            $grantLabel = $grantLabels[$grantId] ?? "Others";
            $status = isset($row["status"]) ? trim((string)$row["status"]) : "Approved";
            if ($status === "") {
                $status = "Approved";
            }
        }
        $approvedApplicants[] = [
            "id" => (int)$row["id"] ?? 0,
            "name" => $row["applicant_name"] ?? "",
            "email" => $row["email_address"] ?? "",
            "grant_id" => $grantId,
            "grant" => $grantLabel,
            "status" => $status,
        ];
    }
    $result->free();
}
$stmt->close();
```



Figure 56. Code Snippet for Approve Applicants

Figure 56 shows the code snippet used for the approved application page of the Institutional Scholarship Management System. It retrieves the list of approved applicants and displays their important details, such as the applicant's name, email address, grant, and application status.

INTERVIEW EVALUATION

Filter: All School Years, All Semesters, All Batches [Print]

Saint Michael College of Caraga
 Brgy. 4, Tagaytay, Agusan del Norte, Philippines
 Tel. No. 083 223-6208
 Website: www.smcsc.edu.ph, Email: communications@smcsc.edu.ph

Office of the Admission & Scholarship
Student Assistance Applicants' Interview Result
 All Semesters, SY: 2025-2026
 All Batches

SEQ	NAME OF APPLICANT	WEIGHTED MEAN	VERBAL DESCRIPTION	VERBAL INTERPRETATION	REMARKS
1	ALDARME, CLARK JUSTIN S.				Discontinued
2	Jhon Jon Tabasco				Discontinued

Prepared by:
 ARLEN B. TUYOGON, MEd
 Head, Admission & Scholarship

Checked by:
 FELMARIE MANLINAS, MACBDS
 Head, Student Affairs & Services

Noted by:
 RICKY E. DESTACAMENTO, RGC, MAED
 Head, BEMDO

Figure 57. Interview Evaluation

Figure 57 shows the Institutional Scholarship Management System interview review page. It shows the interview results of the applicants, including their names, weighted mean, verbal description, verbal interpretation, and remarks. It also provides filter options and a print button for easier viewing and reporting.



```

$applicantIds = array_values(array_unique($applicantIds));
if (!empty($applicantIds)) {
    foreach ($applicantIds as $applicantId) {
        $interviewSummariesByApplicantId[$applicantId] = [
            "assigned_panel_count" => 0,
            "rated_panel_count" => 0,
            "total_points_sum" => 0.0,
        ];
    }

    $placeholders = implode(", ", array_fill(0, count($applicantIds), "?"));
    $types = str_repeat("i", count($applicantIds));
    $panelAssignmentSql = "
        SELECT application_id, COUNT(DISTINCT panelist_username) AS assigned_panel_count
        FROM panelist_queue
        WHERE application_id IN (($placeholders))
        GROUP BY application_id
    ";
    $panelAssignmentStmt = $conn->prepare($panelAssignmentSql);
    if ($panelAssignmentStmt) {
        $panelAssignmentStmt->bind_param($types, ...$applicantIds);
        if ($panelAssignmentStmt->execute()) {
            $panelAssignmentResult = $panelAssignmentStmt->get_result();
            while ($panelAssignmentRow = $panelAssignmentResult->fetch_assoc()) {
                $applicantId = (int)($panelAssignmentRow["application_id"] ?? 0);
                if ($applicantId <= 0 || !isset($interviewSummariesByApplicantId[$applicantId])) {
                    continue;
                }

                $interviewSummariesByApplicantId[$applicantId]["assigned_panel_count"] = (int)($panelAssignmentRow["assigned_panel_count"] ?? 0);
            }
            $panelAssignmentResult->free();
        }
        $panelAssignmentStmt->close();
    }
}

```

Figure 58. Code Snippet for Interview Evaluation

Figure 58 shows the code snippet used for the Institutional Scholarship Management System interview review page. It retrieves applicant records, evaluates assigned panel members, and prepares interview evaluation results to be displayed in the system.

ID	NAME OF APPLICANT	RATING				100% AVERAGE	RANK	REMARKS
		Examination	Interview	Grades				
1	ALDARME CLARK JUSTIN L.	100	100%	5.00	100%	100	100%	
2	Jhon Ivan Tabares							

Prepared by: ARLYN B. TUYOGON, MMMM Head, Admission & Scholarship

Checked by: FERMARIE MANLINAS, MACDROS Head, Student Affairs & Services

Noted by: RICKY E. DESTACAMENTO, RUC, MAED Head, HRD/DO

Figure 59. Ranking Page

Figure 59 shows the ranking page of the Institutional Scholarship Management System. It shows the list of applicants with their test score, interview score, grades, overall average, rank, and sentences. It also provides filter options and buttons for saving and printing the ranking report.



```

if ($_SERVER["REQUEST_METHOD"] === "POST" && ($conn ?? null) instanceof mysqli) {
    $postedSchoolYear = trim((string)($_POST["school_year"] ?? $selectedSchoolYear ?? ""));
    $postedSemester = trim((string)($_POST["semester"] ?? $selectedSemester ?? ""));
    $postedBatch = trim((string)($_POST["batch"] ?? $selectedBatch ?? ""));
    $postedApplicantIds = isset($_POST["applicant_ids"]) && is_array($_POST["applicant_ids"])
        ? array_values(array_unique(array_map("intval", $_POST["applicant_ids"])))
        : [];
    $postedExamRatings = isset($_POST["exam_rating"]) && is_array($_POST["exam_rating"]) ? $_POST["exam_rating"] : [];
    $postedGradesRatings = isset($_POST["grades_rating"]) && is_array($_POST["grades_rating"]) ? $_POST["grades_rating"] : [];
    $postedRemarks = isset($_POST["remarks"]) && is_array($_POST["remarks"]) ? $_POST["remarks"] : [];
    $allowedApplicantIds = [];
    foreach ($panelistSentApplicants as $panelistSentApplicant) {
        $applicantId = (int)($panelistSentApplicant["id"] ?? 0);
        if ($applicantId > 0) {
            $allowedApplicantIds[$applicantId] = true;
        }
    }

    $saveStmt = $conn->prepare(
        "INSERT INTO applicant_rank_inputs (application_id, exam_rating, grades_rating, remarks)
        VALUES (?, ?, ?, ?)
        ON DUPLICATE KEY UPDATE
            exam_rating = VALUES(exam_rating),
            grades_rating = VALUES(grades_rating),
            remarks = VALUES(remarks)"
    );
}

```

Figure 60. Code Snippet for Rank Page

Figure 60 shows the code snippet used for the ranking page of the Institutional Scholarship Management System. It is used to retrieve submitted marks and comments, save applicant ranking details, and update system records.

The screenshot displays the 'List of Qualified Applicants' page. At the top, there are filters for 'All School Years', 'All Semesters', and 'All Batches'. The page header includes the Saint Michael College of Caraga logo and contact information. The main content area features a table with the following columns: #, NAME, ADDRESS, CONTACT NUMBER, PROGRAM ENROLLED, YEAR LEVEL, and ASSIGNED OFFICE. Below the table, there is a section for 'Proposed by: ARLYN B. TUYOGON, MMIM', 'Noted by: FELMARIE MANLUNAS, MACDDS', 'Recommending Approval: RICKY E. DESTACAMENTO, RGC, MAED', and 'Approved by: REV. FR. RONNIEL G. BABANO, STL'.

Figure 61. List of Qualified

Figure 61 shows the list of qualified applicants page of the Institutional Scholarship Management System. It shows applicants who are eligible for the scholarship, along with their basic



details such as name, address, contact number, program enrolled, year level, and assigned office. It also includes filter options and buttons for saving and printing the report.

```
if ($_SERVER["REQUEST_METHOD"] == "POST" && ($conn ?? null) instanceof mysqli) {
    $postedSchoolYear = trim((string)($_POST["school_year"] ?? $selectedSchoolYear));
    $postedSemester = trim((string)($_POST["semester"] ?? $selectedSemester));
    $postedBatch = trim((string)($_POST["batch"] ?? $selectedBatch));
    $postedApplicantIds = isset($_POST["applicant_ids"]) && is_array($_POST["applicant_ids"])
        ? array_values(array_unique(array_map("intval", $_POST["applicant_ids"])))
        : [];
    $postedAssignedOffices = isset($_POST["assigned_office"]) && is_array($_POST["assigned_office"])
        ? $_POST["assigned_office"]
        : [];

    if (!isset($hasRankInputTable) || !isset($hasInstitutionalScholarTable)) {
        header("Location: " . $buildQualifiedUrl($postedSchoolYear, $postedSemester, $postedBatch, "error"));
        exit;
    }

    $allowedApplicants = [];
    $whereClauses = [
        "a.grant_id = 1",
        "LOWER(TRIM(a.status)) = 'approved'",
        "LOWER(TRIM(COALESCE(a1.remarks, ''))) = 'hired'",
    ];
    $params = [];
    $types = "";

    if ($postedSchoolYear != "") {
        $whereClauses[] = "a.school_year = ?";
        $params[] = $postedSchoolYear;
        $types .= "s";
    }
    if ($postedSemester != "") {
        $whereClauses[] = "a.semester = ?";
        $params[] = $postedSemester;
        $types .= "s";
    }
    if ($postedBatch != "" && $hasBatchColumn) {
        $whereClauses[] = "a.batch = ?";
        $params[] = $postedBatch;
        $types .= "s";
    }
}
```

Figure 62. Code Snippet for List of Qualified

Figure 62 shows the code snippet used for the Institutional Scholarship Management System's qualified applicant list page. It obtains the selected filter values, evaluates the qualified applicants, and prepares their records for display in the system.

Figure 63. Evaluation List



Figure 63 shows the department review page of the Scholarship Institutional Management System. It shows the list of student assistants assigned to various offices for examination. It also shows their basic details, review status, and the action or current progress of their review.

```
$hasInstitutionalTable = false;
$institutionalTableResult = $conn->query("SHOW TABLES LIKE 'institutional_scholar_records'");
if ($institutionalTableResult instanceof mysqli_result) {
    $hasInstitutionalTable = $institutionalTableResult->num_rows > 0;
    $institutionalTableResult->free();
}

if ($hasInstitutionalTable) {
    $schoolYearOptions = [];
    $semesterOptions = [];
    $assistantTermResult = $conn->query("
        SELECT DISTINCT
            TRIM(COALESCE(academic_year, '')) AS academic_year,
            TRIM(COALESCE(semester, '')) AS semester
        FROM institutional_scholar_records
        WHERE " . isgDepartmentEvaluationAssistantCategorySql() . "
            AND TRIM(COALESCE(assigned_office, '')) <> ''
        ORDER BY academic_year ASC, semester ASC
    ");
    if ($assistantTermResult instanceof mysqli_result) {
        while ($termRow = $assistantTermResult->fetch_assoc()) {
            $academicYear = trim((string)$termRow["academic_year"] ?? "");
            $semester = trim((string)$termRow["semester"] ?? "");
            if ($academicYear != "" && !in_array($academicYear, $schoolYearOptions, true)) {
                $schoolYearOptions[] = $academicYear;
            }
            if ($semester != "" && !in_array($semester, $semesterOptions, true)) {
                $semesterOptions[] = $semester;
            }
        }
    }
    $assistantTermResult->free();
}
```

Figure 64. Code Snippet for Evaluation List

Figure 64 shows the code snippet used for the Institutional Scholarship Management System department review page. It checks scholarship records and retrieves available academic year and semester options. This helps the system to display and filter the analysis records properly.



Figure 65. Paper Form Evaluation

Figure 65 shows the evaluation form of student assistants in the Institutional Scholarship Management System. It displays the student assistant's completed evaluation, including key details, rating scales, and scores given for each performance indicator. This page helps the administrator to clearly review the submitted review record.

```
<main class="paper">
<h2 class="title">Student Assistants' Evaluation Form</h2>
<section class="info-block">
<div class="info-row">
<span class="info-label">Name of Student Assistant:</span>
<span class="info-value"><?> htmlspecialchars($displayApplicantName) ?></span>
</div>
<div class="info-row">
<span class="info-label">Semester & School Year:</span>
<span class="info-value"><?> htmlspecialchars($displaySemesterSchoolYear) ?></span>
</div>
<div class="info-row">
<span class="info-label">Area of Assignment:</span>
<span class="info-value"><?> htmlspecialchars($displayAreaOfAssignment) ?></span>
</div>
<div class="info-row">
<span class="info-label">Head of Office:</span>
<span class="info-value"><?> htmlspecialchars($displayHeadOfOffice) ?></span>
</div>
<div class="info-row">
<span class="info-label">Date of Evaluation:</span>
<span class="info-value"><?> htmlspecialchars($displayEvaluationDate) ?></span>
</div>
</section>
<section class="direction">
<p>
<span class="font-semibold">Direction:</span> Please rate each item below to determine the performance of the assigned student assistant of
</p>
</section>
<section>
<table class="scale-table">
<thead>
<tr>
<th style="width: 5%;>Scale</th>
<th style="width: 45%;>Verbal Description</th>
<th style="width: 50%;>Verbal Interpretation</th>
</tr>
</thead>
<tbody>
<tr>
<td>4</td>
<td>Excellent</td>
<td>This rating is given to student assistants who consistently exceed expectations and demonstrate outstanding performance in their assigned tasks.</td>
</tr>
<tr>
<td>3</td>
<td>Good</td>
<td>This rating is given to student assistants who consistently meet expectations and perform their assigned tasks satisfactorily.</td>
</tr>
<tr>
<td>2</td>
<td>Fair</td>
<td>This rating is given to student assistants who occasionally meet expectations but may have areas for improvement.</td>
</tr>
<tr>
<td>1</td>
<td>Poor</td>
<td>This rating is given to student assistants who consistently fail to meet expectations and demonstrate unsatisfactory performance in their assigned tasks.</td>
</tr>
</tbody>
</table>
</section>
<table>
<thead>
<tr>
<th colspan="2">PERFORMANCE INDICATORS</th>
<th colspan="4">RATING</th>
</tr>
<tr>
<th colspan="2"></th>
<th>EXCELLENT (4)</th>
<th>GOOD (3)</th>
<th>FAIR (2)</th>
<th>POOR (1)</th>
</tr>
</thead>
<tbody>
<tr>
<td colspan="2">A. QUALITY AND QUANTITY OF WORK</td>
<td></td>
<td></td>
<td></td>
<td></td>
</tr>
<tr>
<td colspan="2">A.1 Accurate at work assigned</td>
<td></td>
<td></td>
<td></td>
<td></td>
</tr>
<tr>
<td colspan="2">A.2 Always completes tasks</td>
<td></td>
<td></td>
<td></td>
<td></td>
</tr>
<tr>
<td colspan="2">A.3 Works in a timely manner</td>
<td></td>
<td></td>
<td></td>
<td></td>
</tr>
<tr>
<td colspan="2">A.4 Adds for more work when assigned tasks are done</td>
<td></td>
<td></td>
<td></td>
<td></td>
</tr>
<tr>
<td colspan="2">A.5 Easily accepts new responsibilities</td>
<td></td>
<td></td>
<td></td>
<td></td>
</tr>
<tr>
<td colspan="2">Total</td>
<td></td>
<td></td>
<td></td>
<td></td>
</tr>
<tr>
<td colspan="2">B. INTERPERSONAL SKILLS</td>
<td></td>
<td></td>
<td></td>
<td></td>
</tr>
<tr>
<td colspan="2">B.1 Answers patron's questions accurately</td>
<td></td>
<td></td>
<td></td>
<td></td>
</tr>
</tbody>
</table>
</div>
```

Figure 66. Code Snippet for Paper Form Evaluation



Figure 66 shows the code snippet used for the student assistants evaluation form in the Institutional Scholarship Management System. It shows the basic details of the student assistant, the directions, and the rating table used in the evaluation form.



Figure 67. Summary Evaluation Report

Figure 67 shows the Institutional Scholarship Management System evaluation summary report page. It presents a summary of the student assistants' evaluation, including their weighted mean, verbal description, strengths or areas for improvement, and evaluator comments or recommendations. It also provides filter options and a print button for easier reporting.

```
$stmt = $conn->prepare($sql);
if ($stmt) {
    if (!empty($params)) {
        $stmt->bind_param($paramTypes, ...$params);
    }

    if ($stmt->execute()) {
        $result = $stmt->get_result();
        $seenApplicationIds = [];

        while ($row = $result->fetch_assoc()) {
            $applicationId = (int)$row["application_id"] ?? 0;
            $applicationKey = (string)$applicationId;
            if (isset($seenApplicationIds[$applicationKey])) {
                continue;
            }
            $seenApplicationIds[$applicationKey] = true;

            $overallTotal = (int)$row["overall_total"] ?? 0;
            $weightedMean = $overallTotal > 0 ? ($overallTotal / 15) : null;

            $summaryRecords[] = [
                "application_id" => $applicationId,
                "name" => trim((string)$row["applicant_name"] ?? ""),
                "weighted_mean" => $weightedMean,
                "verbal_description" => $verbalFromAverage($weightedMean),
                "strengths" => trim((string)$row["strengths"] ?? ""),
                "recommendations" => trim((string)$row["recommendations"] ?? "")
            ];
        }

        if ($result instanceof mysqli_result) {
            $result->free();
        }
        else {
            $summaryLoadError = "Unable to load summary report records.";
        }

        $stmt->close();
    }
    else {
        $summaryLoadError = "Unable to prepare summary report query.";
    }
}
```

Figure 68. Code Snippet for Summary Evaluation Report



Figure 68 shows the code snippet used for the Institutional Scholarship Management System evaluation summary report page. It obtains the analysis records, computes the weighted mean, and prepares the summary details to be displayed in the report.

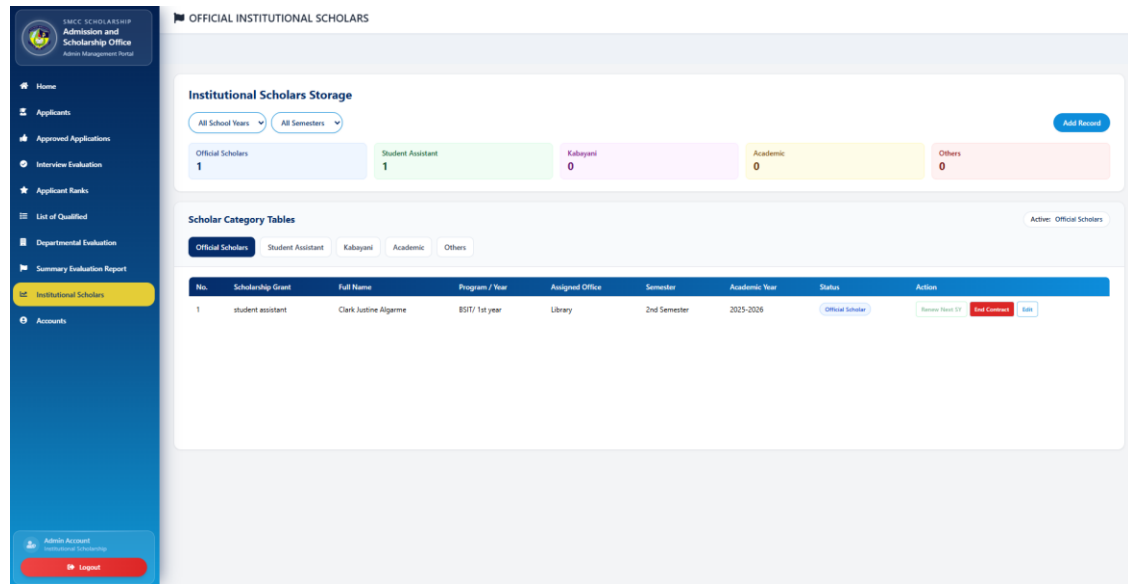


Figure 69. Official ISG Scholars list

Figure 69 shows the institutional scholars page of the Institutional Scholarship Management System. It displays stored records of official institutional scholarships based on scholarship category. It also shows the basic details of the scholar, designated office, semester, academic year, and available actions for record management.



```
function isgLoadScholarRecords(mysqli $conn, array $validCategories): array
{
    $records = array_fill_keys($validCategories, []);
    $dedupedRecords = [];
    $result = $conn->query("
        SELECT
            id,
            category,
            scholar_id,
            grant_applied,
            full_name,
            program_year,
            assigned_office,
            semester,
            academic_year,
            status,
            renewal_status,
            renewal_scope,
            second_semester_renewed,
            contract_ended
        FROM institutional_scholar_records
        ORDER BY id DESC
    ");
    if (!$result instanceof mysqli_result) {
        return $records;
    }

    while ($row = $result->fetch_assoc()) {
        $category = strtolower(trim((string)$row["category"] ?? ""));
        if (in_array($category, $validCategories, true)) {
            continue;
        }
        $record = [
            "id" => (int)$row["id"] ?? 0,
            "scholar_id" => trim((string)$row["scholar_id"] ?? ""),
            "grant_applied" => trim((string)$row["grant_applied"] ?? ""),
            "full_name" => trim((string)$row["full_name"] ?? ""),
            "program_year" => trim((string)$row["program_year"] ?? ""),
            "assigned_office" => trim((string)$row["assigned_office"] ?? ""),
            "semester" => trim((string)$row["semester"] ?? ""),
            "academic_year" => trim((string)$row["academic_year"] ?? ""),
            "status" => trim((string)$row["status"] ?? "official_scholar"),
            "renewal_status" => trim((string)$row["renewal_status"] ?? ""),
            "renewal_scope" => trim((string)$row["renewal_scope"] ?? ""),
            "second_semester_renewed" => (int)$row["second_semester_renewed"] ?? 0 === 1,
            "contract_ended" => (int)$row["contract_ended"] ?? 0 === 1,
            "__category" => $category,
        ];
    }
}
```

Figure 70. Code Snippet for Official ISG Scholars

Figure 70 shows the code snippet used for the institutional scholars page of the Institutional Scholarship Management System. It retrieves scholar records from the system, groups them based on scholarship category, and prepares their details to be displayed on the page.



3.3 ISO 25010 EVALUATION RESULT

TESTING PROCESS

The system's correctness, functionality, and general performance are tested by a variety of testing methodologies, which are described in the testing process. The effectiveness, efficiency, and dependability of the system performance were assessed by the researchers using the system usability, functionality, efficiency scale test as a survey instrument. The ISO 25010 Software Product Quality Standards served as the model for the survey instrument. Respondents were presented with four possibilities on a test scale, with 4 representing the highest system rating and 1 the lowest, based on the question. Data were analyzed and interpreted based on the following parameters:

Table 3

Functional Suitability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Functional
3	2.50 – 3.49	Functional
2	1.50 – 2.49	Moderately Functional
1	1.0 – 1.49	Poor Functional

Table 3 presents the criteria used to evaluate a product or system's functional suitability according to the ISO survey.



Table 4

Performance Efficiency

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Efficient
3	2.50 – 3.49	Efficient
2	1.50 – 2.49	Moderately Efficient
1	1.0 – 1.49	Poor Efficient

Table 4 outlines the scoring parameters used to assess the system's overall performance efficiency in executing its functions and tasks.

Table 5

Usability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Usable
3	2.50 – 3.49	Usable
2	1.50 – 2.49	Moderately Usable
1	1.0 – 1.49	Poor Usable

Table 5 presents the scoring parameters used to the usability of the system.



Table 6

Respondent's Distribution

INSTITUTIONAL SCHOLARSHIP MANAGEMENT SYSTEM FOR SAINT MICHAEL COLLEGE OF CARAGA		
Position	N (Population)	Percentage (%)
Admission Staff	1	12.5%
Panel	1	12.5%
Personnel	2	25%
Student	4	50%
Total	8	100

Table 6 presents the distribution of respondents who participated in the system review within the SMCC Admission and Scholarship Office. Respondents were grouped according to their positions, with 12.5% admissions staff, 12.5% panel members, 25% staff, and 50% students. The results show that most of the feedback comes from students, who serve as the main users of the system. Their insights are particularly important because they interact directly with the platform, providing practical insights into functionality, efficiency, and user-friendliness in real-world scenarios.



DESCRIPTIVE STATISTICS RESULT

Table 7

Functional Suitability

A. Functional Suitability	MEAN	VERBAL INTERPRETATION
1. Functional completeness - Degree to which the set of functions covers all the specified tasks and user objectives	4.0	VF
2. Functional correctness - Degree to which a product or system provides the correct results with the needed degree of precision.	3.75	VF
3. Functional appropriateness - Degree to which the functions facilitate the accomplishment of specified tasks and objectives.	4.0	VF
Weighted Mean	3.91	VF

Legend:

VF – Very Functional (3.50 – 4.0)

MF – Moderately Functional (1.50 – 2.49)

F – Functional (2.50 – 3.49)

PF– Poorly Functional (1.0 – 1.49)

Table 7 shows the results for Functional Suitability, with a weighted mean of 3.91, interpreted as Very Functional (VF). This means that the respondents perceive that the system is working well in meeting its main purpose and required functions.



Among the three indicators, Functional Completeness and Functional Appropriateness both received a mean of 4.00, showing that the system covers the necessary tasks and helps users complete them properly. Functional Correctness obtained a mean of 3.75, which means that the system also provides correct and reliable results.

Overall, the findings show that the system is reliable, useful, and effective for its users. High scores suggest that the system can support scholarship-related tasks well and reliably provide accurate outputs.

Table 8

Performance Efficiency

B. Performance Efficiency	MEAN	VERBAL INTERPRETATION
1. Time behaviour - Degree to which the response and processing times and throughput rates of a system, when performing its functions, meet requirements.	4.0	VE
2. Resource utilization - Degree to which the amounts and types of resources used by a system, when performing its functions, meet requirements.	3.75	VE
3. Capacity - Degree to which the maximum limits of a product or system parameter meet requirements.	3.87	VE
Weighted Mean	3.87	VE
Legend:		
VE – Very Efficient (3.50 – 4.0)	ME – Moderately Efficient (1.50 – 2.49)	
E – Efficient (2.50 – 3.49)	PE – Poorly Efficient (1.0 – 1.49)	



Table 8 shows the results for Performance Efficiency, with a weighted mean of 3.87, interpreted as Very Efficient (VE). This shows that respondents found the system efficient in terms of speed, resource utilization, and overall performance while performing its functions.

Among the three indicators, Time Behavior received the highest mean of 4.00, which means the system responds quickly and processes tasks. The capacity obtained a mean of 3.87, which shows that the system can handle the required workload well. Resource Utilization received a mean of 3.75, which is also under Very Efficient, meaning that the system uses its resources well while performing its tasks.

Overall, the findings show that the system performs well and is capable of meeting user needs efficiently. High ratings suggest that the system is fast, robust, and reliable in handling scholarship-related processes.

Table 9

Usability

C. Usability	MEAN	VERBAL INTERPRETATION
A. Appropriateness recognizability - Degree to which users can recognize whether a system is appropriate for their needs.	3.75	VU
B. Learnability - Degree to which a system can be used by specified users to achieve specified goals of learning to use the product or system with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.	3.87	VU



C. Operability - Degree to which a system has attributes that make it easy to operate and control.	3.87	VU
D. User error protection - Degree to which a system protects users against making errors.	3.62	VU
E. User interface aesthetics - Degree to which a user interface enables pleasing and satisfying interaction for the user.	3.75	VU
F. Accessibility - Degree to which a system can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.	3.87	VU
Weighted Mean	3.78	VU

Legend:

VU – Very Usable (3.50 – 4.0)

MU – Moderately Usable (1.50 – 2.49)

U – Usable (2.50 – 3.49)

PU – Poorly Usable (1.0 – 1.49)

Table 9 shows the results for Usability, with a weighted mean of 3.78, interpreted as Very Usable (VU). This means that the respondents found the system easy to use, easy to understand, and suitable for their needs.

Among the indicators, Learnability, Operability, and Accessibility, each received a mean of 3.87, showing that users can easily learn the system, use its functions, and access it without much difficulty. Appropriateness Recognizability and User Interface Aesthetics both obtained a mean of 3.75, which means that users found the system appropriate for their needs and liked its interface.



User Error Protection received the lowest mean of 3.62, but it is still below Very Usable, showing that the system can still help users avoid mistakes while using it.

Overall, the results show that the system provides a smooth and satisfactory user experience. A high rating means that the system is user-friendly, easy to learn, and useful for its intended users.

Table 10

Summary Table of the Overall Mean and Grand Distribution of the Acceptability Level

Acceptability Level of the System in terms of:	Over-all Mean	Rating
Functional Suitability	3.91	SA
Performance Efficiency	3.87	SA
Usability	3.78	SA
Grand Mean	3.85	SA
Legend:		
SA – Strongly Acceptable (3.50 – 4.0)	U – Unacceptable (1.50 – 2.49)	
A – Acceptable (2.50 – 3.49)	SU – Strongly Unacceptable (1.0 – 1.49)	

Table 10 presents the summary of the overall mean and grand distribution of the system's acceptability level. The table shows that Functional Suitability scored an overall mean of 3.91, Performance Efficiency scored 3.87, and Usability scored 3.78. All three areas were rated Strongly Acceptable (SA). This means that the respondents were highly satisfied with the system in terms of its functions, performance, and ease of use.

The Grand Mean of 3.85 also falls under Strongly Acceptable, which shows that the system was well accepted by the users overall. This means that the system is effective, efficient, and easy to



use based on the evaluation results. Overall, the findings show that the system meets the required standards and can be considered reliable and acceptable for its intended use.



CHAPTER IV

SUMMARY, CONCLUSION, AND RECOMMENDATION

This chapter presents a summary of the study's findings, conclusions, and recommendations. It highlights the main results of the analysis carried out for the system and provides practical suggestions for its further improvement. The discussion is based on data gathered from respondents using the adapted software quality standard ISO/IEC 25010.

4.1 Summary of Findings

Based on the evaluation process, the Institutional Scholarship Management System for Saint Michael College of Caraga was assessed using a survey instrument adapted from ISO/IEC 25010 Software Product Quality Standards. The analysis focuses on three main quality attributes, namely Functional Suitability, Performance Efficiency, and Usability.

The results showed that the system is Very Functional, with a weighted mean of 3.91. This means that the system is able to perform its intended functions properly. It is found to be complete in providing required features, correct in providing results, and suitable in helping users complete their tasks. This shows that the system can effectively support the scholarship process.

In terms of Performance Efficiency, the system received a weighted mean of 3.87, which is interpreted as Very Efficient. This means that the system responds quickly, performs tasks smoothly, and can handle the required workload well. The result also shows that the system uses its resources properly while executing its functions.

For Usability, the system scored a weighted mean of 3.78, which is interpreted as Very Usable. This means that the respondents found the system easy to learn, easy to use, and suitable for their needs. The interface was also seen as clear and useful, which gave users a better experience while using the system.



Overall, the system scored a Grand Mean of 3.85, with a rating of Very Acceptable. This shows that the respondents highly accepted the system in terms of its functions, performance, and usability. The findings suggest that the system is reliable and useful for managing scholarship-related processes in the institution.

4.2 Conclusion

Based on the findings of the study, it can be concluded that the Institutional Scholarship Management System for Saint Michael College of Caraga is an effective and acceptable system for its intended users. The test results showed that the system performed well in three areas measured: functional suitability, performance efficiency, and usability.

The system met the required functions, provided accurate results, and helped users complete scholarship-related tasks more easily. It also showed good performance in terms of response time, workload handling, and proper utilization of resources. In addition, respondents found the system easy to learn and easy to use, which helped make their overall experience better.

Therefore, it can be concluded that the system meets the objectives of the study and is suitable for use in the management of institutional scholarship applications and related processes. The highly acceptable results also show that the system can serve as a useful tool in making scholarship management more organized, efficient, and user-friendly.

4.3 Recommendation/s

Although the system received a Very Acceptable rating, there are still improvements that could help make it even better in the future. It is recommended that the system include an enhanced notification and announcement feature so that applicants, scholars, and administrators are notified immediately of deadlines, updates, interview schedules, and important reminders.

For future development, the researchers recommend adding AI-based features such as scholarly review and an AI chatbot for questions. The scholar review feature can help the institution



review scholar records, academic performance, renewal status, and other relevant data to support better decision making. At the same time, the AI chatbot can help answer common questions about application requirements, schedules, qualifications, and other scholarship concerns, making the system more responsive and helpful to users. Through these features, the system can improve tracking, communication, and overall support for both applicants and scholars.



REFERENCES

- [1] Commission on Higher Education (CHED) / Development Partners Evaluation Unit. (2020). *Thematic Evaluation of the Scholarship Program – Fiscal Year 2020*.
- [2] Dian Prijayanti, Emilya Ullly Artha, & Endah Ratna Arumi. (2020). *Web-Based Monitoring Information System for Scholarship Holder*.
- [3] Kübra Akıllı & Emre İpekçi Çetin. (2020). *Selection of Scholarship Students in Higher Education with VIKOR Method*.
- [4] Umut Tekgüç, Ahmet Adalier, & Kamil Yurtkan. (2020). *ScholarChain: The Scholarship Management Platform with Blockchain and Smart Contracts Technology*.
- [5] Ibrahim. (2021). *Scholarship Monitoring Information System at Ar-Raniry Islamic University*.
- [6] Jacquelyn S. Moon. (2021). *“Overwhelming”: A Case Study Analyzing Scholarship Application Processes*.
- [7] Muhammad Rizki Hidayat, Irsan Jaelani, & Mutiara Andayani Komara. (2021). *Web-Based Scholarship Management Information System Using Rapid Application Development (RAD) Model*.
- [8] T. Sowndhariyaa & T. M. Nithya. (2021). *Advanced Application System for Student Scholarship using Content Based Filtering Technique*.
- [9] Benjamin Wiggins, Cody Hennesy, Brian Vetruba, Alexis Logsdon, & Emily Janisch. (2022). *Digital Scholarship Programs in Practice*.
- [10] Gourab K. Patro, Lorenzo Porcaro, Laura Mitchell, Qiuyue Zhang, Meike Zehlike, & Nikhil Garg. (2022). *Fair Ranking: A Critical Review, Challenges, and Future Directions*.
- [11] Hao Li & Yan Yang. (2022). *Research on the Scholarship Management for International Students in China in the Post-Pandemic Era*.
- [12] Rob Willey, Melanie Knapp, & Ashley Matthews. (2022). *Constructing Inclusive & Current Scholarship Impact Rankings*.



- [13] Sharra Mae B. Fernandez, Sheila Mae S. Pagayonan, & April Rose A. Zaragosa. (2022). *Scholarship Grant Management Information System with SMS Notification*.
- [14] Thi Huyen Ngo, Hong Sinh Nguyen, & Thi Kim Thoa Ninh. (2022). *Examining the Practice of Digital Scholarship Services at Vietnam National University Ho Chi Minh City*.
- [15] Wisdom Tochi Kalu. (2022). *My Scholarship Info Consultancy Management System: Technology Meets the Challenge*.
- [16] Daren Fajutrao Berido, Shem Anthony Agcamaran Pingol, Pauline Duyao Ramil, Kennery Duardo Villacaol, Criselle Jose Centeno, & Diony Soronio Abando. (2023). *Development of Scholarship Automation System for Student Qualification Program Applied to College Universities Using Regression Analysis*.
- [17] Muhammad Akbar, Anita, & Hanifah. (2023). *Scholarship Information Service on UIN Ar-Raniry Banda Aceh Website*.
- [18] Muhammad Salahuddin Al-Ayyubi & Bachtiar Maulana. (2023). *Designing a Web-Based Information System for Scholarship Management: Supporting Access and Rapid Dissemination of Information*.
- [19] Novia Hasdyna, Rozzi Kesuma Dinata, & Sujacka Retno. (2023). *Analysis of the TOPSIS in the Recommendation System of PPA Scholarship Recipients at Universitas Islam Kebangsaan Indonesia*.
- [20] Thomais Gkrimpizi, Vassilios Peristeras, & Ioannis Magnisalis. (2023). *Classification of Barriers to Digital Transformation in Higher Education Institutions: Systematic Literature Review*.
- [21] A. Karunamurthy, C. Lakshmipriya, & M. Priyadarshini. (2024). *Transforming Student Scholarship Application Systems: Integrating NLP with E-Scholarship Platforms*.
- [22] Banaybanay Rose, Bulalaque Claire, Dela Cruz Krishan Aea C., Garzon John Abner, Laraño Ritz, Lloren Kevin Carl T., Patricio Brendyll John J., Cedie E. Gabriel, & Reginald S. Prudente. (2024).



Understanding the Role of OLAP and OLTP in Managing and Interpreting Student Data for SEAIT Scholarship System.

[23] Jannathul Firthous I. A., Soundarya M., Yasini V., Sneka A., & Sreeja G. (2024). *Scholarship Management System(2).*

[24] John A. Balansag, Cherry Mae L. Damasco, & Reynaldo M. Briones. (2024). *A Review of Technical Education Scholarship Program Management System Using Cloud-Based Technology.*

[25] Sahil M. Vaidya, Megha K. Madavi, Khetambari P. Gaidhani, Mohini S. Tambulkar, Bharti Purusharathi, & Deepak S. Bhiogade. (2024). *Scholarship Management System.*

[26] Sri Andriati Asri, I Gusti Ngurah Bagus Caturbawa, Putu Manik Prihatini, Ni Wayan Rasmini, I Made Riyan Adi Nugroho, & Elina Rudiastari. (2024). *Scholarship Application with Decision Support System Feature using Progressive Web App.*

[27] Yan Orgianus, Fijrina R. I. Lapalanti, Halbana Tarmizi, & Hirawati Oemar. (2024). *Optimizing scholarship distribution: a management information system approach.*

[28] Ambika L. G., Nischal Gowda M. N., Nitish T., Sandesh B. K., & Subhash S. (2025). *Paperless Scholarship Disbursement System.*

[29] Ariyani Widyastuti & Diyamon Prasandha. (2025). *Strengthening the Administration System of Scholarship Data Collection Based on Digital Services to Encourage Good Accountability and Program Benefits.*

[30] Florlyn Mae C. Remegio & Ma. Nanette S. Casquejo. (2025). *Development of a Comprehensive Scholarship and Grants Management System (SchoGMS) for State Universities and Colleges.*

[31] Maica D. L. Bagaporo & Marco L. Espinosa. (2025). *Online Scholarship Management System for Camarines Sur Polytechnic Colleges.*

[32] Rachael Hains-Wesson & Fiona Scotney. (2025). *Tracking Scholarship Outcomes to Measure Impact: A White Paper.*



APPENDICES

A. System Source Code

db.php

```
<?php
$host = "localhost";
$user = "root";
$pass = "";
$dbname = "isg_system";

$conn = new mysqli($host, $user, $pass, $dbname);

if ($conn->connect_error) {
    die("Database connection failed: " . $conn->connect_error);
}
?>
```

application-upload.php

```
<?php
require_once __DIR__ . "/includes/admin-auth.php";
adminRequireLogin();
require_once "../db.php";
require_once "../upload-storage.php";

$applicationId = (int)($_GET["application_id"] ?? 0);
$storedPath = trim((string)($_GET["stored_path"] ?? ""));
$mode = trim((string)($_GET["mode"] ?? "view"));

if ($applicationId <= 0 || $storedPath === "") {
    http_response_code(400);
    exit("Invalid upload request.");
}

$stmt = $conn->prepare(
    "SELECT original_file_name, stored_path, mime_type
    FROM application_uploads
    WHERE application_id = ? AND stored_path = ?
    LIMIT 1"
);
```



upload-storage.php

```

if (!function_exists("projectRootPath")) {
    function projectRootPath(): string
    {
        $rootPath = realpath(__DIR__);
        return $rootPath !== false ? $rootPath : __DIR__;
    }
}

if (!function_exists("uploadStorageBasePath")) {
    function uploadStorageBasePath(): string
    {
        $configuredBasePath = trim((string)getenv("ISG_UPLOAD_BASE"));
        if ($configuredBasePath !== "") {
            return rtrim($configuredBasePath, "\\\/");
        }

        $projectRoot = projectRootPath();
        $xamppRoot = dirname(dirname($projectRoot));

        return $xamppRoot . DIRECTORY_SEPARATOR . "isg-system-storage";
    }
}

if (!function_exists("applicationUploadDirectory")) {
    function applicationUploadDirectory(int $applicationId): string
    {
        return uploadStorageBasePath() . DIRECTORY_SEPARATOR . "applications" . DIRECTORY_SEPARATOR . $applicationId;
    }
}

if (!function_exists("storedUploadDbPath")) {
    function storedUploadDbPath(int $applicationId, string $storedFileName): string
    {
        $normalizedFileName = ltrim(str_replace("\\", "/", $storedFileName), "/");
        return "applications/" . $applicationId . "/" . $normalizedFileName;
    }
}

```

add-panelist.php

```

if ($errorMessage === "") {
    $passwordHash = password_hash($password, PASSWORD_DEFAULT);
    if ($passwordHash === false) {
        $errorMessage = "Unable to save the account.";
    } else {
        $insertStmt = $conn->prepare("INSERT INTO panelists (username, full_name, password_hash, status) VALUES (?, ?, ?, ?)");
        if ($insertStmt) {
            $insertStmt->bind_param("ssss", $username, $fullName, $passwordHash, $status);
            if ($insertStmt->execute()) {
                $successMessage = "Panelist account created.";
            } else {
                $errorMessage = "Unable to save the account.";
            }
            $insertStmt->close();
        } else {
            $errorMessage = "Unable to save the account.";
        }
    }
}
}

```



submit_upload.php

```
<?php
session_start();
require_once "../db.php";
require_once "../upload-storage.php";

$errors = [];
$success = false;
$applicationId = null;

function draft_value($draft, $key) {
    return isset($draft[$key]) ? $draft[$key] : "";
}

function draft_int($draft, $key) {
    return isset($draft[$key]) ? (int)$draft[$key] : 0;
}

$draft = $_SESSION["application_draft"] ?? null;

if ($_SERVER["REQUEST_METHOD"] !== "POST") {
    $errors[] = "Invalid request.";
} elseif (empty($draft)) {
    $errors[] = "Missing application draft. Please complete Step 2 first.";
}
```

add-head-office.php

```
if ($errorMessage === "") {
    $passwordHash = password_hash($password, PASSWORD_DEFAULT);
    if ($passwordHash === false) {
        $errorMessage = "Unable to save the account.";
    } else {
        $insertStmt = $conn->prepare("INSERT INTO head_offices (username, full_name, office, password_hash, status) VALUES (?, ?, ?, ?, ?)");
        if ($insertStmt) {
            $insertStmt->bind_param("sssss", $username, $fullName, $office, $passwordHash, $status);
            if ($insertStmt->execute()) {
                $successMessage = "Head of office account created.";
            } else {
                $errorMessage = "Unable to save the account.";
            }
            $insertStmt->close();
        } else {
            $errorMessage = "Unable to save the account.";
        }
    }
}
```



send_message.php

```

$applicantId = (int)($_POST["applicant_id"] ?? 0);
$messageBody = isset($_POST["message_body"]) ? trim((string)$_POST["message_body"]) : "";
$postedRecipientEmail = isset($_POST["recipient_email"]) ? trim((string)$_POST["recipient_email"]) : "";

if ($applicantId <= 0 || $messageBody === "") {
    $_SESSION["message_error"] = "Invalid applicant or empty message.";
    header("Location: approved.php?message_status=error");
    exit;
}

try {
    $recipientEmail = "";
    $lookupStmt = $conn->prepare("SELECT email_address FROM applications WHERE id = ? LIMIT 1");
    if ($lookupStmt) {
        $lookupStmt->bind_param("i", $applicantId);
        $lookupStmt->execute();
        $lookupStmt->bind_result($recipientEmail);
        $lookupStmt->fetch();
        $lookupStmt->close();
    }
}

```

send_to_panelist.php

```

if ((int)$grantId !== 1) {
    $_SESSION["panelist_error"] = "Only Student Assistant applicants can be sent to panelists.";
    header("Location: approved.php?panelist_status=error");
    exit;
}

$sentBy = isset($_SESSION["admin_name"]) && trim((string)$_SESSION["admin_name"]) !== ""
    ? trim((string)$_SESSION["admin_name"])
    : "Admin";

$insertSql = "INSERT INTO panelist_queue (application_id, panelist_username, sent_by, sent_at)
VALUES (?, ?, ?, NOW())
ON DUPLICATE KEY UPDATE sent_by = VALUES(sent_by), sent_at = VALUES(sent_at)";
$insertStmt = $conn->prepare($insertSql);
if (!$insertStmt) {
    $_SESSION["panelist_error"] = "Failed to queue applicant for panelist review.";
    header("Location: approved.php?panelist_status=error");
    exit;
}

```



logout.php

```
$requestedRole = strtolower(trim((string)$_GET["role"] ?? ""));
if ($requestedRole === "panel" || $requestedRole === "panelist") {
    $redirect = "Panelist/panelLogin.php";
} elseif ($requestedRole === "head" || $requestedRole === "departmenthead") {
    $redirect = "DepartmentHead/headLogin.php";
} elseif ($requestedRole === "admin") {
    $redirect = "Admin/adminLogin.php";
} elseif (!empty($_SESSION["panelist_username"])) {
    $redirect = "Panelist/panelLogin.php";
} elseif (!empty($_SESSION["head_username"])) {
    $redirect = "DepartmentHead/headLogin.php";
} elseif (!empty($_SESSION["admin_username"]) || !empty($_SESSION["admin_name"])) {
    $redirect = "Admin/adminLogin.php";
} else {
    // Fallback when no role session exists (e.g., admin pages without role session key).
    $refPath = (string)parse_url((string)$_SERVER["HTTP_REFERER"] ?? ""), PHP_URL_PATH;
    if (stripos($refPath, "/Panelist/") !== false) {
        $redirect = "Panelist/panelLogin.php";
    } elseif (stripos($refPath, "/DepartmentHead/") !== false) {
        $redirect = "DepartmentHead/headLogin.php";
    } elseif (stripos($refPath, "/Admin/") !== false) {
        $redirect = "Admin/adminLogin.php";
    }
}
```

Admin-auth.php

```
if (session_status() !== PHP_SESSION_ACTIVE) {
    session_start();
}

// These helpers are wrapped in function_exists guards so multiple admin includes stay safe.

if (!function_exists("adminBasePath")) {
    function adminBasePath(): string
    {
        $scriptName = str_replace("\\", "/", (string)$_SERVER["SCRIPT_NAME"] ?? "");
        $scriptDirectory = rtrim(dirname($scriptName), "/");

        if ($scriptDirectory === "." || $scriptDirectory === "\\") {
            $scriptDirectory = "";
        }

        if (substr($scriptDirectory, -9) === "/includes") {
            $scriptDirectory = substr($scriptDirectory, 0, -9);
        }

        return $scriptDirectory;
    }
}

if (!function_exists("adminPath")) {
    function adminPath(string $fileName): string
    {
        $basePath = adminBasePath();
        return ($basePath !== "" ? $basePath : "") . "/" . ltrim($fileName, "/");
    }
}

if (!function_exists("adminIsAuthenticated")) {
    function adminIsAuthenticated(): bool
    {
        return isset($_SESSION["admin_username"]) && trim((string)$_SESSION["admin_username"]) !== "";
    }
}
```



mailer.php

```
function isgSendPlainTextMail(string $recipientEmail, string $subject, string $body): void
{
    $recipientEmail = trim($recipientEmail);
    if ($recipientEmail === "" || !filter_var($recipientEmail, FILTER_VALIDATE_EMAIL)) {
        throw new RuntimeException("Recipient email is invalid.");
    }

    $smtpConfigPath = dirname(__DIR__, 2) . "/smtp_config.php";
    if (!file_exists($smtpConfigPath)) {
        throw new RuntimeException("SMTP config missing. Create smtp_config.php in project root.");
    }

    $smtp = require $smtpConfigPath;
    if (!is_array($smtp)) {
        throw new RuntimeException("SMTP config is invalid.");
    }

    $mail = new PHPMailer(true);
    $mail->isSMTP();
    $mail->Host = $smtp["host"] ?? "smtp.gmail.com";
    $mail->SMTPAuth = true;
    $mail->Username = $smtp["username"] ?? "";
    $mail->Password = $smtp["password"] ?? "";
    $mail->Port = (int)($smtp["port"] ?? 587);

    $secure = $smtp["secure"] ?? "tls";
    $mail->SMTPSecure = ($secure === "ssl")
        ? PHPMailer::ENCRYPTION_SMTPS
        : PHPMailer::ENCRYPTION_STARTTLS;

    if (!filter_var($mail->Username, FILTER_VALIDATE_EMAIL)) {
        $maskedUser = isgMaskEmail($mail->Username);
        $hint = $maskedUser !== "" ? " Current: " . $maskedUser : " Current: [empty]";
        if (stripos($mail->Username, "yourgmail@gmail.com") !== false) {
            $hint .= " (still placeholder)";
        }
        throw new RuntimeException("Invalid SMTP username email. Update smtp_config.php (username)." . $hint);
    }
}
```

Panelist-sent-applicants.php

```
$selectedSchoolYear = isset($selectedSchoolYear) ? (string)$selectedSchoolYear : "";
$selectedSemester = isset($selectedSemester) ? (string)$selectedSemester : "";
$selectedBatch = isset($selectedBatch) ? (string)$selectedBatch : "";

$hasBatchColumn = false;
$batchColumnResult = $conn->query("SHOW COLUMNS FROM applications LIKE 'batch'");
if ($batchColumnResult instanceof mysqli_result) {
    $hasBatchColumn = $batchColumnResult->num_rows > 0;
    $batchColumnResult->free();
}

$tableResult = $conn->query("SHOW TABLES LIKE 'panelist_queue'");
if ($tableResult instanceof mysqli_result && $tableResult->num_rows > 0) {
    $tableResult->free();

    $whereClauses = [
        "a.grant_id = 1",
        "LOWER(TRIM(a.status)) = 'approved'",
    ];
    $params = [];
    $types = "";

    if ($selectedSchoolYear !== "") {
        $whereClauses[] = "a.school_year = ?";
        $params[] = $selectedSchoolYear;
        $types .= "s";
    }

    if ($selectedSemester !== "") {
```



School-term-filter.php

```

$currentYear = (int)date("Y");
$currentMonth = (int)date("n");
$currentSchoolYearStart = $currentMonth < 6 ? $currentYear - 1 : $currentYear;
$currentSchoolYear = $currentSchoolYearStart . "-" . ($currentSchoolYearStart + 1);
$currentSemester = $currentMonth < 6 ? "2nd Semester" : "1st Semester";

$schoolYearOptions = [];
$schoolYearResult = $conn->query("SELECT DISTINCT school_year FROM applications WHERE school_year IS NOT NULL AND TRIM(school_year) <> ''");
if ($schoolYearResult instanceof mysqli_result) {
    while ($row = $schoolYearResult->fetch_assoc()) {
        $value = trim((string)($row["school_year"] ?? ""));
        if ($value !== "") {
            $schoolYearOptions[] = $value;
        }
    }
    $schoolYearResult->free();
}

if (!in_array($currentSchoolYear, $schoolYearOptions, true)) {
    $schoolYearOptions[] = $currentSchoolYear;
}

$schoolYearOptions = array_values(array_unique($schoolYearOptions));
usort($schoolYearOptions, function ($a, $b) {
    $aYear = (int)substr((string)$a, 0, 4);
    $bYear = (int)substr((string)$b, 0, 4);
    if ($aYear === $bYear) {
        return strcmp((string)$a, (string)$b);
    }
    return $aYear <=> $bYear;
});

$semesterOptions = ["1st Semester", "2nd Semester"];
$batchOptions = ["Batch 1", "Batch 2", "Batch 3", "Batch 4", "Batch 5"];

$selectedSchoolYear = isset($_GET["school_year"]) ? trim((string)$_GET["school_year"]) : "";
$selectedSemester = isset($_GET["semester"]) ? trim((string)$_GET["semester"]) : "";
$selectedBatch = isset($_GET["batch"]) ? trim((string)$_GET["batch"]) : "";

```

panelist-auth.php

```

if (!function_exists("panelistBasePath")) {
    function panelistBasePath(): string
    {
        $scriptName = str_replace("\\", "/", (string)($_SERVER["SCRIPT_NAME"] ?? ""));
        $needle = "/Panelist";
        $position = strpos($scriptName, $needle);

        if ($position !== false) {
            return substr($scriptName, 0, $position + strlen($needle));
        }

        $scriptDirectory = rtrim(dirname($scriptName), "/");
        return $scriptDirectory === "." ? "" : $scriptDirectory;
    }
}

if (!function_exists("panelistPath")) {
    function panelistPath(string $fileName): string
    {
        $basePath = panelistBasePath();
        return ($basePath !== "" ? $basePath : "") . "/" . ltrim($fileName, "/");
    }
}

if (!function_exists("panelistIsAuthenticated")) {
    function panelistIsAuthenticated(): bool
    {
        return isset($_SESSION["panelist_username"]) && trim((string)$_SESSION["panelist_username"]) !== "";
    }
}

if (!function_exists("panelistRequireLogin")) {
    function panelistRequireLogin(): void

```



Head-auth.php

```
if (!function_exists("headBasePath")) {
    function headBasePath(): string
    {
        $scriptName = str_replace("\\", "/", (string)($_SERVER["SCRIPT_NAME"] ?? ""));
        $needle = "/DepartmentHead";
        $position = strpos($scriptName, $needle);

        if ($position !== false) {
            return substr($scriptName, 0, $position + strlen($needle));
        }

        $scriptDirectory = rtrim(dirname($scriptName), "/");
        return $scriptDirectory === "." ? "" : $scriptDirectory;
    }
}

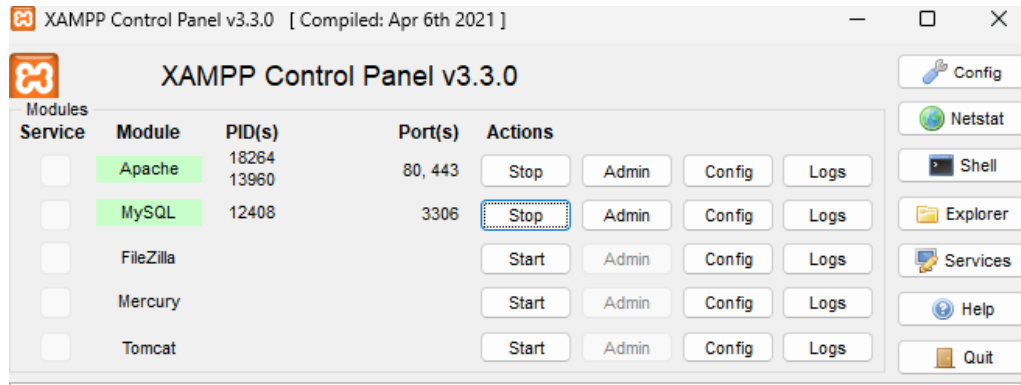
if (!function_exists("headPath")) {
    function headPath(string $fileName): string
    {
        $basePath = headBasePath();
        return ($basePath !== "" ? $basePath : "") . "/" . ltrim($fileName, "/");
    }
}

if (!function_exists("headIsAuthenticated")) {
    function headIsAuthenticated(): bool
    {
        return isset($_SESSION["head_username"]) && trim((string)$_SESSION["head_username"]) !== "";
    }
}

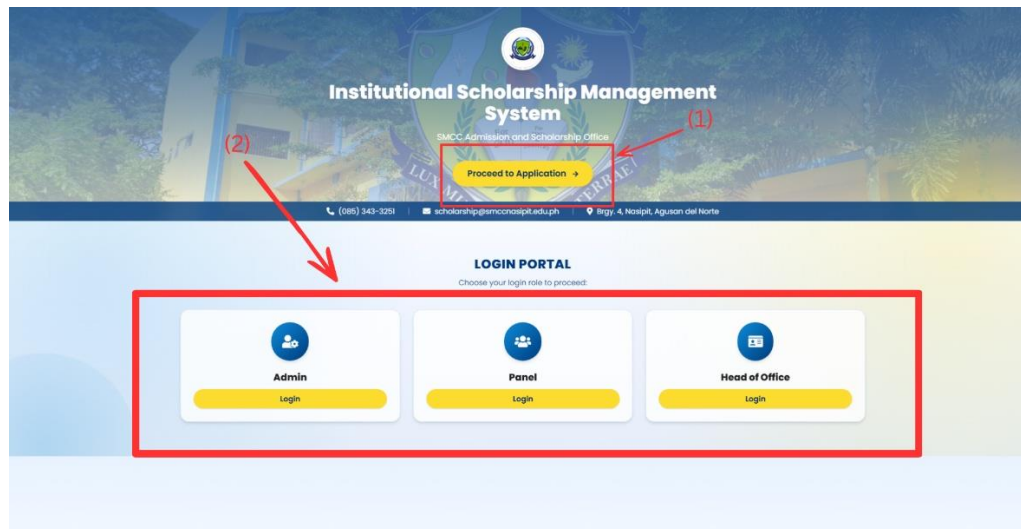
if (!function_exists("headRequireLogin")) {
    function headRequireLogin(): void
    {
        if (headIsAuthenticated()) {
            return;
        }
    }
}
```



B. User's Manual



Ensure that XAMPP's Apache and MySQL services are running (1) before launching the ISMS via the desktop shortcut or by entering <http://localhost/> in your default browser. When the stop button is visible, it indicates that both Apache and MySQL are active. (2) Once these services are running, the system and its database are ready for use.



The Landing Page is the main starting page of the Institutional Scholarship Management System. It is designed to be simple, clear, and easy to use so users can quickly choose the right action. The Proceed to Application button is for applicants who want to start their scholarship application and continue with the application process (1). The Login Portal is for authorized users such as Admin,



Panel, and Head of Office, where each user selects the correct role and clicks the login button to access their assigned part of the system (2).

SMCC ADMISSION AND SCHOLARSHIP OFFICE

Institutional Scholarship Grants Applicant Portal

Track an existing application or start a new scholarship application.

APPLICANT ACCESS

Choose what you need to do today.

Use your application reference number to check the latest status, or continue to the scholarship list and requirements page if you are applying for the first time.

Track Status
Reference-based lookup for submitted applications.

Apply
Review grant requirements before completing the form.

Reference Format
Example: ISG-20260413-7KQ29X

TRACK EXISTING APPLICATION

Open tracking dashboard

Paste the application reference number that was shown after submission.

Application Reference Number

ISG-20260413-7KQ29X

Track Application

NEW APPLICANT

Proceed to ISG list and requirements

Review the available scholarship grants and discounts, then continue to the application form.

Proceed to Application

← Back to homepage

Keep your reference number in a safe place after submitting.

This applicant interface allows the user to track an existing application or start a new scholarship application. The Application Reference Number field is used to enter the reference number of a previously submitted application (1). The Track Application button is used to open and check the current status of the submitted application (2). The Proceed to Application button is used by new applicants to proceed to the scholarship list and requirements page before starting the application process (3).



This applicant interface displays the scholarship category, the list of requirements, and the programs covered by the selected grant. It helps the applicant to check the required information first before moving to the next step. The applicant can click the Apply for this Grant button to proceed to the application form (1).

This applicant interface allows the applicant to enter the required information for the scholarship application. The Mother's Information section is used to provide the necessary details about the applicant's mother (1). The Father's Information section is used to provide the necessary details about the applicant's father (2). After completing the form, the applicant can click Submit



This applicant interface allows the applicant to upload the required documents for the selected scholarship grant. The Select File button is used to select and attach the required document from the device (1). The Clear File button is used to remove the selected file in case the applicant wants to change it (2). Once all required files have been attached, the applicant can click Submit Requirements to complete the submission process (3).

This applicant interface appears after the applicant has successfully submitted the application form and requirements. The Application Reference Number is displayed so that the applicant can save



it and use it later to check the status of the application (1). The Track Application and Back to homepage buttons are used to view the submitted application status or return to the main page (2).

SMCC ADMISSION AND SCHOLARSHIP OFFICE
Institutional Scholarship Grants Track Application
Check the latest application status using your reference number

Application reference number
ISG-20260413-KZ73XT

(1) Search Status

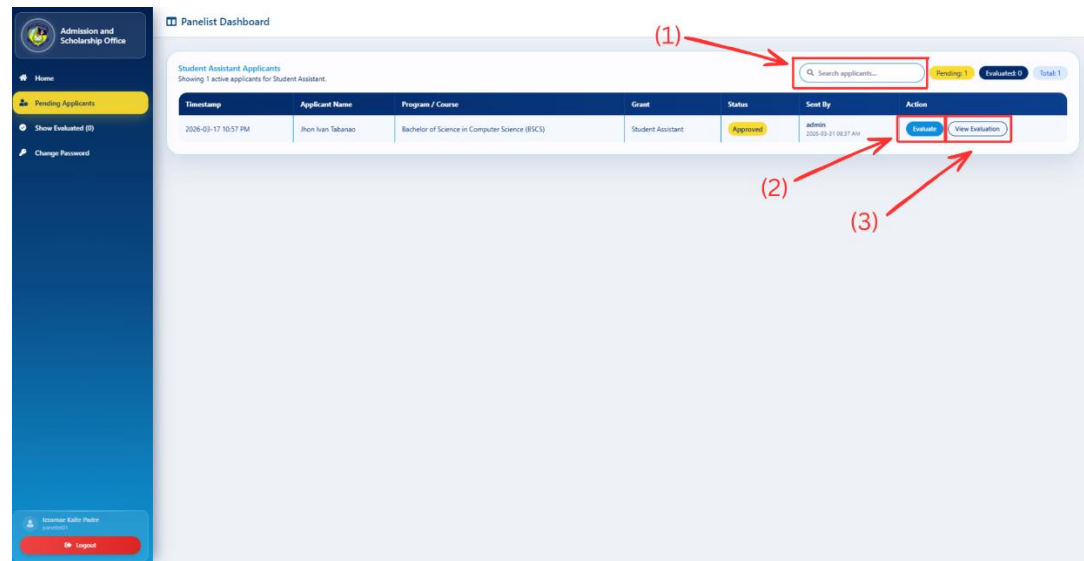
(2) Need to apply instead?
Open the applicant portal to start a new scholarship application or return to the tracking options page.
Applicant Portal Back to Homepage

(3) STUDENT ASSISTANT PROCESS
Student Assistant applicants move through four stages: application, examination, interview, and qualification.

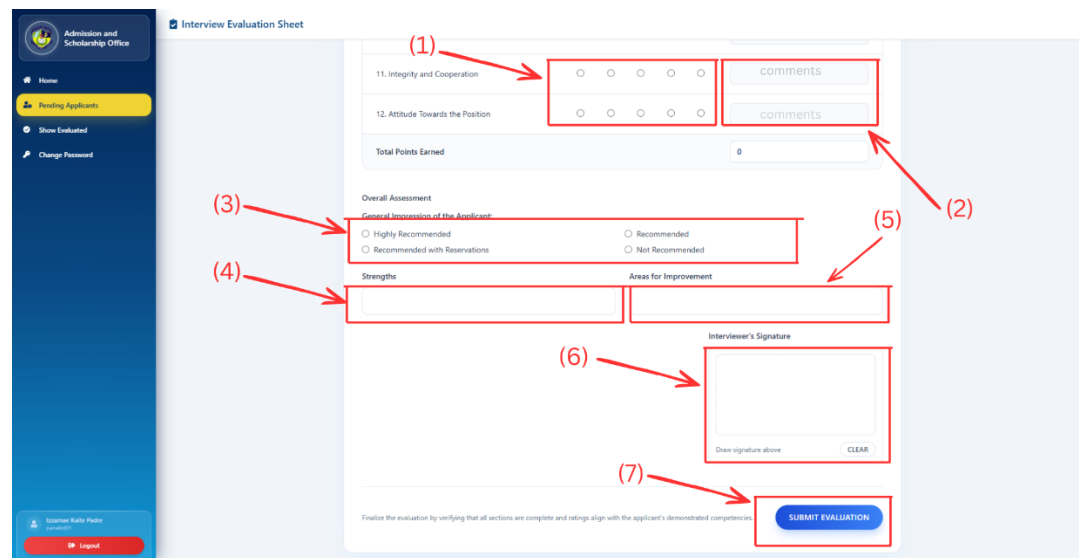
STEP 1	STEP 2	STEP 3	STEP 4
Application Application review has been completed and your submission moved to the next screening stage.	Examination Your application passed the initial review and is now waiting to be forwarded for interview.	Interview Waiting for the examination stage to be completed.	Qualification Waiting for qualification result.

GRANT / DISCOUNT PROGRAM / COURSE SCHOOL TERM
2nd Semester.

This applicant interface is used to check the progress of a submitted scholarship application. The Search Status button is used to display the latest application status based on the entered reference number (1). The Applicant Portal and Back to Homepage buttons are used to start a new application or return to the main page (2). The process tracker shows the current stage of the application, such as Application, Examination, Interview, and Qualification (3).



This panelist interface displays the list of applicants assigned to the panelist for review. The Applicants Search box is used to quickly find a specific applicant record (1). The Evaluate button is used to open the interview evaluation sheet for the selected applicant (2). The View Evaluation button is used to review the applicant's submitted evaluation details (3).



This panelist interface is used by the panelist to rate and evaluate the applicant during the interview. Rating options are used to select the score for each criterion (1). Comment boxes are used to enter comments for rated items (2). The general assessment options are used to select the final



recommendation for the applicant (3). The Strengths box is used to write down the applicant's good points (4). The Areas for Improvement box is used to write down areas that need improvement (5). The signature box is used for the panelist's signature before final submission (6). The Submit Review button is used to save and submit the completed interview review (7).

The screenshot shows the 'My SA's' interface. On the left is a sidebar with 'Home', 'My SA's', 'Show Evaluation', and 'Change Password'. The main content area is titled 'My SA's' and includes a summary of 'TOTAL MY SA'S' (2) and 'READY TO EVALUATE' (2). Below this is a table of student assistants:

Timestamp	Applicant Name	Program / Course	Grant	Status	Action
2026-03-30 12:57 AM	Zharlinda Codina	BSIT / 1st year	Student Assistant	Official Score	Open Evaluation Form
2026-03-30 12:08 AM	Michelle Isong	Bscom/1st Year	Student Assistant	Assessment	Open Evaluation Form

Red arrows indicate: (1) pointing to the 'Search My SA's...' box, and (2) pointing to the 'Open Evaluation Form' button for Michelle Isong.

This department head interface displays the list of student assistants assigned to the office.

The Search My SA box is used to quickly find a specific student assistant record (1). The Open Evaluation Form button is used to open the evaluation form for the selected student assistant (2).

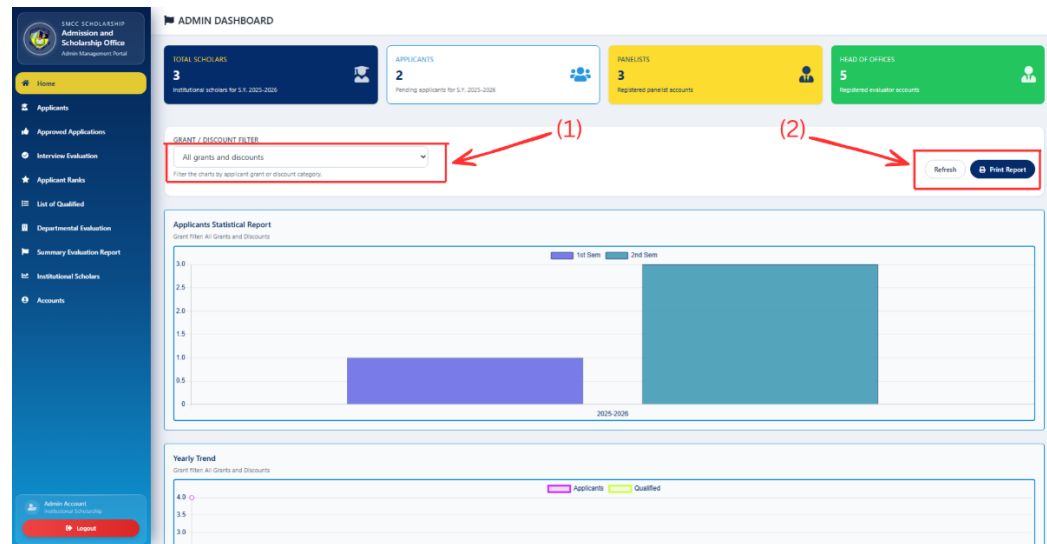
The screenshot shows the 'Student Assistant Evaluation' form. It includes a table for ratings (C.2 to C.5) with a grid of radio buttons. Below the table are sections for 'D. Strength(s)/Areas for Improvement:', 'E. Evaluator's Comment(s)/Recommendation:', and 'Evaluator's Signature'. A 'SAVE EVALUATION' button is at the bottom. Red arrows indicate: (1) pointing to the rating grid, (2) pointing to the 'D. Strength(s)/Areas for Improvement:' text box, (3) pointing to the 'E. Evaluator's Comment(s)/Recommendation:' text box, (4) pointing to the 'Evaluator's Signature' box, and (5) pointing to the 'SAVE EVALUATION' button.



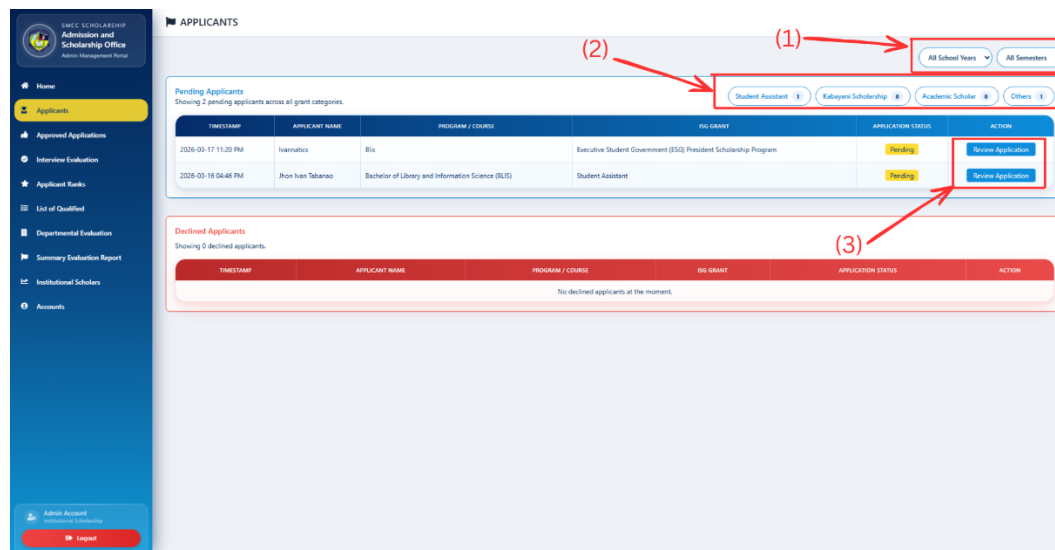
This department head interface is used to evaluate the performance of the assigned student assistant. Rating options are used to select the correct score for each evaluation criterion (1). The Strength(s)/Areas for Improvement box is used to write down the strong points of the student assistant and the areas that still need improvement (2). The Evaluator Comment(s)/Recommendation Box is used to enter the department head's comments or recommendations (3). The Evaluator's Signature box is used for the department head's signature before saving the evaluation (4). The Save Evaluation button is used to save and submit the completed evaluation form (5).

The screenshot shows a web application interface for changing a password. On the left is a dark blue sidebar with the text 'Admission and Scholarship Office' and navigation links: 'Home', 'My SA's', 'Show Evaluation', and 'Change Password' (highlighted in yellow). The main content area is light blue and titled 'Change Password' with a sub-header 'Update your head of office account password.' Below this is a white form box containing three input fields labeled 'Current Password', 'New Password', and 'Confirm Password'. A red box labeled (1) encompasses these three fields. Below the fields is a blue button labeled 'UPDATE PASSWORD', which is pointed to by a red arrow labeled (2). At the bottom of the sidebar, there is a 'Logout' button.

This department head interface is used to update the account password for security. The password fields are used to enter the current password, new password, and confirm password (1). The Update Password button is used to save the new password after completing all fields correctly (2).



This admin interface serves as the main dashboard of the system. The Grant / Discount Filter is used to sort and display the report based on the selected grant or discount category (1). The Refresh and Print Report buttons are used to update the displayed report and print the dashboard report as needed (2).



This admin interface displays the list of scholarship applicants in the system. School Year and Semester filters are used to display applicant records based on the selected period (1). The grant category tabs are used to quickly view applicants under a specific scholarship type (2). The Review Application button is used to open and review the submitted record of the selected applicant (3).



The screenshot shows the 'Admin Interface' for reviewing submitted forms and uploaded documents. The interface includes a sidebar with navigation links: Home, Applicants, Approved Applications, Interview Evaluation, Applicant Rank, List of Qualified, Departmental Evaluation, Summary Evaluation Report, Institutional Scholars, and Accounts. The main content area displays a form for reviewing an applicant's submission. The form includes fields for Contact Number, Company's Name, Company's Address, Father's Name, Contact Number, Company's Name, and Company's Address. It also has sections for 'Name and signature of applicant', 'Name and signature of personnel', 'Recommendation approval', and 'Approved by'. Below the form, there is a table for 'Uploaded Requirements' with columns for 'Requirement' and 'File'. The table lists two requirements: '2x2 ID Picture' and 'Endorsement Letter from OSAS'. The 'File' column shows the uploaded files: 'Yellow Minimalist Business Plan Process Flow Chart.png' and 'Smart Farmbot Feeder (1).png'. At the bottom of the form, there are two buttons: 'Approve Application' (labeled 2) and 'Decline Application' (labeled 3). A red box labeled (1) highlights the 'View' and 'Download' links for the uploaded files.

This admin interface is used to review the submitted form and uploaded documents of the applicant. The View and Download links are used to open or save uploaded prerequisite files for review (1). The Approve Application button is used to accept the applicant's submission (2). The Reject Application button is used to reject the applicant's submission (3).

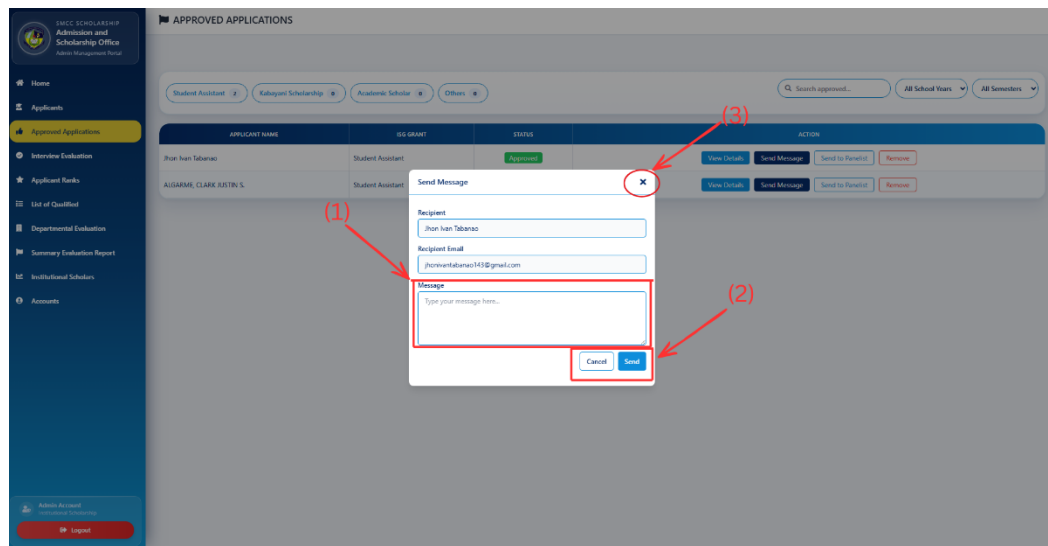
The screenshot shows the 'APPROVED APPLICATIONS' section of the admin interface. It features a sidebar with navigation links: Home, Applicants, Approved Applications, Interview Evaluation, Applicant Rank, List of Qualified, Departmental Evaluation, Summary Evaluation Report, Institutional Scholars, and Accounts. The main content area displays a table of approved applicants. The table has columns for 'APPLICANT NAME', 'ISS GRANT', 'STATUS', and 'ACTION'. The table lists two applicants: 'Jhon Iuan Tabano' and 'ALGARME, CLARK JUSTIN S.'. The 'ISS GRANT' column shows 'Student Assistant' for both. The 'STATUS' column shows 'Approved' for both. The 'ACTION' column contains buttons for 'View Details', 'Send Message', 'Send to Panelist', and 'Remove'. A red box labeled (1) highlights the 'Student Assistant' tab. A red box labeled (2) highlights the search box, 'All School Years' filter, and 'All Semesters' filter. Red arrows labeled (3), (4), (5), and (6) point to the 'View Details', 'Send Message', 'Send to Panelist', and 'Remove' buttons respectively.

APPLICANT NAME	ISS GRANT	STATUS	ACTION
Jhon Iuan Tabano	Student Assistant	Approved	View Details, Send Message, Send to Panelist, Remove
ALGARME, CLARK JUSTIN S.	Student Assistant	Approved	View Details, Send Message, Send to Panelist, Remove

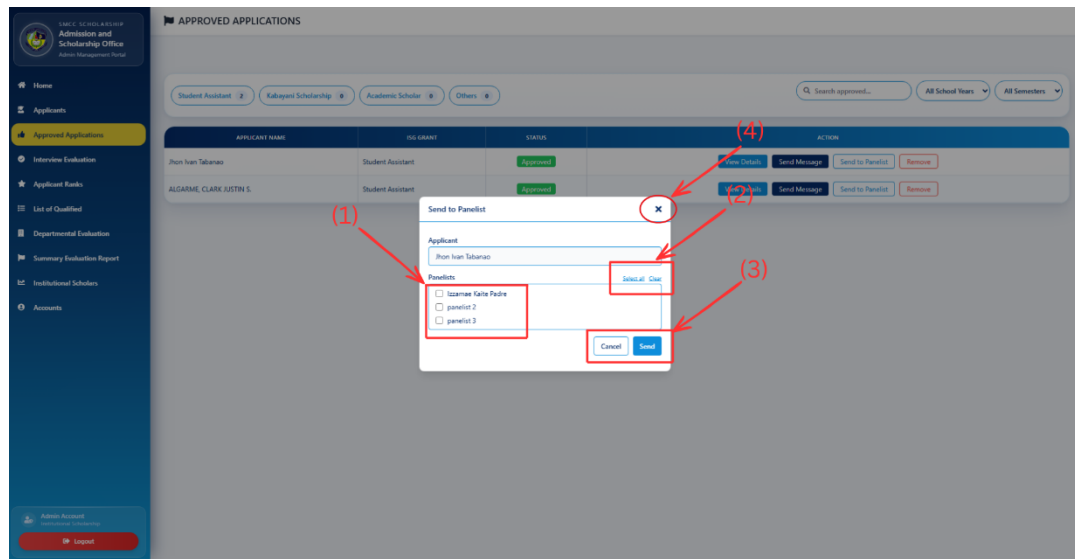
This admin interface displays the list of approved scholarship applicants. The grant category tabs are used to display approved applicants by scholarship type (1). The search box, School Year filter, and Semester filter are used to search and sort approved applicant records (2). The View Details button is used to open the entire record of the approved applicant (3). The Send Message button is



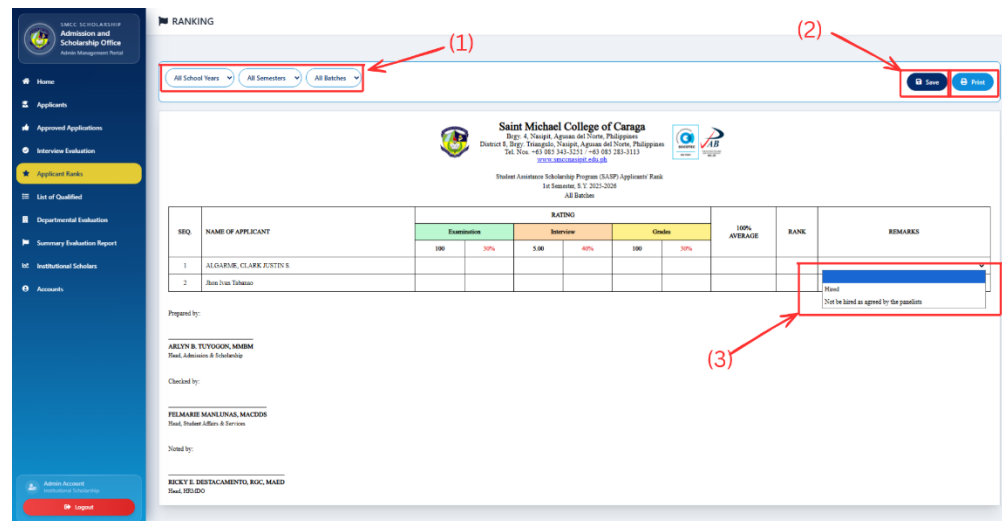
used to send a message to the selected applicant (4). The Send to Panelist button is used to assign the approved applicant to a panelist for interview review (5). The Remove button is used to remove the applicant from the approved list if necessary (6).



This admin interface is used to send a message to the selected applicant. The Message box is used to type the message to be sent to the applicant (1). The Cancel and Send buttons are used to cancel the action or send the written message (2). The Close button is used to exit the message window (3).



This admin interface is used to assign an applicant to one or more panelists. The panelist list with checkboxes is used to select the panelists who will review the applicant (1). The Select all and Clear options are used to quickly select or remove selected panelists (2). The Cancel and Send buttons are used to cancel the action or confirm the assignment (3). The Close button is used to exit the panelist assignment window (4).



This admin interface is used to view and manage the ranking of applicants. School Year, Semester, and Batch filters are used to display the ranking record based on the selected options (1).



The Save and Print buttons are used to save the ranking result and print the report (2). The Remarks dropdown is used to select the last sentence for the applicant, such as hired or not hired (3).

This admin interface is used to display the list of qualified applicants. School Year, Semester, and Batch filters are used to display the eligible list based on the selected period (1). The Save button is used to save the currently qualified applicant record (2). The Print button is used to print the report of qualified applicants (3). The Assigned Office dropdown is used to select the office assignment of the qualified applicant (4).

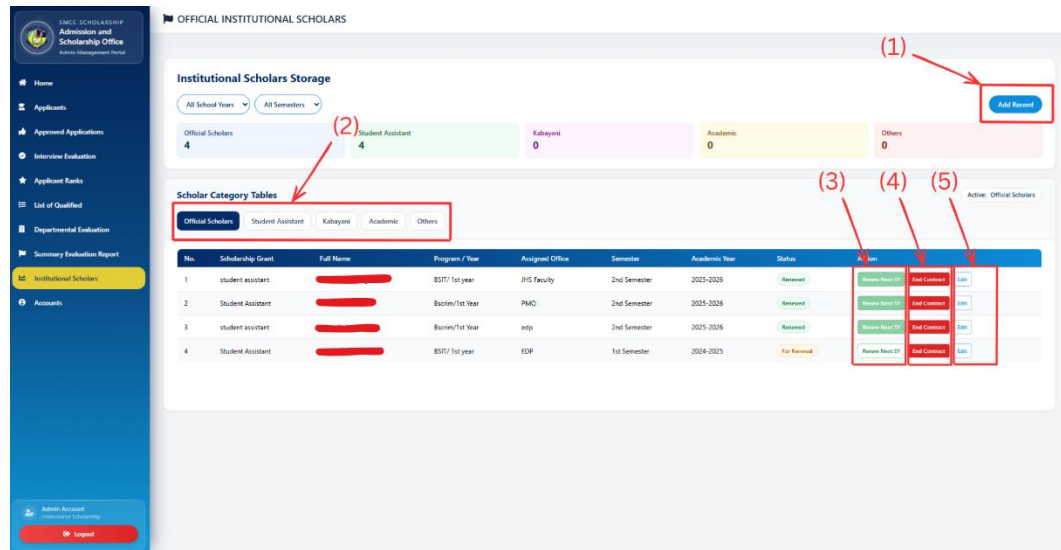


This admin interface is used to monitor the evaluation records of student assistants when the evaluation window is closed. The Open for Evaluation button is used to open the evaluation period for department heads (1). The search box is used to search for a specific student assistant record (2). The Academic Year filter is used to display records for the selected school year (3). The Semester filter is used to display records for the selected semester (4). Action buttons and status labels show whether the review is closed or available for viewing (5).

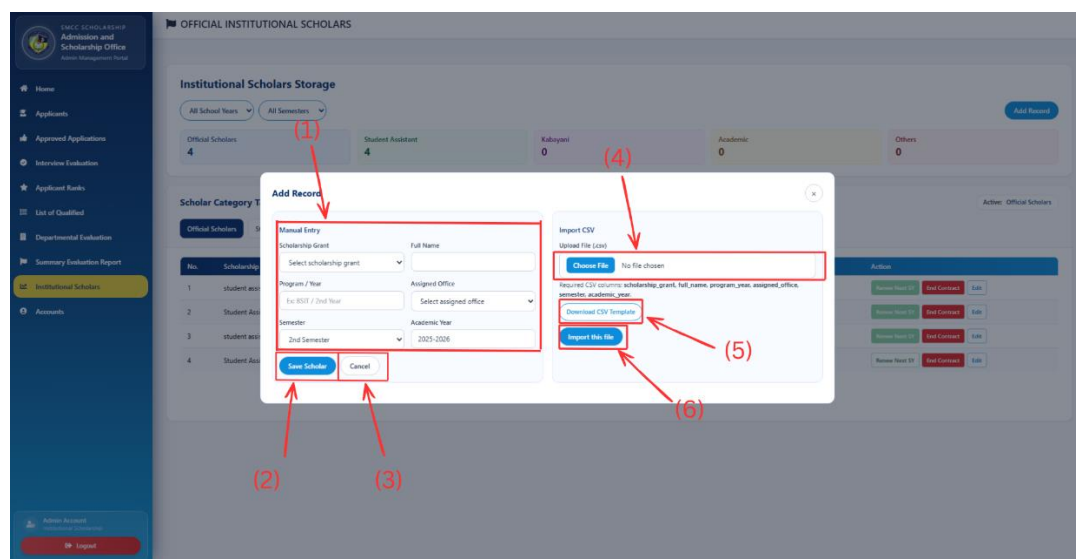
The screenshot shows the 'DEPARTMENTAL EVALUATION' interface. On the left is a sidebar with navigation links: Home, Applicants, Approved Applications, Interview Evaluation, Applicant Ranks, List of Qualified, Departmental Evaluation (highlighted), Summary Evaluation Report, Institutional Scholars, and Accounts. The main content area is titled 'Student Assistants Evaluation List'. It includes a 'Close Evaluation' button (1) in the top right. Below the title, there's a search box (2) and filters for 'Academic Year' (3) and 'Semester' (4). A table lists evaluation records with columns: Timestamp, Name, Course, Year Level, Assigned Office, Academic Year, Semester, Status, and Action. The table shows three records with status labels like 'Not yet evaluated' or 'Evaluated'. The 'Action' column contains buttons like 'Waiting for Head' or 'View Evaluation' (5).

Timestamp	Name	Course	Year Level	Assigned Office	Academic Year	Semester	Status	Action
---	---	BScIn	1st Year	edp	2025-2026	2nd Semester	Not yet evaluated	Waiting for Head
3/24/2026, 6:25:16 AM	---	BSIT	1st year	JHS Faculty	2025-2026	2nd Semester	Evaluated	View Evaluation
---	---	BScIn	1st Year	PHD	2025-2026	2nd Semester	Not yet evaluated	Waiting for Head

This admin interface is used to monitor analysis records once the analysis window is open. The Close Review button is used to close the review period (1). The search box is used to search for a specific student assistant record (2). Academic Year and Semester filters are used to sort review records based on the selected term (3). Action buttons and status labels show whether the record is waiting for the department head or is available for viewing (4).



This admin interface is used to manage the list of official institutional scholars. The Add Record button is used to add a new scholar record to the system (1). The scholar category tabs are used to display scholar records by scholarship type (2). The Renew Next SY button is used to renew the scholar's record for the next school year (3). The End Contract button is used to end the scholar's record or assignment (4). The Edit button is used to update the scholar's information (5).





This admin interface is used to add a new scholar record either through manual entry or CSV import. The Manual Entry fields are used to type the scholar's details individually (1). The Save Scholar button is used to save the manually entered scholar record (2). The Cancel button is used to stop the action and close the entry process (3). The Select File button is used to select a CSV file for upload (4). The Download CSV Template button is used to get the sample format for CSV import (5). The Import this file button is used to upload and save the selected CSV file to the system (6).

The screenshot displays the 'ACCOUNTS' section of the 'SMCC SCHOLARSHIP Administration and Scholarship Office' portal. It features two main tables: 'PANELISTS' and 'HEAD OF OFFICES'. Red arrows and numbers (1-6) highlight specific UI elements:

- (1) Points to the 'ADD PANELIST' button at the top right.
- (2) Points to the 'ADD HEAD OF OFFICE' button at the top right.
- (3) Points to the 'New password' input field in the 'PANELISTS' table.
- (4) Points to the 'Reset' button in the 'PANELISTS' table.
- (5) Points to the 'New password' input field in the 'HEAD OF OFFICES' table.
- (6) Points to the 'Reset' button in the 'HEAD OF OFFICES' table.

The 'PANELISTS' table has columns: Username, Full Name, Password Hash, Status, and Action. The 'HEAD OF OFFICES' table has columns: Username, Name, Last Name, Office, Password Hash, Status, and Action.

This admin interface is used to manage panelist and office head accounts. The Add Panelist button is used to create a new panelist account (1). The Add Head of Office button is used to create a new head of office account (2). The New password field under the panelist table is used to enter a new password for the selected panelist account (3). The Reset button next to it is used to apply the new password for the selected panelist (4). The New password field under the head of office table is used to enter a new password for the selected head of office account (5). The Reset button next to it is used to apply the new password for the selected head of office (6).



C. Letter of Permission/Permit to Conduct the Study



SAINT MICHAEL COLLEGE OF CARAGA

Brgy. 4, Nasipit, Agusan del Norte, Caraga Region

Tel. Nos. (085) 343-2232 / (085) 283-3113

www.smcnasipit.edu.ph



Date: August 2, 2025

To:

The Head, Admission and Scholarship Office

Saint Michael College of Caraga

Nasipit, Agusan del Norte

Subject: Request for Data Gathering and Stakeholder Insights for Capstone Project

Dear Mrs. Tuyogon,

Good day!

We, the undersigned Bachelor of Science in Information Technology students of Saint Michael College of Caraga, are currently conducting our capstone project entitled "*SMCC Institutional Scholars Supervisory System with Performance-Based Retention Decision Mechanism.*"

In line with this, we humbly request permission to gather relevant data, documents, and insights from your office that will help us better understand the current scholarship application, evaluation, and retention processes. The information and feedback you provide will be treated with the highest confidentiality and will be used solely for academic purposes in developing an improved and efficient system for the institution.

We would also like to schedule a brief consultation or interview with your office at your most convenient time to gather your inputs and suggestions regarding the proposed system's design and functionality.

We sincerely appreciate your time and support in this research endeavor, which aims to improve scholarship management at SMCC.

Thank you very much, and we look forward to your positive response.

Respectfully yours,

Jhon Ivan C. Tabanao

Lead Researcher

Saint Michael College of Caraga

Noted by:

Rea Mie Omas-as

Capstone Adviser

Approved by:

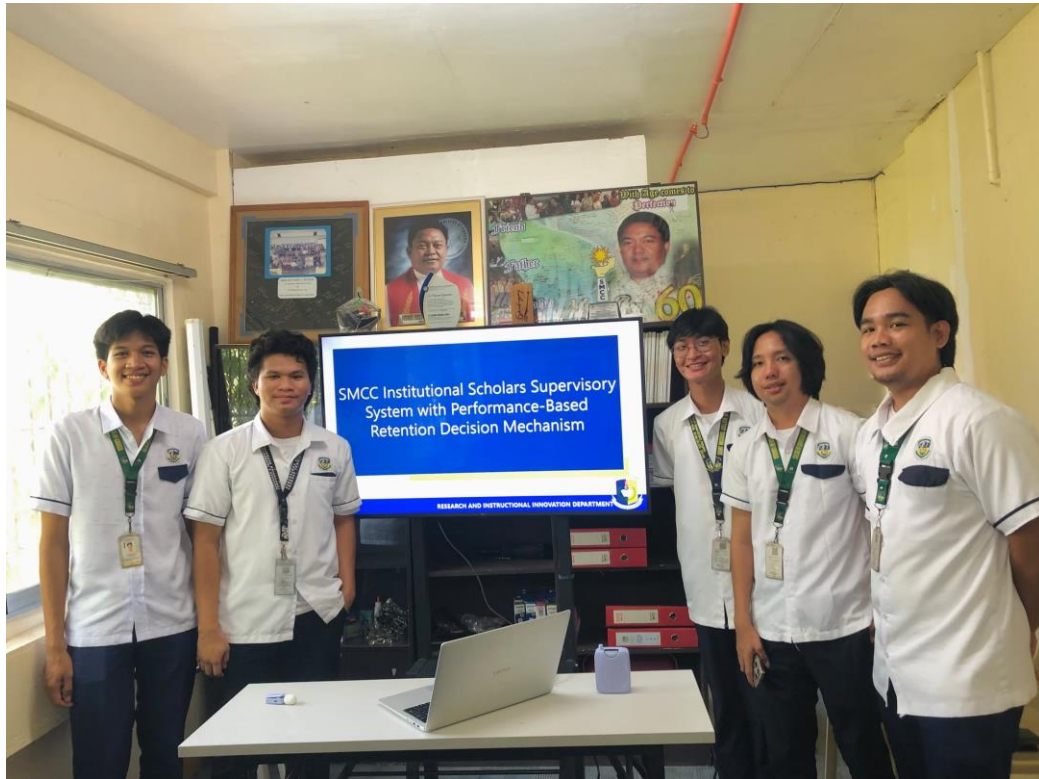
Daisa G. Gupit, MIT

DEAN, CCIS

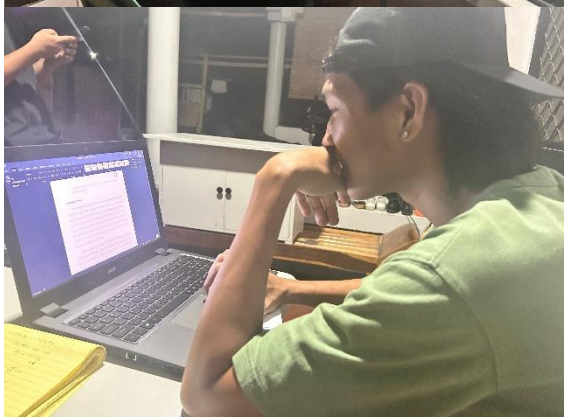
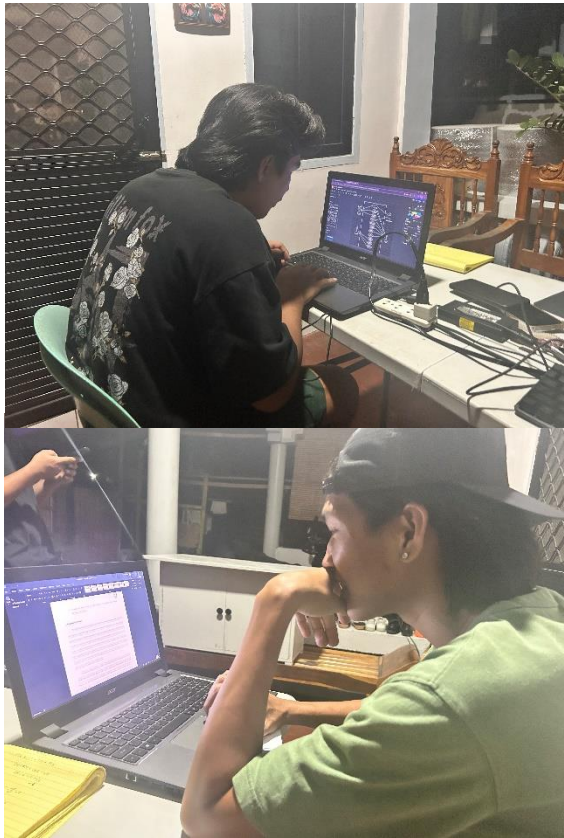


D. Narratives and Photo Documentation

Title Hearing



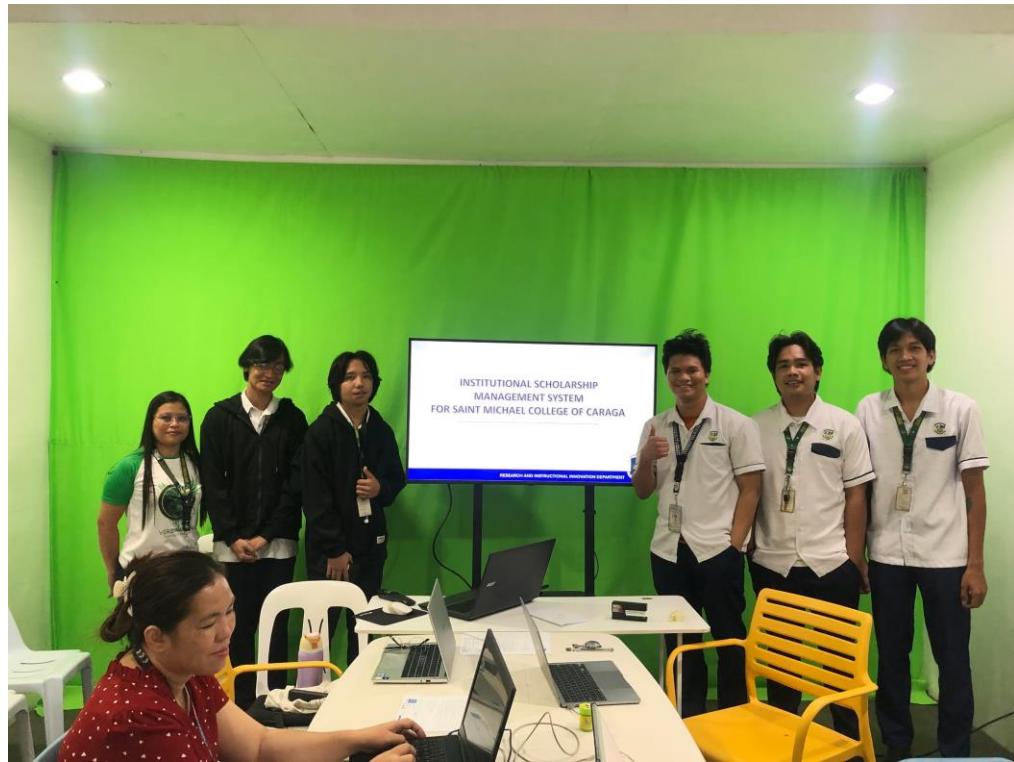
The researchers successfully defended their thesis, entitled “SMCC Institutional Scholars Supervisory System with Performance-Based Retention Decision Mechanism,” during the title hearing, under the guidance of their adviser, Rea Mie A. Omas-As.



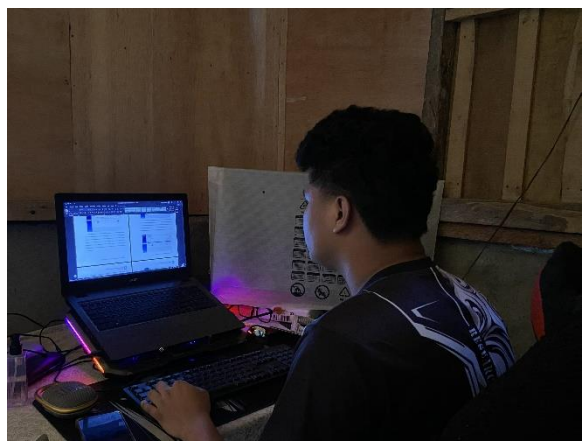
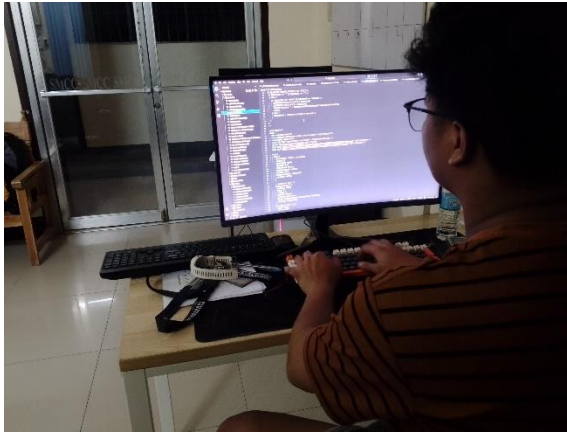
During this phased, the researchers are working on Chapters 1 and 2 of their thesis, which includes writing the introduction, reviewing related literature, and preparing the software and hardware requirements and design specifications of their proposed capstone.



Proposal Defense



The researchers successfully defended their thesis entitled “Institutional Scholarship Management System for Saint Michael College of Caraga” during the proposal defense, with the guidance of their adviser, Rea Mie A. Omas-As.



The researchers are tasked with planning, building, and connecting the system's core features, handling database integration, preparing technical diagrams, and making sure the system satisfied all functional requirements through testing and continuous improvement.



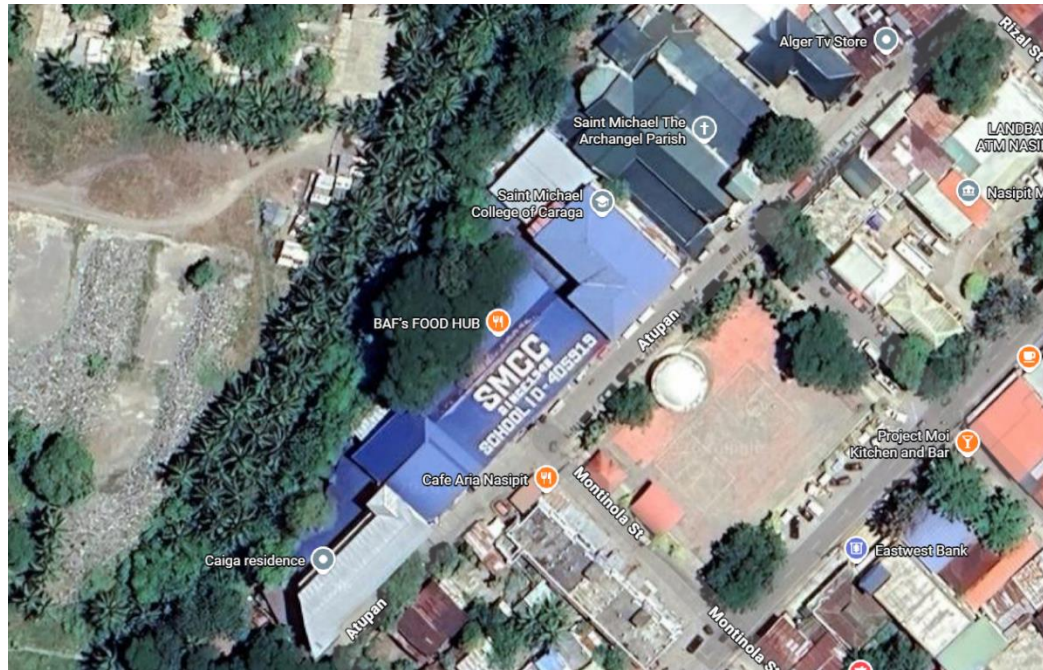
Chapters 3 and 4 – Process of making



F. Certificate of Implementation



F. Maps of the Research locale





CURRICULUM VITAE

NAME: JHON IVAN C. TABANAO

ADDRESS: D-1 Santa Ana, Nasipit , Agusan Del Norte

EMAIL: jhonivantabanao143@gmail.com

Personal Information

Sex : Male

Date of birth : November 14, 2004

Height : 5'7

Weight : 85kg

Civil Status : Single



Educational Attainment

Elementary : Jagumitan Elementary School

Secondary : Northwestern Agusan Colleges

Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
N/A	N/A



CURRICULUM VITAE

NAME: JOHN MARK G. SARABIA

ADDRESS: Barangay 4, Nasipit Agusan del Norte

EMAIL: johnmarkgaran25@gmail.com

Personal Information

Sex : Male

Date of birth : September 3, 2003

Height : 5'7

Weight : 60kg

Civil Status : Single



Educational Attainment

Elementary : Lighthouse Christian School

Secondary : Saint Michael College of Caraga

Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
N/A	N/A



CURRICULUM VITAE

NAME: MARK CHRISTIAN JOVEN BALATAYO

ADDRESS: D-Adelfa, Brgy. 1 Poblacion, Nasipit, Agusan Del Norte

EMAIL: markcjbalatayo@gmail.com

Personal Information

Sex : Male
 Date of birth : December 14, 2000
 Height : 5'11
 Weight : 77kg
 Civil Status : Single



Educational Attainment

Elementary : Medina Central School
 Secondary : St. Isidore School
 Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
N/A	N/A



CURRICULUM VITAE

NAME: RONALD JHON N. REOJA

ADDRESS: Brgy. Cahayagan, Carmen, Agusan Del Norte

EMAIL: reojaronald@gmail.com

Personal Information

Sex : Male
Date of birth : April 27, 2004
Height : 5'6
Weight : 45kg
Civil Status : Single



Educational Attainment

Elementary : Cahayagan Elementary School
Secondary : Cahayagan National High School
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
N/A	N/A



CURRICULUM VITAE

NAME: WARREN C. SALVA

ADDRESS: D-east Cubicubi Nasipit Agusan del Norte

EMAIL: waaween060@gmail.com

Personal Information

Sex : Male
 Date of birth : October 6, 2004
 Height : 5'11
 Weight : 64kg
 Civil Status : Single



Educational Attainment

Elementary : Mateo B. Abao Central Elementary School
 Secondary : Nasipit National Vocational School
 Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
N/A	N/A



APPENDICES

- A - Letter of Approval with Certificate of Appearance and Photo
- B - Instrument with Informed Consent
- C - Map of the Research Locale *(Include Coordinates)*
- D - Curriculum Vitae
- E - Narrative & Photo Documentation
- F- Ethics Committee Approval

CERTIFICATE OF AUTHENTIC AUTHORSHIP

This is to certify that I/we, the undersigned author(s), have solely and honestly written the entire paper titled: [“Click here to enter text.”](#) All contents, ideas, data, and analysis presented in this paper are the results of my/our own diligent work and intellectual effort. The paper has been crafted to the best of my/our abilities, in full compliance with academic integrity standards and ethical research practices.

I/We further declare that:

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